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MINISTRY OF ROADS AND TRANSPORT

RDM 6.1

Road Design Manual

Volume 6: Traffic Control Facilities & Communication Systems Design

Part 1: Road Marking

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Foreword

This manual was developed by the Ministry pursuant to The Fourth Schedule of the Constitution which assigns to the National Government the functions and powers of setting standards for the construction and maintenance of all public roads including those under the County Governments.

It is part of a series of manuals that replace the first generation of road manuals developed in the first and second decades after independence. This second generation of the road manuals were developed to cover the entire road project cycle covering planning, appraisal, design, contracts, construction, maintenance, operations and monitoring. The series incorporates best practices, climate change considerations, and recent technologies to enable the provision of road infrastructure that is safe, secure, and efficient.

Under the Kenya Vision 2030 long term plan, infrastructure expansion and modernisation are some of the foundations for the realisation of economic, social and political transformation of Kenya into a rapidly industrialising middle-income country. The plan envisages an integrated, safe and efficient transport and communication infrastructure network consisting of roads, railways, ports, airports, waterways, and telecommunications infrastructure.

The strategies to be pursued under the Vision 2030 plan to improve infrastructure services and to maximise the economic and social impacts of infrastructure development and management include: Strengthening of the institutional framework for infrastructure development and maintenance; Raising efficiency and quality of infrastructure projects; Enhancing local content of identified infrastructure projects to minimise import content; Benchmarking infrastructure facilities and services provision with globally acceptable performance standards; and, Implementing infrastructure projects that will stimulate demand in hitherto marginalised areas.

The first three 5-year Medium Term Plans (MTP) under the Vision 2030 from 2008 to 2022 targeted construction of 1,950 km, 5,500 km and 10,000 km of new paved roads under MTP I, II and III, respectively, totalling 17,450 km. This was a massive infrastructure development program intended to double the paved road network in 10 years compared to 8,600 km developed from independence in 1963 to 2008.

Implementation of MTP I to III resulted in the construction of 14,000 km of paved roads, which extended the paved road coverage to Arid and Semi-Arid regions, that had been previously neglected. However, some key milestones of the Vision 2030 goals have not been realised. This has been due to internal and external challenges. External challenges included: climate change – prolonged droughts; the emergence of COVID-19 pandemic; global supply chain disruptions; exchange rate volatility; and rising interest rates in the leading economies.

The internal challenges included: inadequate road maintenance equipment; pavement overloading by heavy goods vehicles; huge maintenance backlog of the road network; low contracting and supervision capacity particularly in the Counties; poor quality control and assurance of works; congestion in urban areas; encroachment on road reserves; high costs and delays in payments of land acquisition; lack of harmonisation of cross-border transport regulation and operational procedures; rapid urbanisation; increased traffic volume with exponential growth of motorcycle traffic; high cost/delays in relocation of utilities and services along and across road reserves; inadequate funding of projects and programs; and, delay or default in payments for goods, services and works.

The implementation of MTP III came to an end on 30th June 2023, ushering in the implementation of the Fourth Medium Term Plan (MTP IV), which has been aligned to the aspirations of the Kenya Vision 2030 and the Kenya Kwanza Government's Bottom-Up Economic Transformation Agenda (BETA) planning approach and its key priorities.

BETA is the Government's transformation agenda geared towards economic turnaround through a value chain approach. BETA has targeted sectors with the highest impact to drive economic recovery and growth. This will be achieved through bringing down the cost of living; eradicating hunger; creating jobs; expanding the tax base; improving foreign exchange balances; and inclusive growth. BETA ensures rational resource allocation by eliminating wastage of resources occasioned by duplication, overlaps, fragmentation and ineffective coordination in the implementation of programmes and projects.

The Fourth Medium Term Plan key priorities are clustered under five key sectors, namely: Finance and Production; Infrastructure; Social; Environment and Natural Resources; and Governance and Public Administration. The infrastructure sector seeks to: enhance transport connectivity by constructing 6,000 km of new roads, maintaining rural and urban roads, rail, air and seaport facilities and services; expand communication and broadcasting systems; and promote the development of energy generation and distribution by increasing investments in green energy (geothermal, wind, solar and hydro). The infrastructure gap is expected to be bridged by promoting economic participation of the private sector through public private partnerships in the financing, construction, development, operation, and maintenance of infrastructure.

The plan entails a shift of focus to fundamentals in project planning and implementation which include: respect for technical input, regulations and standard practices; adherence to project life cycle i.e., planning, feasibility studies and design before procurement of works; public and stakeholder consultation; procurement within budgetary ceilings; shifting focus during project implementation from the finished product 'black top' to the construction of the foundation; building local capacity particularly MSMEs by ensuring prompt payments; and capacity building at all levels to enable internalisation of policies and processes.

The first generation of the road manuals were used for 35 to 45 years. It is my sincere hope that the second generation of the road standards which have been developed in alignment with the Government's strategy will provide guidance in solving most of the above challenges and those expected to emerge in the next 50 years. Implementation of the manuals will enable achievement of the Government aspirations which include inclusive growth; creation of sustainable employment; building of MSMEs; climate change adaptation and realisation of the UN SDGs; enhanced efficiency in management of infrastructure and transport system; and, laying the foundation for the next national long-term plan at the end of the Vision 2030.

On behalf of the Government of Kenya, I would wish to thank the European Union for financing the development of the first drafts of the manuals in 2009 and the African Development Bank for the financial support in the review and updating of the manuals. I would also like to thank the members of the National Steering Committee and the Technical Task Force for their input. The Technical Administrators, and the Kenya National Highways Authority (KeNHA) for the procurement and able administration of the consultancy Contract. I also thank the Consultant, TRL Limited for their role in providing technical expertise that was essential for the success of the manuals updating exercise. I also wish to express my deepest appreciation to our stakeholders and all those who have contributed to this process and the staff of the Ministry for their continued input.

Hon. Davis K. Chirchir, E.G.H

Cabinet Secretary, Ministry of Roads and Transport

Preface

The Manual is intended to assist users to use proper road marking in order to safely and efficiently control traffic movement. The various types of road marking material are described along with the colour pattern, for use on flexible and rigid pavements. The major types of marking such as: longitudinal, transverse, block, arrows, word messages are explained in detail. Usage of different types of road studs are explained. Marking for at-grade and grade separated junctions are elaborated. Marking for bus lane, bus stop, truck lay-by, toll plaza and parking spaces are presented in detail. Object marking for obstructions on or near carriageway are explained. Performance measures for wear, durability, night visibility and skid resistance are detailed. Testing methods for marking and maintenance are explained.

Eng. Joseph M. Mbugua, CBS

Principal Secretary, State Department for Roads

Document Management

Document Status

This document has the status of a Manual. Users shall apply the contents there-in to fully satisfy the requirements set out. The content of the manual is based on current practice in Kenya and latest practices in the road sector, both regionally and internationally.

Sources of the Document

Copies of the document can be obtained from:

The Principal Secretary, State Department for Roads, Ministry of Roads and Transport, Works Building, Ngong Road, P.O. Box 30260 - 00100, NAIROBI Email: ps@road.go.ke

A secured PDF copy maybe downloaded from: www.roads.go.ke/downloads

Notification of Errors and Requests for Amendments

While all care and consideration has been applied in the compilation of this document, the Ministry accepts no responsibility for failure in any way related to the application of this manual or any reference documents cited in it.

Requests for edits and corrections can be freely sent to the following address:

The Principal Secretary, State Department for Roads, Ministry of Roads and Transport, Works Building, Ngong Road, P.O. Box 30260 - 00100, NAIROBI Email: ps@road.go.ke

Amendments Request Form

Request No.	Name	Organisation	Chapter	Page	Section/ Clause	Ref. to: Figure/ Table/	Type of request	Request

Type of request: General – G; Editorial – E; Technical - T

Amendments to Date

Amendment No.	Description	Amendment Effective Date	Amendment Approved by

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A National Steering Committee was set up and chaired by the Permanent Secretary, Ministry of Roads and Transport, with the following membership: Principal Secretary for Devolution, Office of the Deputy President; Chief Executive Officer, Inter-Governmental Relations Technical Committee; Chief Executive Officer, Council of Governors; Managing Director and Council Secretary, Kenya Bureau of Standards; Director, National Transport and Safety Authority; Director General, Kenya Roads Board; Director General, Kenya Wildlife Services; Chief Executive Officer, Engineers Board of Kenya; Director General, Kenya Rural Roads Authority; Director General, Kenya Urban Roads Authority; President, Institution of Engineers Kenya; Director Policy, Strategy and Compliance; Kenya National Highways Authority; Chief Engineer, Roads Division, State Department for Roads; Chief Engineer, Materials Testing and Research Division, State Department for Roads.

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Abbreviations

AASHTO	American Association of State Highway and Transportation Officials
ASTM	American Society for Testing and Materials
BPN	British Pendulum Number
BS	British Standard
CLS	Combined Line System
EAS	East Africa Standard
EN	European Standards
EP	Edge of Pavement
FRP	Fibre Reinforced Plastic
IRDH	International Rubber Hardness Degree
IRC	Indian Roads Congress
KeNHA	Kenya National Highways Authority
KeRRA	Kenya Rural Roads Authority
Kmph, km/h	Kilometres per hour
KS	Kenya Standard
KURA	Kenya Urban Roads Authority
LED	Light Emission Diode
LOS	Level of Service
MMA	Methyl Methacrylate road markings
MoR	Ministry of Roads and Transport
NMT	Non-Motorised Traffic
PSD	Passing Sight Distance
RC	Reverse Camber
SBL	Start of Barrier Line
SE	Super-Elevation
TRL	Transport Research Laboratory
USD	United States Dollar
vph	Vehicles per Hour
VRU	Vulnerable road user
VUP	Vehicle Under Pass
WB	World Bank
WSD	Warning Site Distance

Glossary of Terms

Advance Direction Sign	A directional sign giving information about a junction ahead.
Bus Lane	A traffic lane reserved for buses either in transporting, discharging or loading passengers.
Bus Bay	A lay-by reserved for public service vehicles.
Capacity	The maximum number of vehicles which has a reasonable expectation of passing over a given section of a lane or a roadway in one direction (or in both directions for a two-lane highway) during a given time period under prevailing roadway and traffic conditions.
Central Reserve	An area separating the carriageways of a dual carriageway road.
Channelisation	A system of controlling traffic by the introduction of a traffic island or median or markings on a carriageway in order to direct traffic into predetermined paths usually at an intersection or junction.
Cycle Path	A path physically separated from motorised traffic by an open space or barrier and located either within the road reserve or an independent reserve.
Deceleration Lane	An extra width of carriageway to enable a vehicle leaving the through traffic stream to reduce speed without interfering with other traffic.
Delay	The time lost while traffic is impeded by some element over which the driver or pedestrian has no control.
Dual Carriageway	A road in which there are two physically separated carriageways reserved for travelling in opposite directions.
Edge Line	Line marking used to indicate the edge of pavement pedestrian crossing, etc. For the pedestrian crossing, the inside edge is the one closest to the stop line.
Expressway	A road having dual carriageways and hard shoulders with complete grade separation and completely fenced in. It is for the exclusive use of prescribed classes of motor vehicles.
Footpath	That portion of a road reserved exclusively for pedestrians.
Gap	The distance between solid lines on longitudinal line marking.
Highway	A way over which traffic may lawfully pass, but such traffic may be limited to a special class
Island	See Traffic Island
Kerb	A raised border of rigid material formed at the edge of a carriageway.
Lane Line	A painted line other than the separation line which separates parallel vehicle movements in the same direction.
Lay-By	A part of the highway set aside for vehicles to draw out of the traffic lanes for short periods.
Median	An island in the roadway which separates opposing traffic streams, usually on approaches to intersection.
Mountable Kerb	The kerb designed so that it can be mounted without damage to a vehicle.
Pedestrian Crossing	A transverse strip marked on a carriageway for the guidance of pedestrians crossing the road.

Glossary of Terms *(continued)*

Pedestrian Refuge	An area set aside on roadway for protection of pedestrians.
Queue	Line of stationary vehicles waiting to be given right-of-way through an intersection.
Road Authority	The Highways Authority, the Rural Roads Authority or the Urban Roads Authority in terms of the Kenya Roads Act, 2007
Separation line	A painted line marking the division of a roadway to separate opposing traffic movements.
Sight Distance	The distance over which a driver or pedestrian has an unobstructed view of conditions which affect his behaviour in traffic.
Signals	Message conveyed to motorists or pedestrians by illumination of a particular aspect of a lantern (narrow definition). The installation comprising signal lanterns, posts, controller, and associated cables, as installed in a particular street situation (broad definition).
STOP Line	A transverse pavement marking behind which vehicles should stand when stopped by traffic light signals or regulatory signs (also referred to as a stop-bar)
Tangent Point	The point where straight alignment ceases and curvature begins; also, a point of change of hand or type of curvature.
Traffic Island	A defined area within a roadway, usually at an intersection, from which traffic is intended to be excluded and which is used to guide vehicular movements and for a pedestrian refuge
Traffic Lane	A strip of carriageway intended to accommodate a single line of moving vehicles.
Traffic Sign	A sign, signal or other device, or line or mark on the carriageway, for the purpose of regulating, warning, guiding or informing traffic.
Traffic Stream	Vehicles in one or more lanes on the same approach to the controlled area which, when they have the right-of-way, will move in the same direction
Uncontrolled Pedestrian Crossing	A pedestrian crossing marked by studs, alternate black and white strips and flashing beacons where a pedestrian has priority over all vehicles by which he would be impeded.
Verge	That part of the highway outside the carriageway and generally at substantially the same level. It may contain walkways, cycle path or ditches.
Volume of Traffic	The number of vehicles passing a given point on a lane or carriageway during a specified period of time
Walkway	That portion of a road reserved exclusively for pedestrian. Often referred to as a footpath.
Warning Sign	A traffic sign, often, but not necessarily, bounded by a red triangle, warning of a particular hazard ahead requiring caution.
Wig-Wag	A type of signal control used at level crossings, swing bridges, and fire and ambulance stations. It has two red aspects arranged horizontally and flashing alternately and a steady amber aspect placed centrally between the red lamps and displayed before the red lights begin to operate.
Zig-Zag Line	A series of short, sharp angles or turns in alternate directions. This marking is used near pedestrian crossing to alert drivers and to convey that no parking is allowed.

Contents

	Foreword	iii
	Preface	iv
	Document Management	v
	Acknowledgements	vi
	Abbreviations	viii
	Glossary of Terms	ix
	Contents	xi
	List of Figures	xiv
	List of Tables	xvi
	List of Equations	xiii
1	Introduction	1
	1.1 General	1
	1.2 Objective of this Part	3
	1.3 Scope	3
	1.4 Organisation of the Manual	3
	1.5 Design Process	4
2	Road Marking Materials and General Features	5
	2.1 Background	5
	2.2 Definition	5
	2.3 Authority	5
	2.4 Thermoplastic Markings	6
	2.5 Solvent-based and Water-based Road Marking Paints	6
	2.6 Cold Applied Plastics	6
	2.7 Preformed Adhesives Tapes	7
	2.8 New Materials	7
	2.9 Retro Reflectivity	7
	2.10 Colour Pattern for Road Markings	8
	2.11 Visibility Related to Speeds	8
	2.12 External Factors Influencing Performance of Road Marking	9
	2.13 Pavement Marking on Cement Concrete Roads	9
3	Classification of Pavement Markings	11
	3.1 Classes of Markings	11
	3.2 Longitudinal Markings	11
	3.3 Transverse Markings	11
	3.4 Hazard Markings	11
	3.5 Block Markings	11
	3.6 Arrows and Word Markings	11
	3.7 Junction Markings	12
	3.8 Pedestrian & Cycle Markings	12
	3.9 Bus, Truck Layby and Toll Plaza Markings	12
	3.10 Parking	13
	3.11 Object Markings	13
4	Longitudinal Markings	15
	4.1 General	15
	4.2 Centre Line and Edge Line	16
	4.3 Lane Line Markings	19
	4.4 Dimensions of Marking	20
	4.5 Combined Line System – Visibility and Speeds	20
	4.6 Other Longitudinal Markings	23
	4.7 Raised Profile Line Marking	23

5	Traverse Markings	25
	5.1 General	25
	5.2 STOP Marking	26
	5.3 GIVE WAY Marking	27
	5.4 NO ENTRY Markings	29
	5.5 Transverse Rumble Strips	30
6	Hazard, Block, Arrow Markings & Word Messages	31
	6.1 General	31
	6.2 Hazard Markings	31
	6.3 Block Markings	31
	6.4 Arrow Markings	31
	6.5 Word Messages	34
7	Markings for At-Grade Intersections	35
	7.1 General	35
	7.2 Simple Junction	36
	7.3 Skew or Y-junction	38
	7.4 Ghost Island Junction	38
	7.5 Right Turn Protected T-Junction	41
	7.6 Staggered Junction	41
	7.7 Dual Carriageway Junctions	41
	7.8 Signalised Intersection	41
	7.9 Roundabouts	44
	7.10 Double Roundabout	47
	7.11 Mini Roundabout	48
8	Grade Separated Intersections	49
	8.1 General	49
	8.2 Merging/Diverging Without Lane Gain or Lane Drop	49
	8.3 Merging/Diverging With Lane Gains and Lane Drop	49
9	Markings for Pedestrians and Cyclists	51
	9.1 General	51
	9.2 Pedestrian Crossings	51
	9.3 Marking for Vulnerable Road Section	52
	9.4 Cycle Lane Marking	53
	9.5 Railway Level Crossing	54
10	Markings for Buses, Truck Lay-Bys and Toll Plaza	55
	10.1 General	55
	10.2 Bus Lane	55
	10.3 Bus Stop	56
	10.4 Bus Bay Marking	56
	10.5 Truck Lay-by Marking	56
	10.6 Toll Plaza Marking	58
11	Parking and Restrictions	59
	11.1 General	59
	11.2 Parking Spaces	59
	11.3 Restrictions	60
12	Object Marking	63
	12.1 General	63
	12.2 Markings for Objects Within the Carriageway	63
	12.3 Markings for Objects adjacent to Carriageway	64

13	Road Studs	67
	13.1 General	67
	13.2 Types of Road Studs	67
	13.3 Siting and Placing of Road Studs	67
	13.4 Colour of Road Studs	68
	13.5 Spacing of Road Studs	68
	13.6 Solar Studs	72
	13.7 Studs on Concrete Pavement	72
	13.8 Studs on Guard Rails	72
14	Performance Assessment and Monitoring	73
	14.1 General	73
	14.2 Wear Durability	73
	14.3 Daytime Visibility	73
	14.4 Night Visibility	73
	14.5 Performance Criteria	74
	14.6 Skid Resistance	74
15	Warranty and Testing Methods	75
	15.1 General	75
	15.2 Testing Methods	75
	15.3 An Assessment of Wear	76
	15.4 Road Marking Measurements	76
	15.5 Road Marking Evaluation	76
	15.6 Category of Defect and Maintenance	77
	15.7 Records	77
	15.8 Inventory	77
16	Bibliography	79
17	Appendices	81
	Appendix A Pavement Markings	81
	Appendix B Details of Arrow Marking	89
	Appendix C Details of Letter and Numbering	91
	Appendix D Measurement Method for the Luminance Coefficient	101
	Appendix E Procedure for the In-Situ Testing of Retroreflection Using Handheld Devices	103
	Appendix F Flow Diagram for Inspection Method	105
	Appendix G Measuring Method for Skid Resistance	107

List of Figures

Figure 2.1	Retro reflectivity in pavement-marking using glass beads.	7
Figure 4.1	Longitudinal Markings	15
Figure 4.2	Longitudinal Markings - Slow Lane Arrangement	15
Figure 4.3	Typical Network Marking Layouts	17
Figure 4.4	Typical Road and Junction Marking Layout	18
Figure 4.5	Edge Lines and Junctions	19
Figure 4.6	Splayed Marking	22
Figure 4.7	Raised Profile Edge Lines	23
Figure 5.1	Typical STOP Marking Layout	26
Figure 5.2	Typical 'GIVE WAY' Marking Layout	28
Figure 5.3	Triangular GIVE WAY Marking Ratios	28
Figure 5.4	NO ENTRY LINE and Markings for use with NO ENTRY Sign	29
Figure 5.5	Details of Rumble Strips	30
Figure 6.1	Deflection Arrow	32
Figure 6.2	Bifurcation and Lane Indication Arrows (All dimensions in m)	32
Figure 6.3	Mandatory Turn Arrow	33
Figure 7.1	Dimensions for Minor Road Approaches	35
Figure 7.2	Prohibitory Marking at Intersections	36
Figure 7.3	Typical Marking for Simple T-Junction	37
Figure 7.4	Typical Marking for Simple Skew or Y-Junction	37
Figure 7.5	Hatch Marking and Ghost Island	38
Figure 7.6	Junction with Ghost Islands	39
Figure 7.7	Marking for Refuge for Physical Island	39
Figure 7.8	Typical Marking T-Junction with Physical Islands	40
Figure 7.9	Layout for Staggered Intersection (with Right Turn Protected Arrangement)	40
Figure 7.10	Typical Marking for Junction with a Divided Carriageway	42
Figure 7.11	Marking in Signal Controlled Junction	42
Figure 7.12	Typical Marking for Signalised Junction	43
Figure 7.13	Typical Marking for Signalised Junction (with free left turn)	44
Figure 7.14	Give Way Marking for Priority in Roundabout Approach	45
Figure 7.15	Typical Marking for Roundabout	45
Figure 7.16	Typical Marking for Roundabout (Approach Roads of Different Category)	46
Figure 7.17	Signal Controlled Roundabout	47
Figure 7.18	Typical Marking for Double Roundabout	47
Figure 7.19	Mini Roundabout (Traffic Control by Give Way)	48

List of Figures

Figure 8.1	Merging/Diverging (Without Lane Gain/Lane Drop)	49
Figure 8.2	Merging/Diverging (With Lane Gain/Lane Drop)	50
Figure 8.3	Merging/ Diverging with Ghost Island (With Lane Gain/ Drop)	50
Figure 9.1	Pedestrian Crossing	51
Figure 9.2	Pedestrian Crossing Marking on Priority Junction	52
Figure 9.3	Pedestrian Crossing in Vulnerable Reach (Special Treatment)	52
Figure 9.4	Cycle Lane Marking	53
Figure 9.5	Rail Crossing	54
Figure 10.1	Bus Lane Marking	55
Figure 10.2	Bus Stop Marking	55
Figure 10.3	Bus Bay Markings	56
Figure 10.4	Truck Lay-by Markings	57
Figure 10.5	Toll Plaza Markings	57
Figure 11.1	On Street Parking	59
Figure 11.2	Off Street Parking	60
Figure 11.3	Keep Clear Markings	60
Figure 11.4	No Parking for Certain Hour	61
Figure 11.5	No-Parking	61
Figure 11.6	No Parking for Stricter Compliance	62
Figure 12.1	Object and Diagonal Marking	63
Figure 12.2	Object Marking Adjacent to Carriageway	65
Figure 12.3	Kerb Marking	65
Figure 12.4	Kerb Marking Where Restrictions Apply	66
Figure 13.1	Road Studs for Two directional Road	69
Figure 13.2	Road Studs for a Divided Carriageway	70
Figure 13.3	Road Studs for Diagonal and Chevron Markings	71
Figure 13.4	Road Studs for Crossable Continuity Lines	72
Figure 14.1	Above 95 % of area intact (left) and below 70 % of area intact (right)	73
Figure B.1	Details of Arrow Marking, AM07, AM02 and AM03	89
Figure B.2	Details of Arrow Marking, AM04, AM05 and AM06	89
Figure B.3	Details of Arrow Marking, AM07, AM18 and AM24	90
Figure B.4	Details of Arrow Marking, AM2, AM26, AM27 and AM28	90
Figure C.1	Details of Letters and Numerals (speed up to 50 mph)	91
Figure C.2	Details of Letters and Numerals (speed 51 to 100 mph)	94
Figure C.3	Details of Letters and Numerals (speed > 100 mph)	97
Figure F.1	Flow Diagram A: Assessment of Longitudinal Road Markings	105
Figure F.2	Flow Diagram B: Assessment on areas not Accessible for Assessment by HSM	106

List of Tables

Table 1.1	Road Design Manual (RDM) Coding Structure	2
Table 2.1	Minimum preview distance for a driver to react	8
Table 4.1	Longitudinal Markings - Dimensions and Spaces	16
Table 4.2	'RURAL' – Operating Speed of 80 km/h and above	20
Table 4.3	'URBAN' – Operating Speed under 80 km/h	20
Table 4.4	Speed and Visibility Distances for Road Markings	21
Table 5.1	Transverse Markings - Dimensions and Mark Spaces	25
Table 5.2	Marking Dimensions	27
Table 6.1	Size of Arrows	33
Table 6.2	Size of Letterings	34
Table 13.1	Road Studs Spacing Based on Approach Speed	67
Table 13.2	Road Studs Spacing for Undivided Roads	69
Table 13.3	Road Studs Spacing for Divided Carriageways	70
Table 13.4	Road Studs for Ramp / Slip Road/One Way Street	71
Table 14.1	Initial and Threshold Reflectivity (Dry Retro Reflectivity during Night-time)	74
Table 14.2	Initial and Minimum Performance for Night Visibility in Wet Conditions	74
Table 14.3	Initial and Minimum Performance for Skid Resistance	74
Table A.1	Longitudinal Markings	81
Table A.2	Transverse Marking	81
Table A.3	Longitudinal Markings	82
Table A.4	Transverse Marking	83
Table A.5	Arrow Marking	84
Table A.6	Directional Marking	86
Table A.7	Facility Marking	87
Table G.1	Properties of the Rubber Slider	107
Table G.2	Threshold Levels of Skid Resistance	108

1 Introduction

1.1 General

This Manual was prepared by the Ministry of Roads and Transport as part of a series of manuals that cover the entire project cycle. The series incorporate best practices, climate change considerations, and recent technologies thereby enabling the provision of road infrastructure that is safe, secure, and resilient.

The Kenya road manual series is as follows:

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
A. General	Procedures and Standards Manual	PSM
	1. General	
	2. Policies	
	3. Procedures Guidance	
	4. Codes of Practice	
	5. Guidelines	
B. Planning	Network and Project Planning Manual	NPM
	1. Road Classification	
	2. Route/Corridor Planning	
	3. Route/Corridor Planning	
	4. Highway Capacity	
C. Appraisal	Project Appraisal Manual	PAM
	1. Environmental Impact Assessment and Audit	
	2. Social Impact Assessment	
	3. Traffic Impact Assessment	
	4. Road Safety Audits	
	5. Project Appraisal	
D. Design	Road Design Manual	RDM
	1. Geometric Design	
	2. Hydrology and Drainage Design	
	3. Materials and Pavement Design for New Roads	
	4. Bridges and Retaining Structures Design	
	5. Pavement Maintenance, Rehabilitation and Overlay Design	
	6. Traffic Control Facilities and Communication Systems Design	
E. Contracts	Works and Services Contracts Manual	WSCM
	1. Forms of contracts	
	2. Standard Specification for Road and Bridge Construction	
	3. Bills of Quantities	
F. Construction	Road Construction Manual	RCM
	1. Construction Management	
	2. Project Management	
	3. Site Supervision	
	4. Quality Assurance	
	5. Quality Control	

This table continues onto the next page...

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
G. Maintenance	Road Asset Management Manual	RAAM
	1. Maintenance Management	
	2. General Maintenance	
	3. Pavement Maintenance	
	4. Bridges and Structures Maintenance	
H. Operations	Road Operation Manual	ROM
	1. Traffic Management	
	2. Vehicle Load Control	
	3. Emergency Services	
	4. Tolling	
I. Monitoring & Evaluation	Road Design Manual	MEM
	1. Performance Monitoring Manual	
	2. Technical Audits	
	3. Poverty, Gender Equality and Social Inclusion Monitoring	

Volume 6, Part 1 – Road Marking is part of the Roads Design Manual (RDM) whose volumes, parts and coding system are as follows:

Table 1.1 Road Design Manual (RDM) Coding Structure

Vol.	Manual Title	Part Name	Code
1	Road Design Manual: Vol. 1 Geometric Design	Part 1 - Topographic Survey	RDM 1.1
		Part 2 – Traffic Surveys	RDM 1.2
		Part 3 – Geometric Design of Highways, Rural and Urban Roads	RDM 1.3
2	Road Design Manual: Vol. 2 Hydrology & Drainage Design	Part 1 – Hydrological Surveys	RDM 2.1
		Part 2 – Drainage Design	RDM 2.2
3	Road Design Manual: Vol. 3 Materials & Pavement Design for New Roads	Part 1 – Ground Investigations and Material Prospecting	RDM 3.1
		Part 2 – Materials Field and Laboratory Testing	RDM 3.2
		Part 3 – Pavement Foundation and Materials Design	RDM 3.3
		Part 4 – Flexible Pavement Design	RDM 3.4
		Part 5 – Rigid Pavement Design	RDM 3.5
4	Road Design Manual: Vol. 4 Bridges & Retaining Structures Design	Part 1 – Geotechnical Investigation and Design	RDM 4.1
		Part 2 – Bridge and Culvert Design	RDM 4.2
		Part 3 – Retaining Structures Design	RDM 4.3
		Part 4 – Reinforced Fill Structures Design	RDM 4.4
		Part 5 – Bridges and Structures Condition Survey	RDM 4.5
		Part 6 – Bridge Maintenance Design	RDM 4.6
5	Road Design Manual: Vol. 5 Pavement Maintenance, Rehabilitation & Overlay Design	Part 1 – Pavement Condition Survey	RDM 5.1
		Part 2 – Pavement Maintenance, Rehabilitation and Overlay Design	RDM 5.2
6	Road Design Manual: Vol. 6 Traffic Control Facilities & Communication Systems Design	Part 1 – Road Marking	RDM 6.1
		Part 2 – Traffic Signs	RDM 6.2
		Part 3 – Traffic Signals and Communication System	RDM 6.3
		Part 4 – Other Traffic Control Devices	RDM 6.4
7	Road Design Manual: Vol. 7 Road Lighting Design	Part 1 – Grid-connected Road Lighting	RDM 7.1
		Part 2 – Solar Road Lighting	RDM 7.2

This Manual must be applied sensibly and flexibly in conjunction with the skill and judgement of the designer. Compliance with the guidance given in the Manual does not relieve designers of the responsibility for establishing that their design is suitable, appropriate, safe, and adequate for the purpose stated in the project requirements.

1.2 Objectives of this Part

This Manual will be used on the national and county roads to assist users in proper road marking for safely and efficient control of traffic movement.

The standards are intended to meet three important objectives:

1. Design of road markings
2. Implementation of road markings
3. Operation and enforcement of road marking

1.3 Scope

The scope includes detailing the various types of road marking material along with the colour pattern, for use on flexible and rigid pavement. The major types of marking such as longitudinal, transverse, block, arrows, word messages are explained in detail. Usage of different types of road studs are explained. Marking for at-grade and grade separated junctions are elaborated. Marking for speed reduction measures, bus lane, bus stop, truck layby, toll plaza and parking spaces are presented in detail. Object marking for obstructions on or near carriageway are explained. Performance measures for wear, durability, night visibility and skid resistance are detailed. Testing methods for marking and maintenance are explained.

1.4 Organisation of the Manual

The Manual is organised in 15 chapters.

- **Chapter 1** describes the objectives and the scope of manual, organisation of the manual and the design process for pavement marking.
- **Chapter 2** describes the background of signs, definition, and authority. It then describes the road marking materials, retro-reflectivity, colour patterns, visibility, performance, and road marking on concrete pavement.
- **Chapter 3** describes the various classes of marking.
- **Chapter 4** details longitudinal markings such as centre line, edge line, combined line system and raised profile edge lines.
- **Chapter 5** describes transverse markings including STOP, GIVE WAY and NO ENTRY markings as well as transverse rumble strips.
- **Chapter 6** describes the hazard markings, block markings, arrow markings and word messages.
- **Chapter 7** describes the markings used at various types of at-grade intersections, including roundabouts.
- **Chapter 8** presents merging and diverging road markings for approaches to grade separated intersections.
- **Chapter 9** deals with markings for pedestrians, vulnerable road sections, cycles, and railway level crossings.
- **Chapter 10** details the markings used for bus lanes, truck layby, bus stops and toll plazas.
- **Chapter 11** deals with parking and restriction markings.
- **Chapter 12** details object markings.
- **Chapter 13** deals with road studs.
- **Chapters 14 and 15** detail the performance assessment, warranty, and testing methods for markings.

This manual is to be read in conjunction with RDM Vol 1, Part 3: Geometric Design of Highways, Rural Roads and Urban Roads, RDM Vol 6, Part 1: Traffic Signs and RDM Vol 6, Part 3: Traffic Signals and Communication Systems.

1.5 Design Process

The process for design and installation of road signs is:

1. Study the need for road marking.
2. Select the type of marking to be used based on purpose and type of facility.
3. Choose single / double/ dashed line to be used based on visibility, geometry, approach speed of vehicles.
4. Determine the siting location, spacing based on the approach speed of vehicles, and road geometry.
5. Prepare plans, specification, Bill of Quantities for installation.

2 Road Marking Materials and General Features

2.1. Background

Properly designated and placed highway signs and markings are essential for safe and efficient movement of traffic. The adequacy of road signs and markings is a function of their location, number, accuracy, legibility, and uniformity.

Road Markings perform an important function of guiding and controlling traffic on a highway. The markings serve as a psychological barrier and help to delineate the traffic path. Further, the road markings channelise pedestrians and cyclists to safe locations. Road markings are indispensable for smooth and orderly flow of traffic and for promoting road safety.

Road markings have the limitation that they may be obscured by mud, sand and dust. Their conspicuousness is impaired when they are wet. Their durability depends on the exposure to traffic wear. The markings have an advantage that they can often be seen when an erected sign is obscured. Road marking helps in enhancing lane discipline, improves traffic flow and safety.

The Vienna Convention on Road Signs and Signals, 1968 and subsequent amendments of 1993 and 2006 helped to achieve a degree of uniformity in traffic signing and road marking in various countries. Countries all over the world follow the signing convention, to a great extent, to avoid confusion of drivers.

The legal framework for road marking (including signs, devices, notices, and markings) is contained in the Kenya Traffic Act, Chapter 403, revised 2015. Road marking has been in operation in Kenya, based on the Manual for Traffic Signs in Kenya Part 1, Road Markings, with some variations.

The objective of this manual is to establish a uniform system for road markings in Kenya. The manual aims to provide guidance, procedures, and recommendations on the type of road markings with details on dimensions including selection, installation and inspection procedures for different road marking materials.

The recommendations in this manual are applicable to all categories of road located in rural and urban areas.

2.2 Definition

Road markings are defined as lines, patterns and words (except road signs) which are applied or attached to the road surface or kerbs or to objects within or adjacent to the roadway for the control, warning, guidance or provision of information to road users.

Markings may be used to supplement kerb side or overhead signs, or they may be used independently.

2.3 Marking Materials

The commonly used materials for road markings are:

1. Hot applied thermoplastic paint.
2. Solvent-based and Water-based Road Marking Paints.
3. Other special material for road marking includes Cold Applied Plastics and Preformed Adhesive Tapes.

Newer material like preformed thermoplastic marking and Methyl Methacrylate (MMA) are getting used as durable material under heavy traffic.

2.4 Thermoplastic Markings

Thermoplastic is commonly used pavement marking material on roadways. It is a mixture of plasticizer and resins that holds all ingredients together. Thermoplastic paint is to be used for transverse markings, block and hazard marking, word messages. For longitudinal markings, thermoplastic is to be used in areas where wear of solvent-based paint is high.

When thermoplastic is applied in a molten state, it adheres to the pavement and solidifies immediately at ambient temperature. They have better retro reflective performance and service lives than other types of road marking paint. Thermoplastic marking should be applied on smooth bituminous surface for better service life. The service life of one application of thermoplastic can be for 2 to 10 years depending on the quality of paint, volume and composition of traffic, type of surface and workmanship.

Thermoplastic markings last longer than paint and is resistant to fading, chipping, peeling and cracking. It can withstand heavy traffic and extreme weather conditions and is also more reflective. Thermoplastic marking can be used for stop bars, transverse markings, pedestrian crossings and also in areas with heavy rainfall.

Thermoplastic paint shall be as per Kenya standards KS EAS 928-1. The markings shall comply to KS EAS 928 – 2. For glass beads specifications, refer KS 1610. Reference can be made to BS EN 1871 / AASHTO M249 and for glass beads to AASHTO M247.

Refer to specifications for proper application of thermoplastic. List of standards and specifications are given in **Appendix H**.

2.5 Solvent-based and Water-based Road Marking Paints

Road marking paint is the oldest form of pavement marking material. It can be used for longitudinal lines in temporary work zone markings. It can be used for marking that are required for a short period (1 -2 years) and can be removed easily thereafter. Paints are suitable for use for all longitudinal marking. At locations where wear is high, thermoplastic can be used.

The two types of road marking paint are: solvent-based; and water-based. Water-based paint is environmentally friendly and is easier to handle compared to solvent-based paints and poses less safety hazards to workers. The road surface painted with water-based paints can be opened to traffic faster than the road surface painted with solvent-based paints.

Material shall be white ready mixed traffic paint conforming to Kenya standard KS EAS 927. Reference is to be made to AASHTO M248 – Specifications for Ready Mix White and Yellow Traffic Paints. Glass beads shall be applied to the surface material and shall conform to AASHTO M247 (Type 2) or equivalent. Specifications are to be followed for proper application of paint.

2.6 Cold Applied Plastics

Cold applied plastic is a better choice for coloured pavement marking than paint and thermoplastic. Cold applied plastics are the best means to provide audible raised pavement marking for edge lines.

Cold applied plastic is more durable than thermoplastic markings in retaining the original colour and luminance values. It can be applied to the surface in a variety of ways. It has superior finishing and has no need of large application equipment. It can be easily carried in a medium sized van and trailer. Cold applied paints shall be suitable for brush application or other mechanical means. Application shall conform to AASHTO M237 Standard Specification for Epoxy Resin Adhesives for Bonding Traffic Markers to Hardened Portland Cement and Asphalt Concrete.

2.7 Preformed Adhesives Tapes

Preformed adhesive tapes are available in continuous rolls of various lengths and widths, conforming to ASTM D4505-17. The service life of one application can be up to 1 year depending on the quality of the tape, volume and composition of traffic, type of surface and workmanship. For improved service life, tape should be applied on clean, dry, and smooth pavement surface.

Unlike road marking with sprayed or extruded materials, preformed tapes do not require application equipment or experienced operators and do not require a drying or curing period. Preformed tapes are also available with more limited-service lives, conforming to ASTM D4592-17. They are designed for a service life of 3 to 6 months depending on wear and tear.

Preformed tapes offer longer service life and better performance in locations with high traffic volumes on trunk routes. These tapes can be used at critical sections like accident black spots, bridge decks, road tunnels, school zones, and high-speed traffic lanes etc. The tapes have high reflectivity during inclement weather conditions.

Tapes are suitable for graphical pavement marking for non-motorised traffic (NMT), for indoor and outdoor application, and on various surfaces like asphalt, concrete, tiles, paver block, etc. It is a synthetic rubber-based sheet of coloured/graphic/symbol marking less than 2 mm thick with an embossed top PVC film; it has a skid resistance of up to 40 British Pendulum Number when tested according to ASTM E 303.

2.8 New Materials

Preformed Thermoplastic Markings

These are a newer type of road marking material that is made by heating pre-cut plastic shapes and applying them to the road surface. This method offers many of the same benefits as traditional thermoplastic markings, including durability and reflectivity, but with the added benefit of easy and efficient installation. Preformed thermoplastic markings are available in a variety of shapes and sizes, making them a versatile option for a wide range of road marking applications.

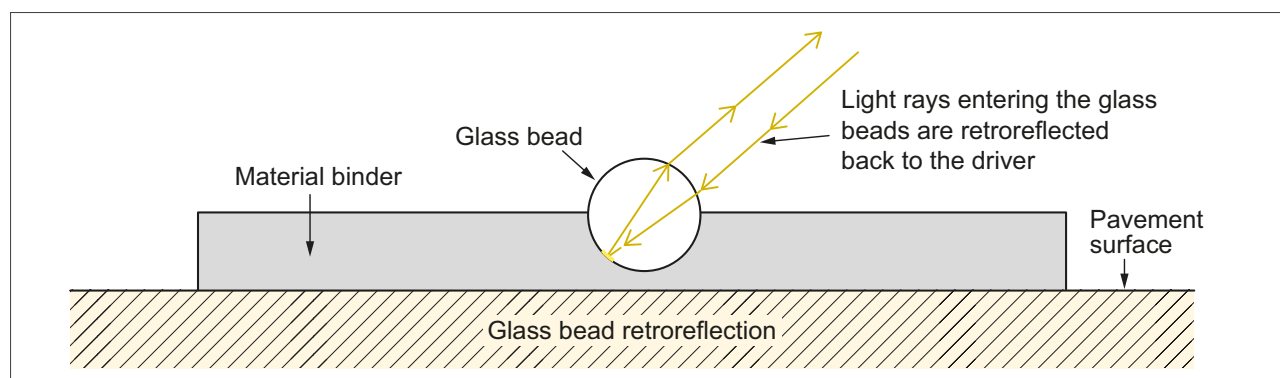
Methyl Methacrylate (MMA) Markings

It is a type of road marking material that is rapidly gaining popularity, MMA road markings are made from a two-part material that is mixed together and applied to the road surface. This material sets quickly, providing a durable and long-lasting road marking that is resistant to wear and tear.

2.9 Retro Reflectivity

Retro reflection is accomplished in pavement marking through the use of glass beads, partially embedded on the surface of the marking as binder material and also spread externally during application time. The quality of glass beads plays an important role in the retro reflectivity of pavement-marking as the glass bead returns light from vehicle headlamp to a driver. Figure 2.1 explains the working of retro reflection.

Figure 2.1 Retro Reflectivity in Pavement-marking Using Glass Beads.



Visibility distance is adversely affected by glare from oncoming vehicles, or dirty head lamps/windcreens, especially during the rainy season. The size of a glass bead can affect retro reflectivity, especially under wet conditions. Larger glass beads have slightly higher retro reflectivity than standard beads under dry conditions. When markings are wet, glass beads can often be ineffective because a film of water can cover the marking, causing the light to scatter before it can enter the bead. This causes the wet markings to be nearly invisible at night. Large bead (minimum 1 mm thickness) is more effective when roads are slightly wet because the higher profile protrudes through the film of water better than small glass beads.

2.10 Colour Pattern for Road Markings

The Traffic Act CAP 403, 2015 sections 15 and 16 says yellow coloured longitudinal marking is to be used for centre line. It can be single continuous line, double line or broken line. The edge lines are to be of white colour, of at least 100 mm width.

This manual recommends yellow centre line and white edge line marking. The colour pattern to be followed for road marking is:

White: The white colour should be widely used for road markings. It has good visibility and contrast against the road surface.

Yellow: Longitudinal marking in yellow colour is to be used to demarcate between opposing traffic flows such as centrelines of undivided carriageways.

Blue: The blue colour should be used to indicate new and special markings which are not conventional. Blue is the colour of public transportation including three wheelers and dedicated bus lanes in the Bus Rapid Transit (BRT) corridor on urban streets. When bus lanes are marked across an intersection, the blue marking should be used to inform the road users that the lane is specifically meant for buses and other vehicles should not drive or stop on this lane.

Green: The green colour should be used to mark cycle and NMT facilities. Green colour background should be marked at the intersection to give priority to cyclists and pedestrians in crossing the road. When the motorised and NMT share the same carriageway, green background lanes are preferred.

When dedicated non-motorised lanes are built, special green cycle boxes should also be used.

Red: Where multiple road users share the road space at hazardous locations, the red colour marking is primarily used to help people understand the danger. Red marking is recommended on hazardous intersections and at places where pedestrian traffic conflicts with the motorised traffic.

2.11 Visibility Related to Speeds

Road markings must be clearly visible both in the day and night-time. Road marking is an important aid to drivers on unlit roads at night where the visual cues on roadsides are absent.

As a general requirement, the drivers need to be able to detect guidance marking at a distance equivalent to a minimum of two seconds of travel time. Table 2.1 gives the minimum preview distance based on this criterion.

Table 2.1 Minimum Preview Distance for a Driver to React.

Speed (km/h)	30	40	50	65	70	80	90	100	110	120
Preview Distance (m) 2 seconds of travel time	17	22	28	36	39	44	50	56	61	67

Variety of factors influence the visibility distance of road marking. The visibility of the road marking gets enhanced when the line thickness is wider and with a higher mark to gap ratio and also with a higher coefficient of retro-reflective luminance. Wider pavement marking combined with higher coefficients of retro reflectivity are used to help drivers detect the lines according to the design speed of the roadway.

2.12 External Factors Influencing Performance of Road Marking

Many factors influence the performance of a given pavement marking material. The major factors can be grouped into three categories:

1. Roadway surface.
2. Traffic.
3. Environmental.

The factors influencing the performance of different pavement marking materials should be considered in the material selection process. Prior to marking, the pavement surface shall be made clean and dry to achieve proper pavement marking performance.

All types of material are very sensitive to the variables governing their application and hence this warrants strict quality control during application. The key variables that influence the durability and retro reflectivity performance of pavement markings are:

1. Material composition.
2. Application procedure.
3. Application machines.
4. Roadway surface.
5. Traffic on the road.

2.13 Pavement Marking on Cement Concrete Roads

Concrete is used sparingly for pavements in rural areas where hot-mix asphalt and surface treatments are more economical and more easily constructed. Concrete is much more likely to be used for roadway surfaces in urban areas and on higher class of roadways, where traffic volumes are higher.

2.13.1 Colour of Painting

Colour of marking on concrete shall be same as on flexible pavement.

2.13.2 Material

Thermoplastic material shall be used for marking. The thickness shall be 1.5 – 2.3 mm. Epoxy material can also be used. The thickness shall be 0.6 – 0.63 mm. Pre-formed tapes can be used. The same performance criteria as in flexible pavement shall be valid.

Road studs can be used with the same spacing as on flexible pavement. Placing of studs at the joints shall be avoided. An allowance of +/- 100 mm shall be used if a stud is required to be placed at a joint.

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Road Marking Materials and General Features

3 Classification of Pavement Markings

3.1 Classes of Markings

The various types of pavement markings that are used are described in this chapter. Details on each type of marking are given in individual chapters.

3.2 Longitudinal Markings

These markings are provided along the length of the road and are the most widely seen on urban and rural roads. Lane, centre, and edge of carriageway lines are used for guiding drivers to drive in the correct lanes. These markings are either continuous, dashed, single or double lines.

Other longitudinal markings can be used to separate vehicular traffic from cycle path and to separate cycle paths from walkway. Longitudinal marking is used to designate bus bay or bus lane.

3.3 Transverse Markings

Markings in the direction transverse to the travel direction, are provided to instruct drivers to stop or give way to other vehicles or not to enter some road. Examples are: Stop line and stop marking, Give Way marking, No Entry marking.

3.4 Hazard Markings

These markings facilitate merging/diverging and warn of hazardous areas. These are usually by chevron, diagonal marking and hatch marking.

3.5 Block Markings

Pedestrian crossings, triangular and chequered markings for speed breakers, and Give Way symbols are the block markings.

3.6 Arrows and Word Markings

Marking of arrows on the pavement directs the drivers in proper direction. Arrows are deflection arrows, bifurcation arrows and lane indication arrows.

Deflection arrows are used in advance to:

1. Warn of an approaching restriction.
2. Direct traffic to the correct lane.
3. Warn of a hazard or change of direction.
4. Indicate the side on which traffic should pass.

Bifurcation arrows are provided at the commencement of deceleration lanes on the approach to junctions. These are used to guide vehicles and to ensure that the full length of the lane is used to slow down for the junction without impeding the through vehicles on the main carriageway.

Lane indication arrows guide drivers to take left turn, right turn or to go straight. On a side road approach to a main road, arrows are marked on the side road.

On approaches to major intersections on a multi-lane highway with more than two lanes, arrows are marked on the lanes to assist drivers to take the correct lane for turning.

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Classification of Pavement Markings

3.7 Junction Markings

Markings in the vicinity of an intersection ensures orderly movement of traffic. The type of marking depends on the approach speed of the traffic and availability of space. The different types of marking used are:

Marking for appropriate traffic control (Stop, Give Way, signalised junction)

1. Lane demarcating markings.
2. Diagonal and Chevron Markings.
3. Directional Arrow Markings and Word messages.
4. Hatch Markings/Ghost Island.
5. Prohibitory Markings.
6. Pedestrian crossings.

At roundabouts, drivers give way to any traffic on their immediate right. Conventional roundabouts always operate based on the 'priority from right' rule and markings assist drivers to follow this rule in the circular carriageway. Signal-controlled roundabouts have traffic signals on one or more approach arms. The Stop line should be laid at right angles to the carriageway edge. Lane markings should be provided on the circulatory path of the roundabout.

At grade separated junctions, lane changes from the main carriageway to the auxiliary lanes or vice versa takes place at high speed and hence have to be dealt with carefully. The through lanes and the auxiliary lanes should be clearly demarcated.

Merging and diverging can be provided either with lane gain /drop or without lane gain / drop, depending upon the volume of traffic diverging or merging.

3.8 Pedestrian and Cycle Markings

Pedestrian crossings are to be provided at important intersections and are essential locations where there is conflict between vehicles and pedestrians. The site should be selected such that the pedestrians have minimum inconvenience and have enough visibility to see the oncoming vehicles. At intersections, the pedestrian crossings should be preceded by a stop line. In an unsignalised pedestrian crossing, marking shall be provided. Pedestrians should be given sufficient space on the walkway to wait. Obstructions such as trees, signposts, lamp posts etc. shall be cleared in the path of pedestrians. At bus stops, pedestrian crossings should be marked slightly behind the standing position of buses in order to avoid conflicts.

Cycle lane markings shall be provided when a portion of the carriageway is to be used exclusive by cyclists. The cycle symbol shall be marked on cycle lanes. At signalised crossing, space should be provided for cycles to cross safely.

3.9 Bus, Truck Layby and Toll Plaza Markings

Lanes reserved for buses (without physical separation) shall be provided with white lines as bus-lane markings. The legend 'BUS LANE' should be marked on the carriageway across the lane at its commencement and repeated after each junction. Pavement markings at bus stops should be provided with the word 'BUS STOP' written prominently on the pavement. The bus bay marking should be provided with proper chevron marking and can be either with physical island or with ghost island.

The truck bay marking should be provided with proper chevron marking and with advance informatory signs.

A toll plaza area shall be equipped with proper chevron marking and with advance signs.

3.10 Parking

Markings are to be provided to indicate where vehicles can park. The markings shall be solid white lines 100 mm wide. The words 'TAXI', 'CARS', 'MOTORCYCLE', should be written if the parking area is specific for any particular type of vehicle.

The stretches where parking is to be restricted is to be delineated with yellow kerb painting coupled with the provision of yellow edge line. Restrictions like 'Keep Clear' marking can be placed on road stretches located in front of schools and hospitals by providing zigzag marking.

3.11 Object Markings

Objects within /adjacent to the carriageway are physical obstructions which constitute a serious hazard to the traffic flow and shall be adequately marked. Objects within the carriageway shall be marked with vertical black and yellow stripes. Structures that have limited clearance, will also be marked with yellow and black stripes on the top. Objects adjacent to carriageway shall be marked by black and white vertical stripes.

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Classification of Pavement Markings

4 Longitudinal Markings

4.1 General

Longitudinal markings comprise of centre line, edge line, combined line system. Other longitudinal markings are cycle lane, bus lane and bus bay marking. Lane, centre, and edge lines in both urban and rural areas are used for guiding and confining traffic to its correct lane.

Details of the lines and its usage are shown in Figure 4.1, Figure 4.2 and Table 4.1.

The standard edge of carriageway marking is the continuous white line, which is broken or discontinued through junctions before resuming again.

Figure 4.1 Longitudinal Markings

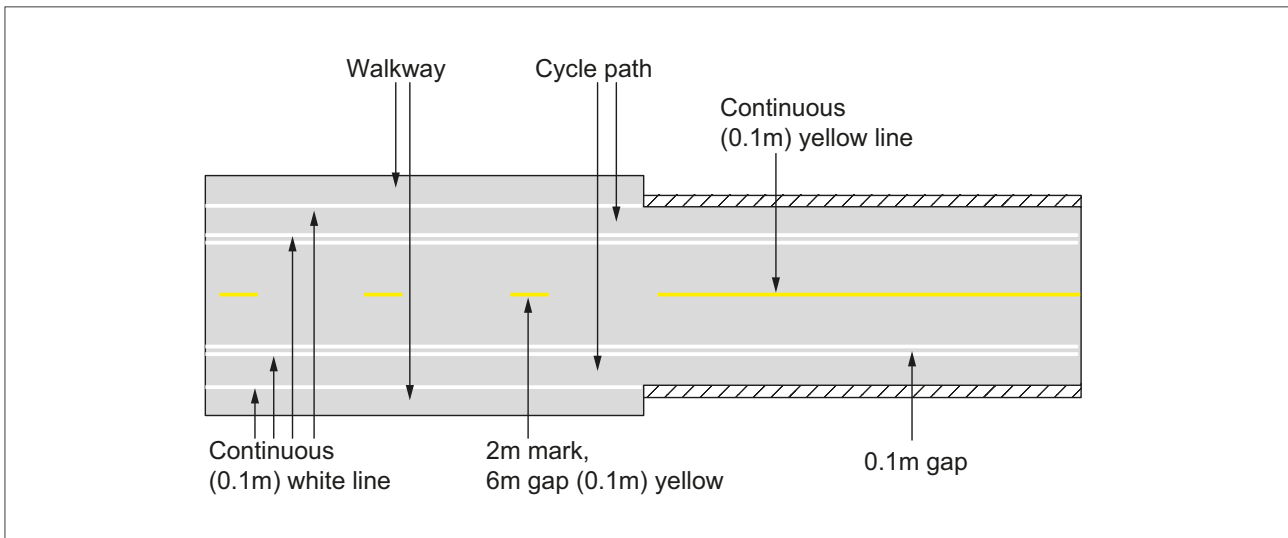
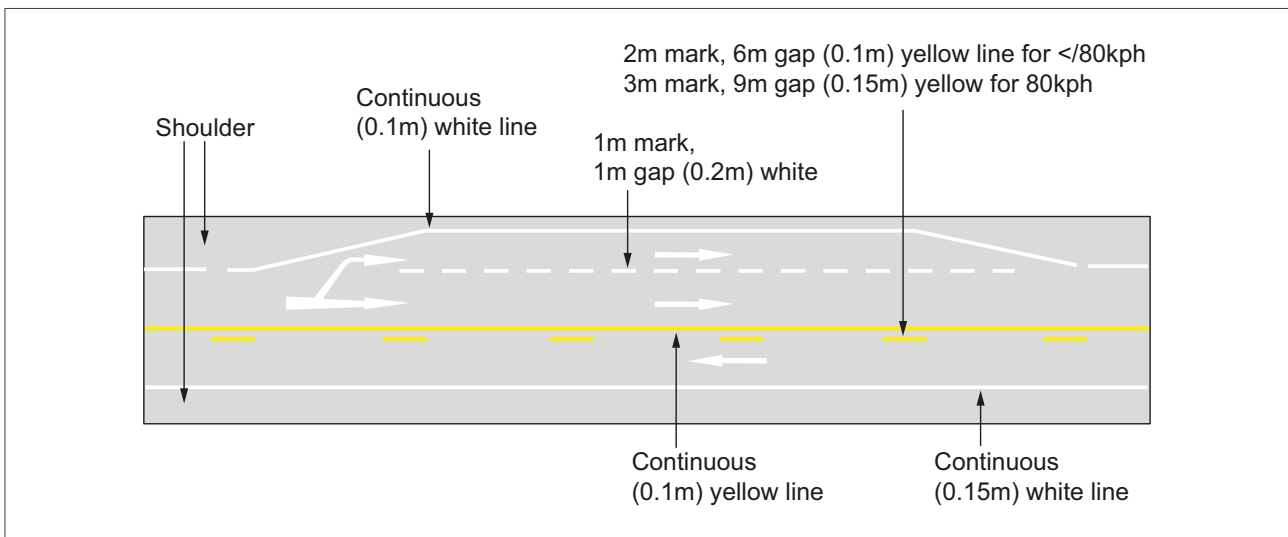


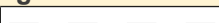
Figure 4.2 Longitudinal Markings - Slow Lane Arrangement



Notes:

1. The taper on the approach, and at the end of the slow lane, should be gradual (min, 1:25) and should be approximate to the natural path of the vehicle.
2. The slow lane marking at the end of the lane should be discontinued when the taper width reduces to 2.5m.
3. The line on the single lane side of the centre line should be dashed, or continuous depending on the sight distance characteristics of the section.

Table 4.1 Longitudinal Markings - Dimensions and Spaces

No.			< 80 km/h				Ref.	>/ 80km/h			
			a (m)	b (m)	c (m)	d (m)	Appendix A1	a (m)	b (m)	c (m)	d (m)
1	Yellow line between opposing traffic flow						-				
1.1	Continuous centreline				0.1		LM24			0.15	
1.2	Double continuous line				0.1	0.1	LM32			0.1	0.1
1.3	Broken line		2.0	6.0	0.1		LM01	3.0	9.0	0.15	
1.4	Combined line		2.0	6.0	0.1	0.1	LM28	3.0	9.0	0.1	0.1
2	White line for other marking						-				
2.1	Broken line		2.0	6.0	0.1		LM03	3.0	9.0	0.15	
2.2	Continuous line				0.1		LM26			0.15	
3	Edge marking						-				
3.1	Continuous line				0.1		LM26			0.15	
3.2	Broken line		2.0	2.0	0.1		LM07	2.0	2.0	0.15	
4	Separation line between traffic lane and cycle path and walkway				0.1	0.1	LM30			0.1	0.1
5	Separation line between cycle path and walkway						LM26			0.15	
6	Separation of bus bay and lane for slow moving traffic		1.0	1.0	0.2		LM05	1.0	1.0	0.2	

Notes:

a is the length of the mark, *b* is the length of the gap, *c* is the width of the line and *d* is the distance between the lane lines.

4.2 Centre Line and Edge Line

When a road has at least two lanes in each direction, with lane widths greater than 3 m and no central reserve, it shall be marked with a double continuous yellow line as compensation for a central reserve.

The double continuous line is formed as shown in Figure 4.3, Figure 4.4 and Table 4.1. The double continuous line shall be interrupted through junctions.

4.2.1 Broken Centre Line

On a two-lane road of 5.5 m width or greater, the centre line must be indicated by a yellow marking. This marking shall normally consist of a broken line. The broken line shall be interrupted through junctions.

The broken line is formed as shown in Figure 4.1, Figure 4.2, Figure 4.3 and Table 4.1

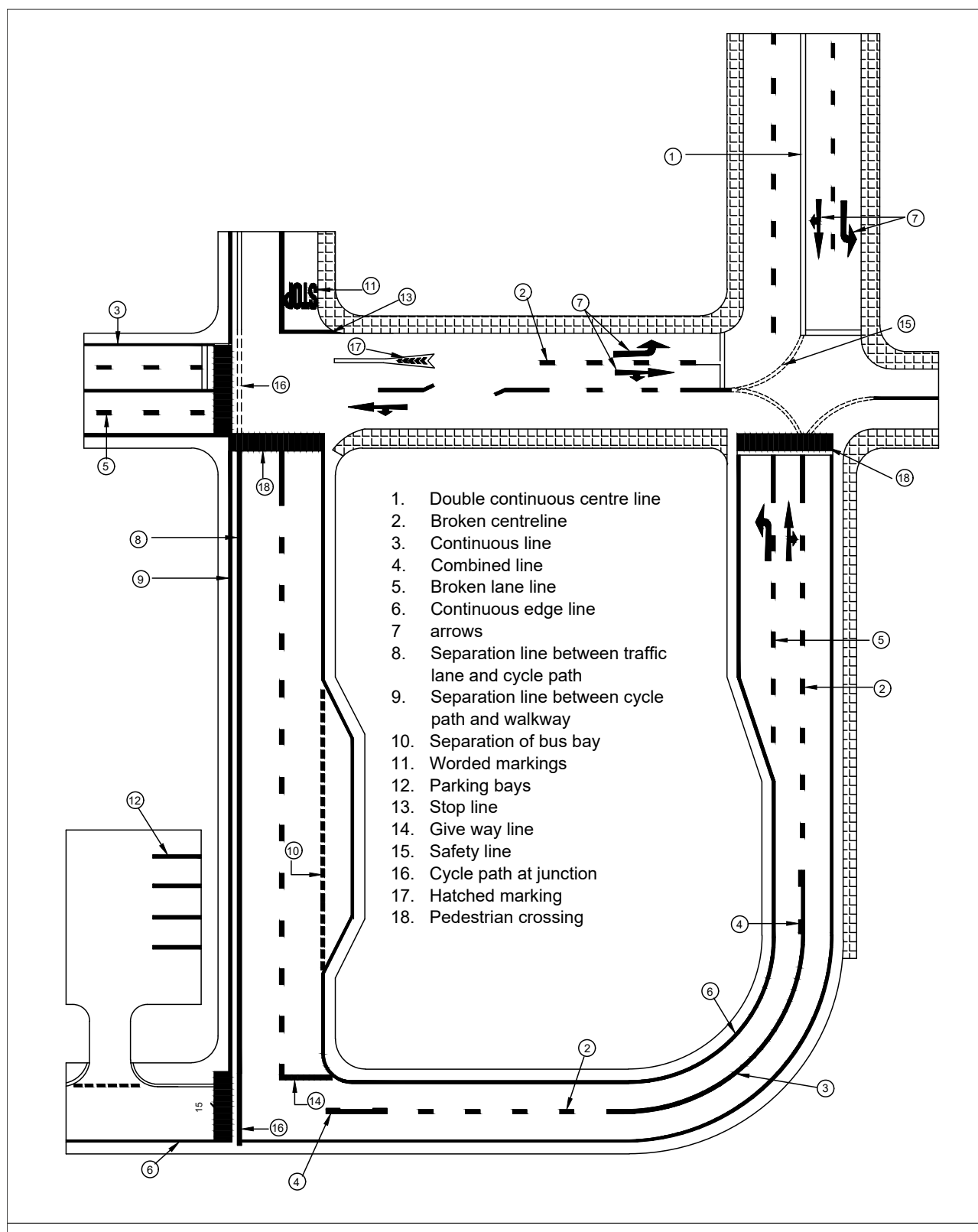
4.2.2 Continuous Centre Line

When it is necessary to prohibit vehicles to cross the centre line, continuous yellow line marking should be used.

The continuous line shall be interrupted through junctions. The continuous line is shown in Figure 4.1, Figure 4.2, Figure 4.3 and Table 4.1.

When overtaking is not allowed for vehicles travelling in one direction, the Combined Line System (CLS) is used. It has a continuous line on the left-hand side of the centre line as seen from the driving direction.

Figure 4.3 Typical Network Marking Layouts



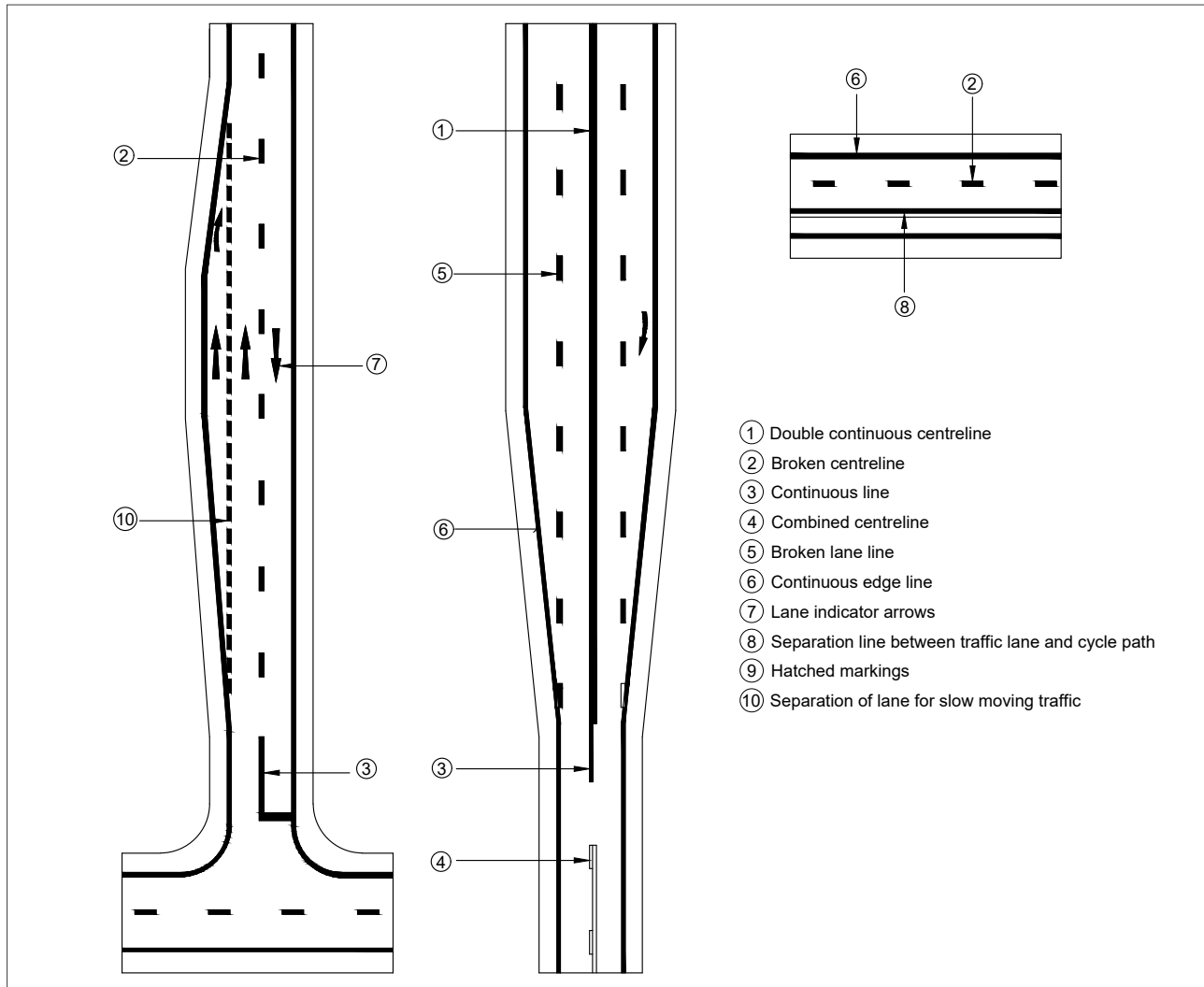
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Longitudinal Markings

Figure 4.4 Typical Road and Junction Marking Layout

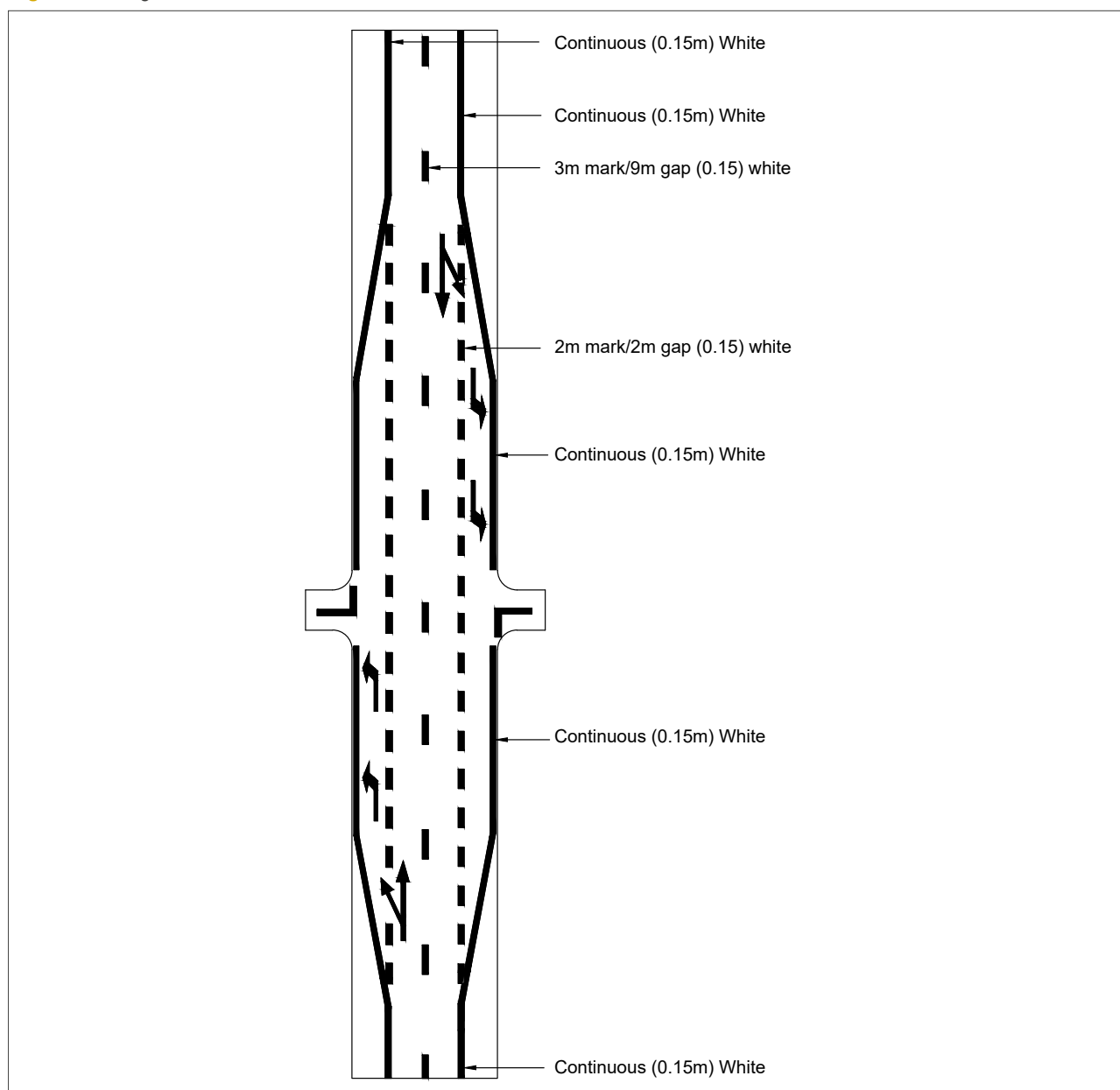
4.2.3 Edge Lines

Where the carriageway is divided into lanes, the edge must be marked with a continuous white line, except where there are kerbs or raised walkways.

This line shall be discontinued or broken through junctions depending on the size of the minor roads, or to denote the continuation of a cycle path through junctions. Continuous and broken edge lines are shown in [Figure 4.5](#) and in [Table 4.1](#).

Double continuous edge line is to be provided if a cycle lane is placed next to the carriageway, for separation between cycle and vehicular traffic. This marking is not covered under Traffic Act but is included in this Manual for additional safety of cycle users.

For expressways, the width of the edge line shall be 200 mm. This is not covered under Traffic Act. It is to be provided at high-speed roads like expressways to reduce run-away crashes.

Figure 4.5 Edge Lines and Junctions

4.3 Lane Line Markings

When a road has sufficient width for at least two lanes in the same direction (each lane of at least 3 m width) a broken white line must be used to divide the lanes going in the same direction.

The broken line shall be interrupted through junctions. The broken line is shown in [Figure 4.3](#), [Figure 4.4](#) and [Table 4.1](#).

If movement between lanes is not to be allowed, a continuous lane line should be provided.

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Longitudinal Markings

4.4 Dimensions of Marking

The length of the mark, gap, width of the mark is shown in Table 4.2 and Table 4.3. The dimensions vary on rural and urban roads. The two tables show the requirement in the two situations.

Table 4.2 'RURAL' – Operating Speed of 80 km/h and above

Marking	Mark (m)	Gap (m)	Width (m)	Use	Ref. Appendix A1
Lane Lines (white)	3	9	0.15	Division of Carriageway into Traffic Lanes	LM04
	2	2	0.15	Edge of Speed Change Lane of an Approach to Junctions	LM08
Centre of Carriageway Lines (yellow)	3	9	0.15	For use on roads of higher category	LM02
			0.1	For use on roads of lower category	LM01
	Continuous	-	0.15	Four or more lanes	LM25
Edge of Carriageway (white)	2	2	0.15	For use on roads of higher category	LM27
			0.1	For use on roads of lower category	LM26

Table 4.3 'URBAN' – Operating Speed under 80 km/h

Marking	Mark (m)	Gap (m)	Width (m)	Use	Ref. Appendix A1
Lane Lines (white)					
a	2	6	0.1	Dual carriageway lanes	LM03
b	2	2	0.1	All other Roads and Approaches to junctions	LM07
Centre of Carriageway Lines (yellow)	2	6	0.1	Two lane carriageway	LM01
	Continuous	-	0.15	Four or more lanes	LM24
Edge of Carriageway (white)	2	2	0.15	For use on roads of higher category	LM27
			0.1	For use on roads of lower category	LM26

4.5 Combined Line System – Visibility and Speeds

Combined line system (CLS) provides a means of prohibiting overtaking on lengths of road where visibility is restricted. It consists of a 0.1 m wide continuous prohibitory line (yellow) accompanied either by another continuous line or a broken line.

The CLS permits each direction of travel to be separately marked according to the visibility available in that direction. Visibility is dependent on the speeds of vehicles on the road. Where visibility is above a minimum standard, but overtaking may be risky, or where it is not possible to use CLS (e.g., because of restricted carriageway width) a single continuous line is recommended.

Where visibility standards are not satisfied, it does not automatically mean that double lines must be laid down. Judgment should be exercised in deciding where it is reasonable to impose restrictions, considering the topographical and traffic characteristics on the road.

Continuous line shall be used till a point where visibility exceeds the critical value. Visibility distance and critical value for different speeds and class of roads are given in Table 4.4.

Table 4.4 Speed and Visibility Distances for Road Markings

Road Class	Average 85 Percentile (Speed km/h)	Visibility Distance (m)	Critical Value (m)
E, UL, and other minor roads	40	40	65
C, D, UC	60	80	75
B, UA	80	130	95
A	> 90	300	135

When the length required for a continuous line is between 20 and 50 m, a 50 m line shall be used. When the distance between two continuous lines is less than the critical value, the two lines shall be joined to make one line.

The length of a broken line element within a combined line system should not be less than the visibility distance for the appropriate 85 percentile speed. This will prevent overtaking where the distance is insufficient.

4.5.1 Use of Combined Line System

CLS is 0.1 m wide continuous yellow line, accompanied by either a continuous or a broken line. On roads with more than two lanes, where a slow lane or hard shoulder is provided, the lines should be 0.15 m wide. Continuous yellow line is used where the visibility on bends or crests is less than the critical value given in Table 4.4. Visibility distance is measured from an eye-height of 1.05 m to a target of the same height above the road level.

All CLS should be marked with a single row of bi-directional yellow reflecting studs spaced at 6 m intervals. These should be laid between the lines, except where the lines are splayed to form a hatched area. The directional studs should be laid symmetrically in each longitudinal line at 6 m intervals arranged so that only the studs on the line nearest the driver reflect back.

4.5.2 Combined Line System – Bends, Crests, and Dips

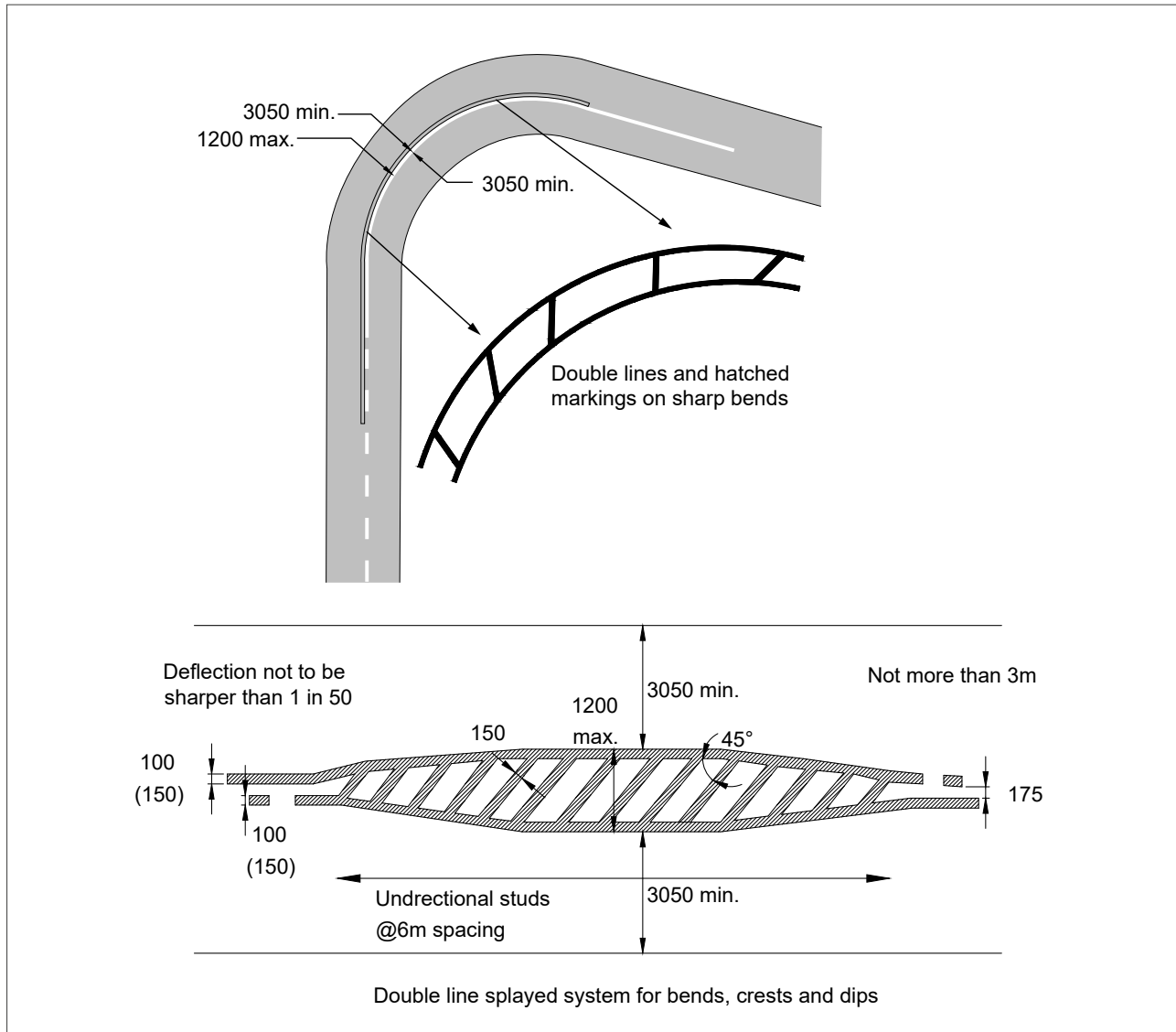
Depending on the physical characteristics of a bend or crest, the solid lines for traffic in opposing directions can either overlap or have a gap between them.

If there is a gap, less than the visibility distance, the two solid lines should be joined. If the gap is greater than the visibility distance, a single line should be used to join the two sections of double line.

4.5.3 Bends

The CLS section should be located along the centre of the carriageway. Double continuous lines should be located symmetrically about the centre line of the carriageway.

On sharp bends where double continuous lines are required, they can be splayed to form a central island. The maximum outside width can be 1.2 m, if there is sufficient space. The area between the lines must be hatched with inclined 0.15 m wide lines at 2 m spacing. Splayed marking is shown in Figure 4.6.

Figure 4.6 Splayed Marking Double Lines and Hatched Markings on Sharp Bends

4.5.4 Crests and Dips

A crest should be treated in the same way as a horizontal bend, regarding visibility criteria and line markings. Where splayed double continuous lines are required, the tapering should not be sharper than 1 in 50. The maximum outside width of the central island should be 1.2 m. The area in the island should be hatched as shown in Figure 4.6. The crests on either side of a dip should be treated individually.

4.5.5 Road Widths

Double line markings should not be used where the carriageway is less than 5.5 m in width. A solid centre line should be used as required.

4.5.6 Exceptional Use of Double Centre Lines

In exceptional circumstances, double centre lines may be used, even if the visibility conditions are more than critical value given in Table 4.4. Such circumstances include markings carried out in conjunction with traffic calming measures.

Where a slow lane is provided, a double centre line should be used in a continuous manner. The width should be 0.15 m.

4.6 Other Longitudinal Markings

4.6.1 Separation Line between Traffic Lane and Cycle Path / Walkway

A double continuous white line shall be used as the separation line between lanes for vehicular traffic and cycle path/walkway. The double continuous line shall be interrupted through junctions and replaced by a double broken line.

4.6.2 Separation Line between Cycle Path and Walkway

A single continuous line shall be used as the separation line between cycle path and a walkway.

4.6.3 Separation of Bus Bay and Lane for Slow Moving Traffic

A specially formed broken line is used in order to separate bus bays and lanes for slow moving traffic. This broken line is 0.2 m in width and formed as shown in Figure 4.2.

Detailed dimensions for each type of longitudinal marking are shown in Table A.1 of Appendix A to this Manual. Different classes of longitudinal markings are expressed as LM01, LM02 etc.

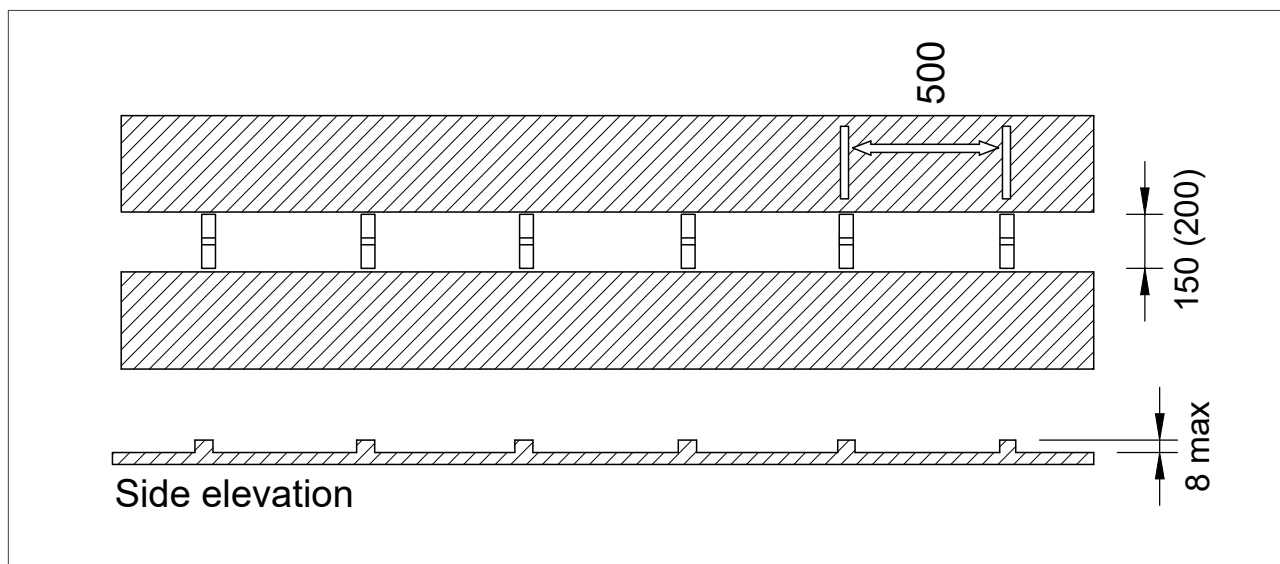
4.7 Raised Profile Line Marking

A raised profile line is an alternative to edge markings. It is a continuous line marking with ribs across the line at regular intervals. The advantage of ribs is that the vertical edges of the raised ribs are clearly visible above the water film in wet conditions. The other advantage is that they provide an audible warning to drivers as well as vibration when vehicles pass over the ribs. The types and patterns of raised profile lines are shown in Figure 4.7.

In general, spacing between two ribs should not exceed 500 mm. However, for expressways the recommended spacing is 250 mm. Closely spaced ribs help in maintaining the rumble effect but can be hazardous to cyclists. The width of raised profile edge should be 150 to 200 mm. The height should be between 4 and 8 mm. For an expressway, the height can be a maximum of 11 mm. Raised profile markings should be discontinued when edge lines pass through a crossway meant for pedestrians and cyclists.

On curved roads with radius less than 1000 m, raised edge lines should not be provided as the ribs of the raised edge line markings destabilise two wheelers. To facilitate drainage, a gap of 100 to 150 mm should be provided between each set of marking, at an interval of 36 m.

Figure 4.7 Raised Profile Edge Lines



1

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Longitudinal Markings

5 Transverse Markings

5.1 General

Transverse markings are placed on the pavement in transverse direction to the vehicular movement. These markings are provided to instruct drivers to stop or give way to other vehicles, or not to enter some roads.

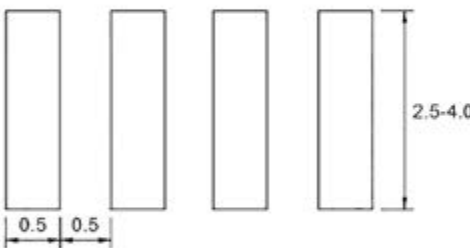
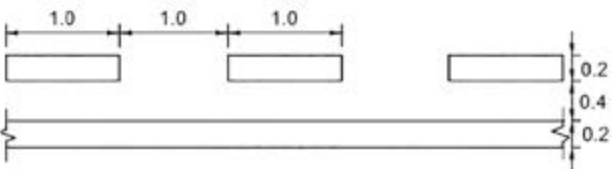
The transverse markings are:

1. STOP Line.
2. Junction STOP Line.
3. Traffic Signal STOP Line.
4. GIVE WAY Line.
5. Triangular GIVE WAY Approach Marking.
6. NO ENTRY Marking.

Alternative dimensions for line widths and letter heights are provided for use in different situations. The smaller size dimensions should be used in urban areas. The larger dimensions should be used in rural areas and at major urban junctions.

The dimensions of different transverse marking are shown in Table 5.1.

Table 5.1 Transverse Markings - Dimensions and Mark Spaces

No.	Type	Dimensions (m)
1	STOP LINE	
2	GIVE WAY LINE	
3	SAFETY LINE	
4	PEDESTRIAN CROSSING	
5	CYCLE PATH IN JUNCTION	
6	NO ENTRY	

5.2 STOP Marking

5.2.1 STOP Line

A stop line is a broad continuous transverse white line of width 0.4 m. The wide white line indicates the position beyond which a driver must not proceed when required to STOP by signs, or by traffic signals, including signals at level crossings.

Figure 5.1 show details of the marking.

5.2.2 Junction STOP Line

The marking consists of a single continuous white line 0.4 m in width. The line indicates the position at which a vehicle must stop when this is required by the road sign STOP, or by traffic signals, or by level crossing controls.

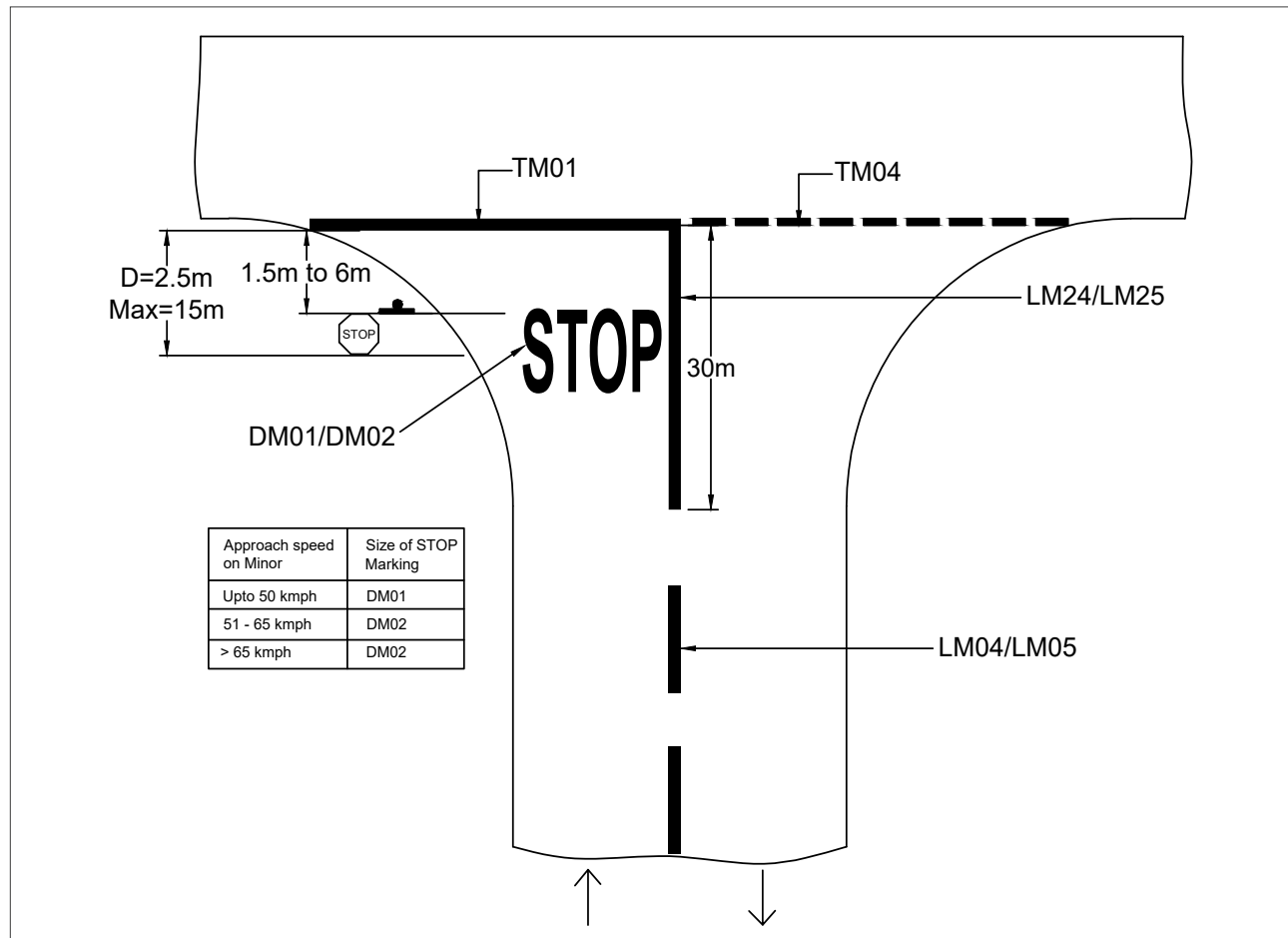
Where the Stop line is used in connection with the road sign STOP, the line must always be accompanied by the worded STOP marking on the carriageway.

The edge of the STOP line nearest to the major road should not be closer than 0.6 m to the path normally followed by vehicles on the major road. If there is a marked pedestrian crossing, the Stop line should be placed back by 1 m from the crossing.

At level crossings, the Stop line is located 5 m before the nearest rail or from the primary signal.

On a two-way road the STOP transverse marking should always be accompanied by a central continuous longitudinal line extending back from the junction by a distance of 30m. The STOP worded marking, where used, should have letters of height of 1.8 m.

Figure 5.1 Typical STOP Marking Layout



5.2.3 Traffic Signal STOP Line

The marking consists of a single continuous white line 0.4 m in width.

At traffic signals, including pedestrian signals, the STOP line is located 1 m before the nearside primary signal. Site conditions may necessitate adjustment of this distance. Figure 5.1 shows details of this marking.

The Stop line is laid down across the lanes and is located at that place which gives the driver the best possible general view of the junction. If a junction has a marked pedestrian crossing, the stop line must be placed at least 1m before the crossing.

5.3 GIVE WAY Marking

5.3.1 GIVE WAY Line

A GIVE WAY line is a broad broken white line laid across traffic lanes at junctions or along the inner edge of acceleration lanes, when the traffic in these lanes is required to give way by the road-sign 'GIVE WAY'. The give way line indicates the stopping location of the vehicle, when stopping is necessary.

Figure 5.2, Figure 5.3 and Table 5.1 shows for details of this marking.

The sign is located at a place which gives the driver the best possible general view of the junction.

When the give way sign is used at a junction, it must be accompanied by a longitudinal continuous line extending back from the junction by 30 m.

The recommended marking consists of a broken line 0.4 m wide comprising 0.5 m marks and 0.5 m spaces. Where the width of the approach lane is not sufficient to display the pattern, segments of 0.5 m marks and 0.5 m spaces may be used instead.

The precise location of the marking nearest to the edge of the major road is governed by the same considerations as the Stop line.

5.3.2 Triangular GIVE WAY Approach Marking

The hollow triangular marking is be used when a transverse 'GIVE WAY' line is provided and should be accompanied by a GIVE WAY upright sign. It must not be used elsewhere. The marking should be located with its lower base 2.1 to 2.75 m from the transverse marking. This distance may be increased to a maximum of 15 m depending on the visibility at the junction, its layout, and the speed of traffic on the minor road. A suitable place to locate it would be near the tangent point of the kerb-line. The sign should be placed with the triangle pointing downwards.

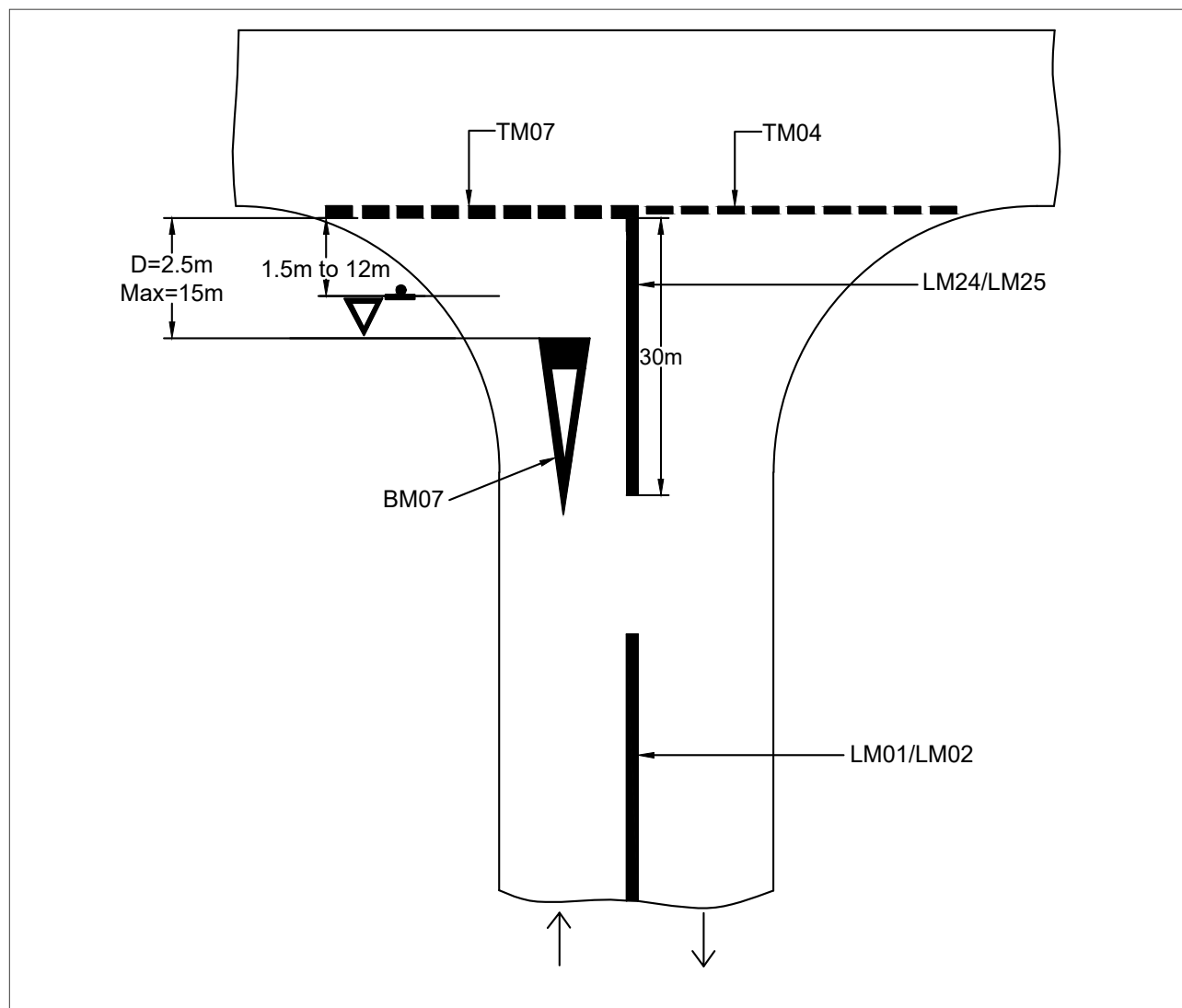
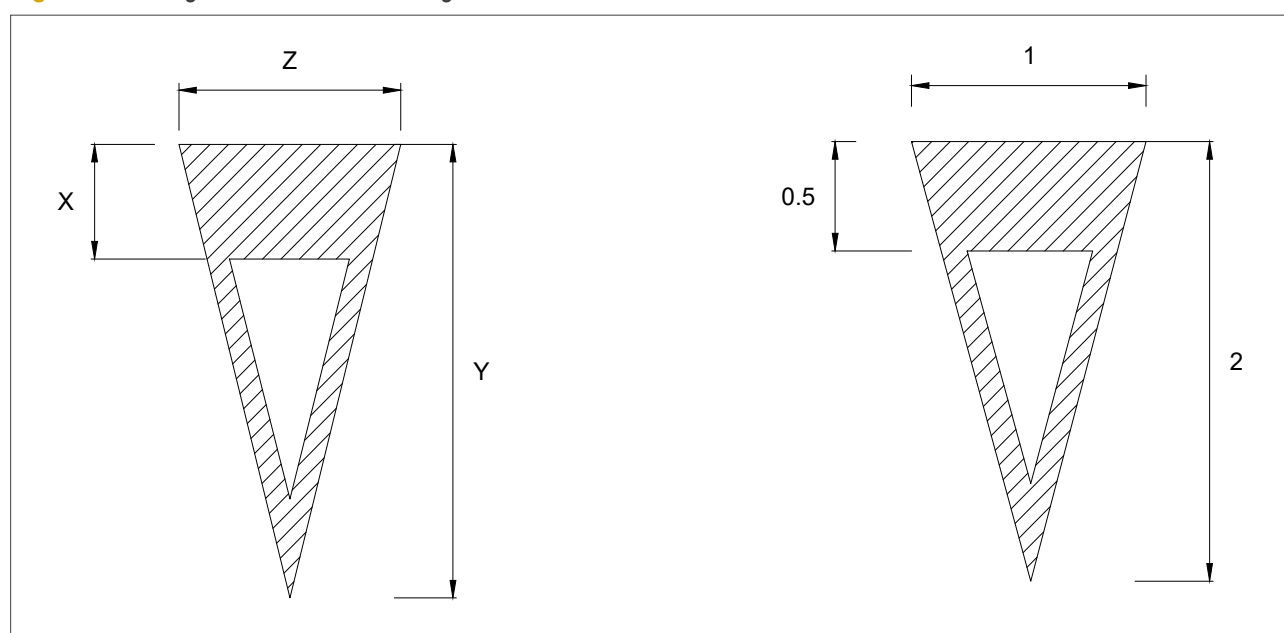
The Triangular GIVE WAY line is shown in Figure 5.2 and in Table 5.1.

International convention on Give Way markings suggests the ratios shown in Figure 5.3.

Where triangular markings are used, they should be positioned approximately in the centre of the traffic lane. Where the approach to the junction is divided into two or more lanes, then a triangular marking should be provided in each lane.

Table 5.2 Marking Dimensions

Road Speed (km/h)	X (m)	Y (m)	Z(m)
≥60	1	6x	2x
<60	0.5	2x	2x

Figure 5.2 Typical 'GIVE WAY' Marking Layout**Figure 5.3** Triangular GIVE WAY Marking Ratios

5.4 NO ENTRY Markings

The recommended marking consists of one a continuous white line combined with one broken line of 1 m marks and 1 m gaps. The lines are 0.2 m wide and are spaced 0.4 m apart. The precise location of the marking nearest to the major road is governed by the same conditions as those of the STOP line.

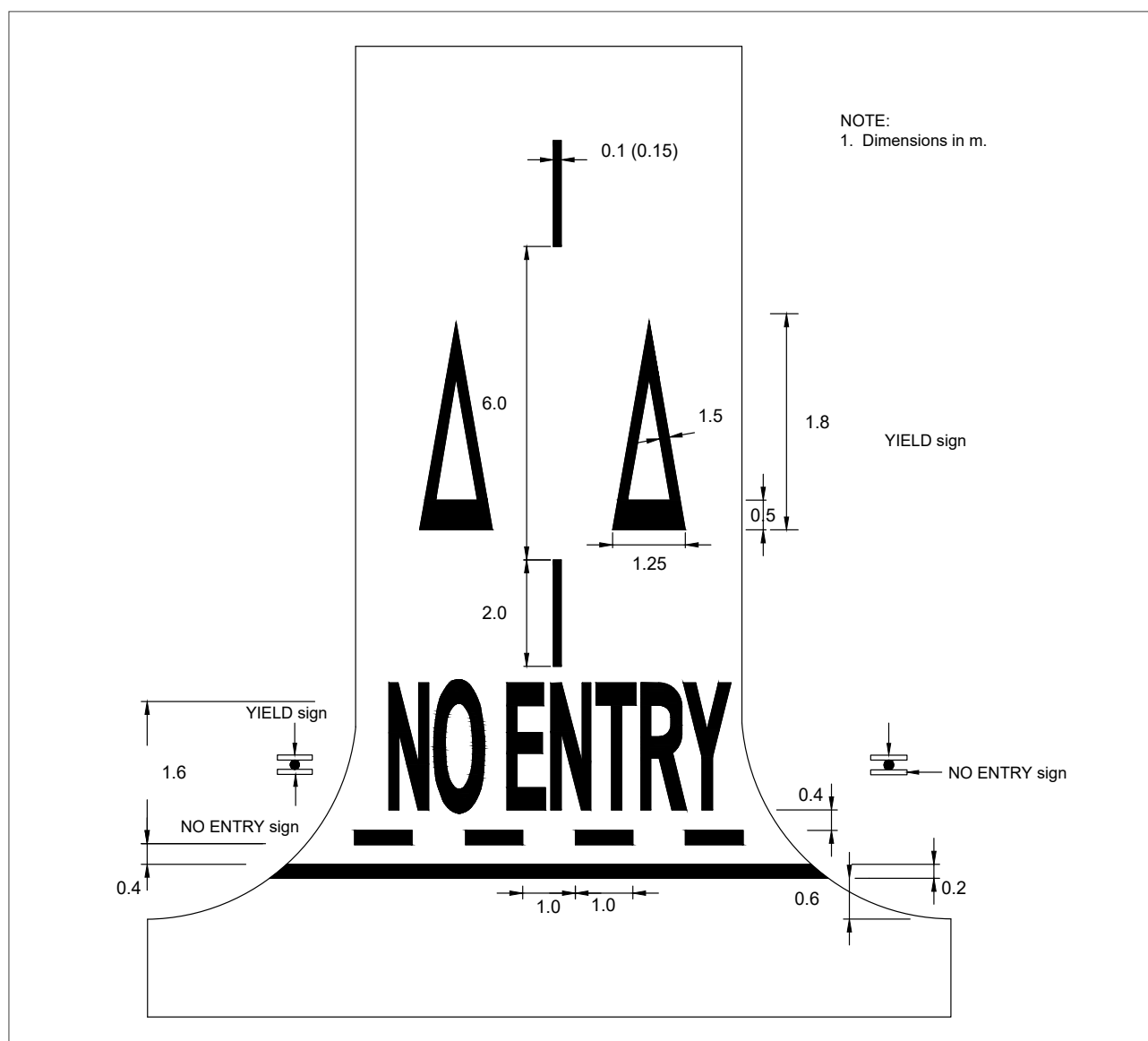
The wording NO ENTRY may accompany the lines. The spacing between the broken line and the lettering is 0.4 m. The height of lettering is either 1.8 m or 2.8 m, depending upon the location or speed of the road.

The NO ENTRY line is shown in Figure 5.4.

Where two or more lanes exit across a NO ENTRY 'combined line' the approach lane division lines are usually broken into 2 m marks with 6 m spaces.

The road markings must be accompanied by the appropriate regulatory signs, i.e., the 'NO LEFT TURN' and 'NO RIGHT TURN' signs on the approaches to the junction on the intersecting road, and the 'NO ENTRY' sign on both sides of the one-way road at its junction with the intersecting road.

Figure 5.4 NO ENTRY LINE and Markings for use with NO ENTRY Sign



The NO ENTRY sign (usually 2 signs) may be placed on dual carriageways at junctions and U-turn openings to remind road users that vehicles may not enter a one-way road or the 'wrong' side of a dual carriageway respectively.

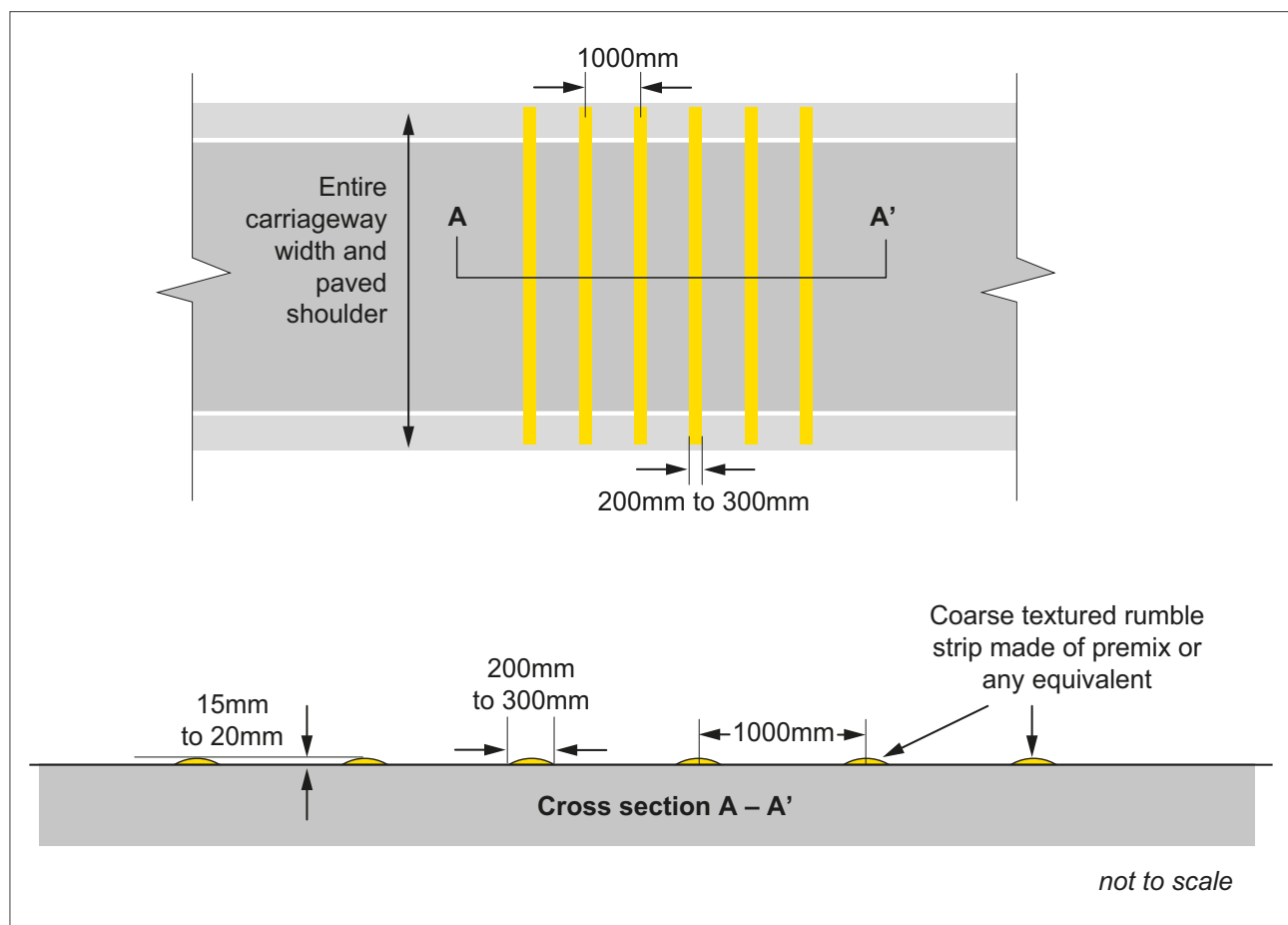
Detailed dimensions for each type of transverse marking are shown in Table A.2 of Appendix A to this Manual. Different classes of longitudinal markings are expressed as TM01, TM02 etc.

5.5 Transverse Rumble Strips

Transverse rumble strips are provided at locations where speed control is required. Rumble strips can be cast in situ with cement concrete or premix bituminous materials. Rumble strips can also be made of rubber, polyurethane or thermoplastic paint. Rumble strips may be provided across the entire width of carriageway and paved shoulders. Their raised section should be 15 to 20 mm high, 200 to 300 mm wide and spaced about one metre (centre to centre) with about 6 strips at one location. Proper workmanship must be exercised to achieve the required height. Figure 5.5 shows details of the rumble strip.

On approaches to narrow bridges, transverse rumble strips can be used in the shoulder to force the driver to slow down and drive on the restricted width of pavement. In areas where crashes have occurred due to vehicles running off the carriageway, rumble strips can be used on shoulder side as well.

Figure 5.5 Details of Rumble Strips



6 Hazard, Block, Arrow Markings and Word Messages

6.1 General

Hazard Markings facilitate merging /diverging, warning of hazardous location. These are usually by chevron and diagonal marking and hatch marking.

Block Markings are used for pedestrian crossings, Give Way symbols. Triangular and chequered markings are used for speed breakers.

Arrows and Word Marking on the pavement directs the drivers in the proper direction. The types of arrows are deflection arrows, bifurcation arrows and lane indication arrows.

Bifurcation arrows are provided at the commencement of deceleration lanes on the approach to junctions. It is used to guide vehicles and to ensure that the full length of the lane is used to slow down without impeding the through vehicles on the main carriageway.

Lane indication arrows guide drivers to take left turn, right turn or to go straight. On a side road approach to a main road, arrows are marked on the side road.

On approaches to major intersections on a multi-lane highway, arrows are marked in the lanes to assist drivers to take the correct lane for turning.

6.2 Hazard Markings

These are pavement markings that facilitate merging/diverging and warn of hazardous locations. These are usually by chevron and diagonal marking and hatch marking. The hazard marking should always be accompanied by an appropriate sign.

Detailed dimensions for each type of hazard marking are shown in [Table A.3](#) of Appendix A to this Manual. Different classes of longitudinal markings are expressed as HM10, HM12 etc.

6.3 Block Markings

These are: crossings for pedestrians, triangular and chequered markings for speed breakers, and Give Way symbols.

The application of thermoplastic paint for block marking is generally different from longitudinal marking. The quality of block marking with adequate visibility is of utmost importance.

Detailed dimensions for each type of block marking are shown in [Table A.4](#) of Appendix A to this Manual. Different classes of block markings are expressed as BM01, BM02 etc.

6.4 Arrow Markings

6.4.1 Deflection Arrows

Deflection arrows are used in advance to warn of an approaching restriction, to direct traffic to the correct lane, to warn of a hazard or change of direction, and to indicate the side on which traffic should pass.

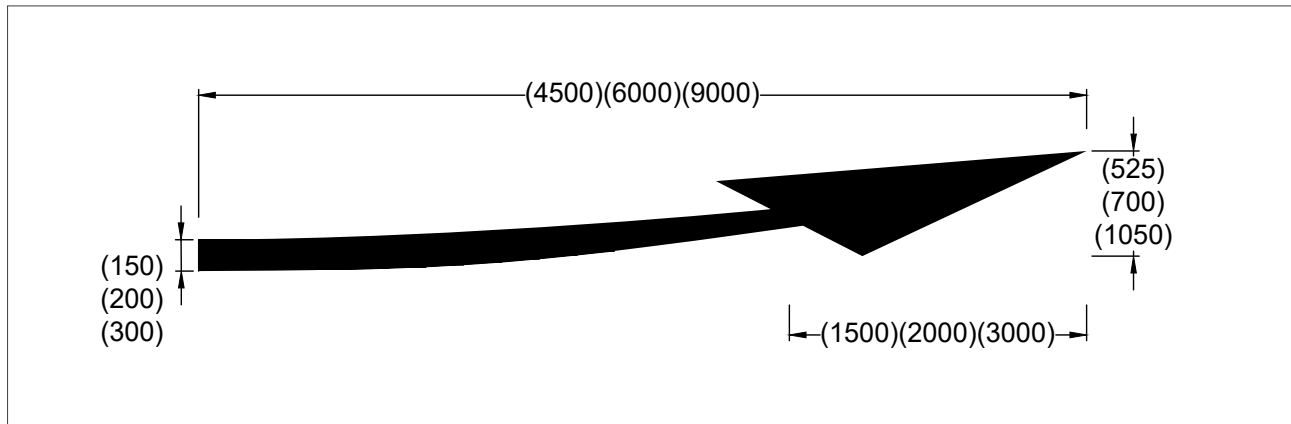
A deflection arrow of 4.5 m length shall be used for speeds up to 65 km/h. Arrows of 6 m length shall be used for speeds ranging between 65 km/h and 100 km/h. Arrows of 9 m shall be used for speeds greater than 100km/h. Arrows are shown in [Figure 6.1](#).

Deflection arrows are also used in cycle or bus lane boundary markings to indicate the side on which other vehicles should pass them.

In bi-directional carriageways, ahead of no-overtaking stretches, the deflection arrow shall be used to direct the traffic to the proper lane.

The arrow may be reversed so that it points to the right in appropriate circumstances.

Figure 6.1 Deflection Arrow



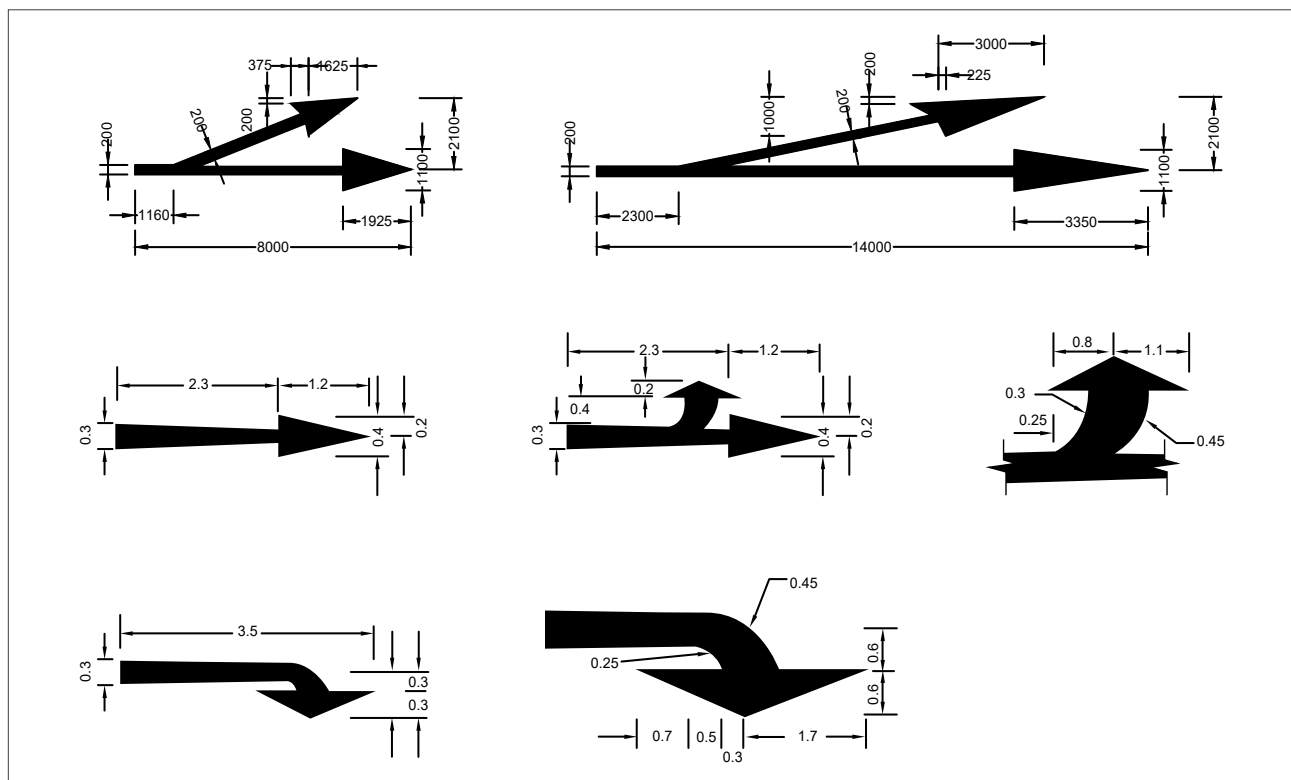
6.4.2 Bifurcation Arrows

Bifurcation arrows are provided at the commencement of deceleration lanes on the approach to junctions. It is used to guide vehicles and to ensure that the full length of the lane is used to slow down without impeding the through vehicles on the main carriageway. Arrow markings should be placed in the middle of the lane.

Two sizes are prescribed. The 16 m arrow is to be used on roads with approach speed from 65 to 100 km/h. The 8 m arrow is to be used on roads with lesser approach speed.

The arrow marking may be reversed to suit right turn movements into deceleration lanes in the central reservation of dual carriageways and dedicated right turn lanes on other roads. Bifurcation arrows are shown in Figure 6.2.

Figure 6.2 Bifurcation and Lane Indication Arrows (All dimensions in m)



6.4.3 Lane Indication Arrows

Lane Indication or Directional arrows should be used to guide drivers to the correct lane when approaching busy intersections. It can be used in signal-controlled or un-signalised intersection. Directional arrows are shown in Figure 6.2. The details are given in Appendix A of this Manual. Table 6.1 shows the length of arrow required for approach speed of vehicles.

Table 6.1 Size of Arrows

Speed (km/h)	Length of Arrows (m)
Up to 50	3.5
51 -100	5
>100 & expressway	9

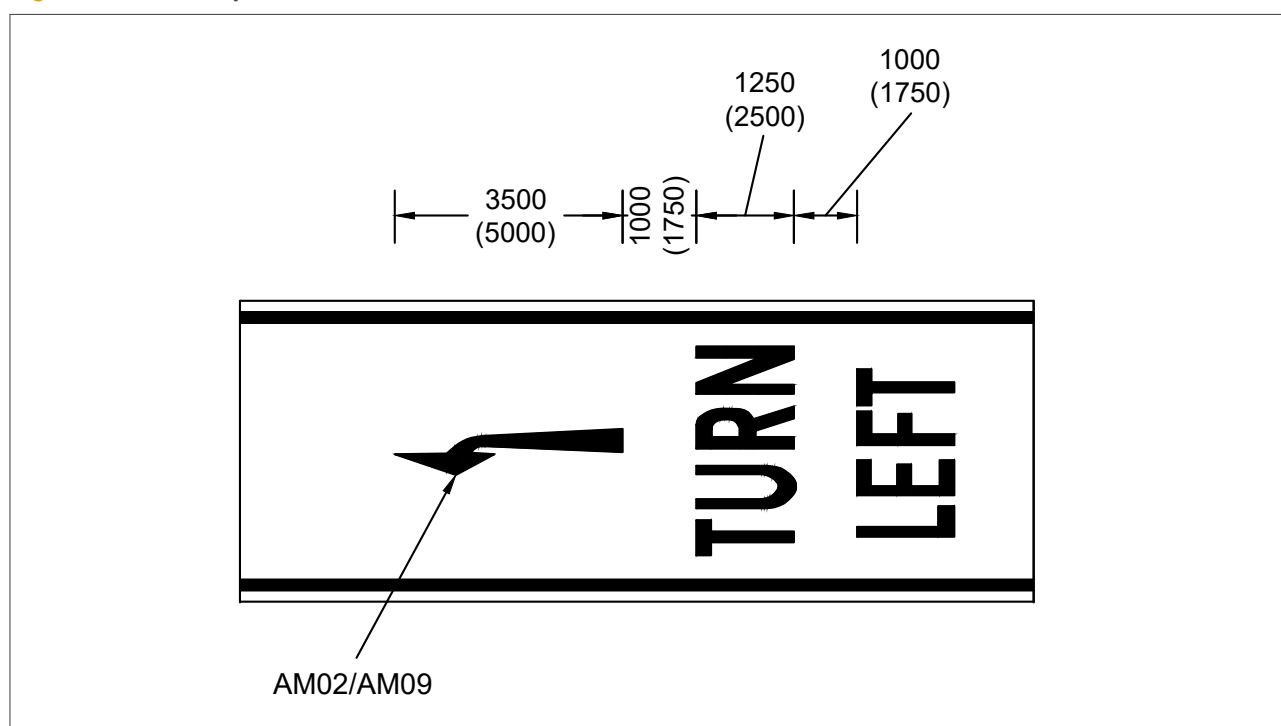
On two lane approaches to an intersection, the arrangement of arrows indicating the lanes for straight ahead, left turn, and right turn will depend on the relative turning volumes and on the site conditions. For a two-lane approach road having heavy right turn movements, the straight ahead and left turn arrow can be provided by combining both of them and the right turn arrow can be provided on the extreme right lane, signifying exclusive right turning movements. Similarly, where there is a left filter lane, this should be marked with a left arrow marking alone, in order to exclude non-filtering traffic.

Mandatory turn arrows: lane arrows supplemented with the legend 'TURN LEFT', 'TURN RIGHT' and 'AHEAD ONLY' are prescribed. These versions may be used only where they indicate the effect of a statutory prohibition. Figure 6.3 shows mandatory turn arrow.

Guidance arrows: these are marked in the junction area where some guidance to traffic is considered to be helpful. Care should be taken that the meaning is clear to drivers on all approaches. In order to avoid confusion and to ensure compliance, dedicated arrow markings are mandatory in signal-controlled junctions.

Detailed dimensions of the arrows are shown in Table A.5 of Appendix A of this Manual. The various types of arrows are shown as AM1, AM2, etc. Dimensions of arrows are given in Appendix B of this manual as AM01, AM02 etc.

Figure 6.3 Mandatory Turn Arrow



6.5 Word Messages

Information to guide, warn or regulate traffic are conveyed by the inscription of word messages on the road surface. Some of these augment kerbside signs, others indicate the areas of the carriageway intended for a particular function (e.g., Bus Stop) or to be kept clear (e.g., school).

The size of the basic alphabets and numerals shall conform to the speed range presented in Table 6.2. The alphabets and numerals shall be reflectorised. The letters should be elongated in the direction of traffic and details are given in Appendix A of this Manual.

Words and arrows shall be white in colour. Word messages should not exceed three lines of information and symbol markings should be no more than one lane in width.

Because of the necessity for brevity, the following typical legends are recommended: STOP, SLOW, BUS, BUS LANE, KEEP CLEAR, SCHOOL, RIGHT TURN ONLY, EXIT ONLY. The message should be marked in advance in case visibility is restricted.

The legends RIGHT TURN ONLY, EXIT ONLY should be used to supplement the directional arrows and should not be used without them. The Legends SCHOOL, SLOW and STOP should be used only in support of standard signs.

Table 6.2 Size of Letterings

Speed (km/h)	Length of Letters (m)
Up to 50	1.25
51 -100	2.5
>100 & expressways	5

6.5.1 Lane Destination Markings

Worded lane destination markings reinforce the advance direction sign in conformity. The letter size depends on the speed limit, with lengths of 1.25 m for speeds up to 50 km/h and 2.5 m for speeds of more than 50 km/h. These marking should be used at a distance of equal to longest peak hour queue.

STOP word message must be used to supplement a Stop sign and stop line and may not be used in any other circumstances. The marking should normally be located so that the top of the word is 2 to 3 m from the nearest part of the double stop line.

The word message SLOW may be used to supplement any warning sign on the approach to a hazard or road intersection, including the advance signs giving warning of STOP or GIVE WAY. It should not be used to supplement the roadside Give Way sign alone for which the hollow triangular marking has been prescribed.

The word message KEEP CLEAR indicates that part of the carriageway at a road intersection which should be left clear.

6.5.2 Directional Markings

Word messages which are directional in nature are classified under Directional Marking. These are STOP, SLOW, KEEP CLEAR markings and are abbreviated as DM01, DM02, DM03 etc. For easy referencing as given in Table A.6 of Appendix A of this Manual.

6.5.3 Facility Markings

The words and markings used for parking for buses, cyclists, and disabled people are classified under Facility Marking and are abbreviated as FM 01, FM 02, FM03 etc. For easy referencing as given in Table A.7 of Appendix A of this Manual.

Dimensions of words and letters are given in Appendix C of this Manual.

7 Markings for At-Grade Intersections

7.1 General

Carriageway markings at an intersection are used to ensure orderly movement of traffic. It depends on the approach speed of the traffic and availability of space. The choice of layout for a particular location depends upon the conditions at site. The markings for the various intersection types illustrated in this section are typical examples only. The precise layout may be adjusted to suit the design of the intersection under consideration.

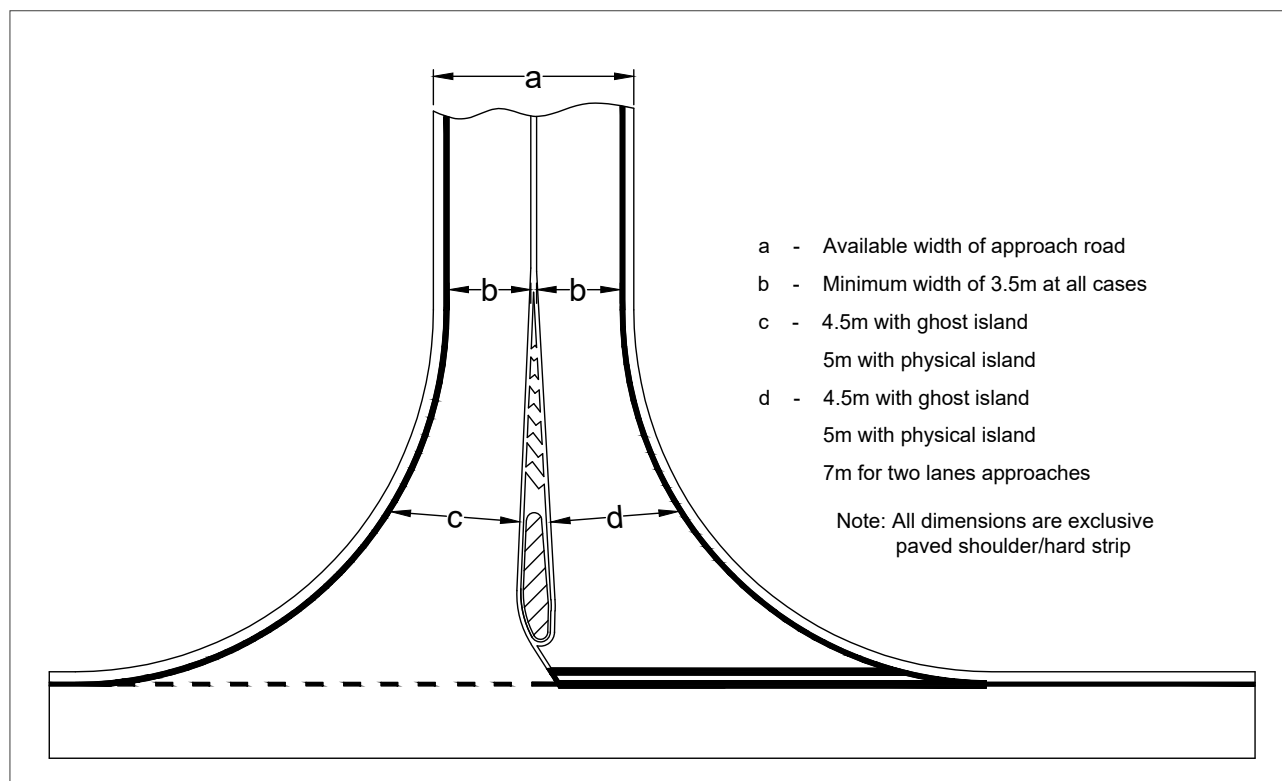
At-grade intersection marking involves:

1. Marking for traffic control.
2. Lane demarcation markings.
3. Diagonal and chevron markings.
4. Directional arrow markings and word messages.
5. Hatch markings / Ghost Island.
6. Prohibitory markings.
7. Pedestrian crossings.

The markings on a minor road approach is shown in Figure 7.1.

Lane demarcating markings, such as centre line, traffic lane line and edge line are explained in Chapter 4. The Stop and Give Way markings are explained in Chapter 5. Directional arrow markings and word messages for junctions are explained in Chapter 6.

Figure 7.1 Dimensions for Minor Road Approaches

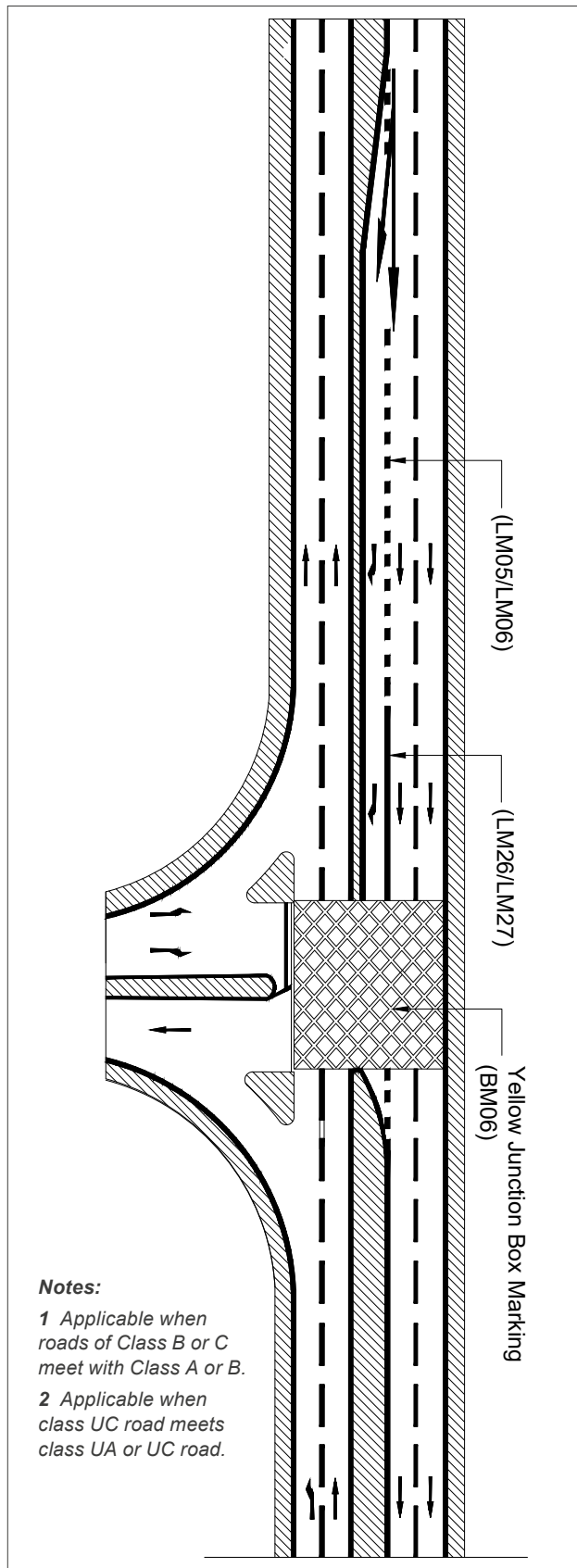


Notes:

- 1 Applicable when roads of Class D, E or other minor roads meet with Class A, B and C road.
- 2 Applicable when class UL road meets class UC road.

Yellow Box Markings: Figure 7.2 shows prohibitory marking to convey the message that no vehicle shall be stopped or stationary in the designated box junction area. Regular maintenance is required. Re-painting is needed when the marking wears out or is not clear.

Figure 7.2 Prohibitory Marking at Intersections



In order to prevent the unauthorised use of the right turn storage lane by the through traffic, the broken marking can be solid line in Figure 7.2.

General guidelines to be followed while providing pavement marking are given for the following typical junction layouts:

1. Simple Junction.
2. Skew or Y-junction.
3. Ghost Island Junction.
4. Right Turn Protected T- Junction.
5. Staggered Junction.
6. Signalised Intersection.
7. Conventional Roundabout.
8. Signal Controlled Roundabout.
9. Double Roundabout.
10. Mini Roundabout.

7.2 Simple Junction

A simple junction is a T or staggered junction without any ghost or physical island in either the major road or minor road. Simple junctions are applicable at locations where the side road traffic is very low and there is no evidence of right turning traffic involved in road crashes. The marking for a simple junction is given in Figure 7.3.

If the forward visibility is clear, broken centre line can be continued through the junction influence area.

If the main road has visibility problems due to horizontal or vertical geometry or any other external factors, the centre line shall be provided for no-overtaking.

The provision of edge lines and the width of the centre line are governed by carriageway width. A Stop line marking along with Stop sign shall be provided. A barrier line (of about 30 m length) shall be provided for the side road, preceded by the warning line.

Figure 7.3 Typical Marking for Simple T-Junction

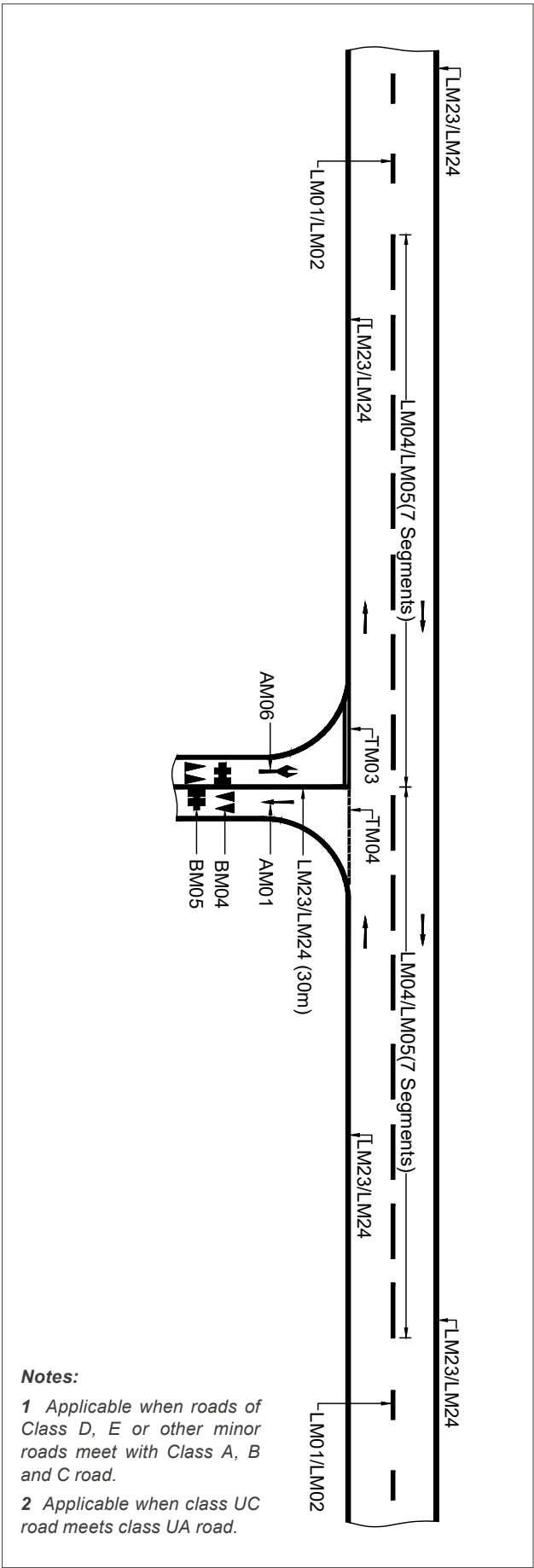
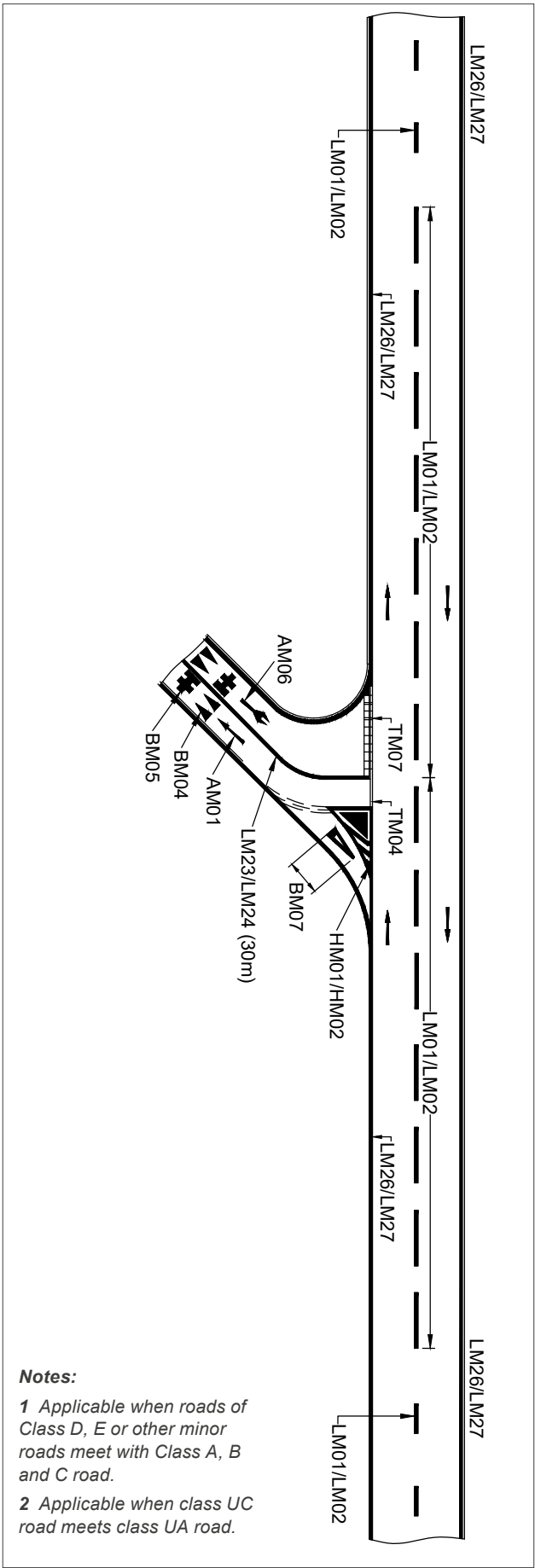


Figure 7.4 Typical Marking for Simple Skew or Y-Junction



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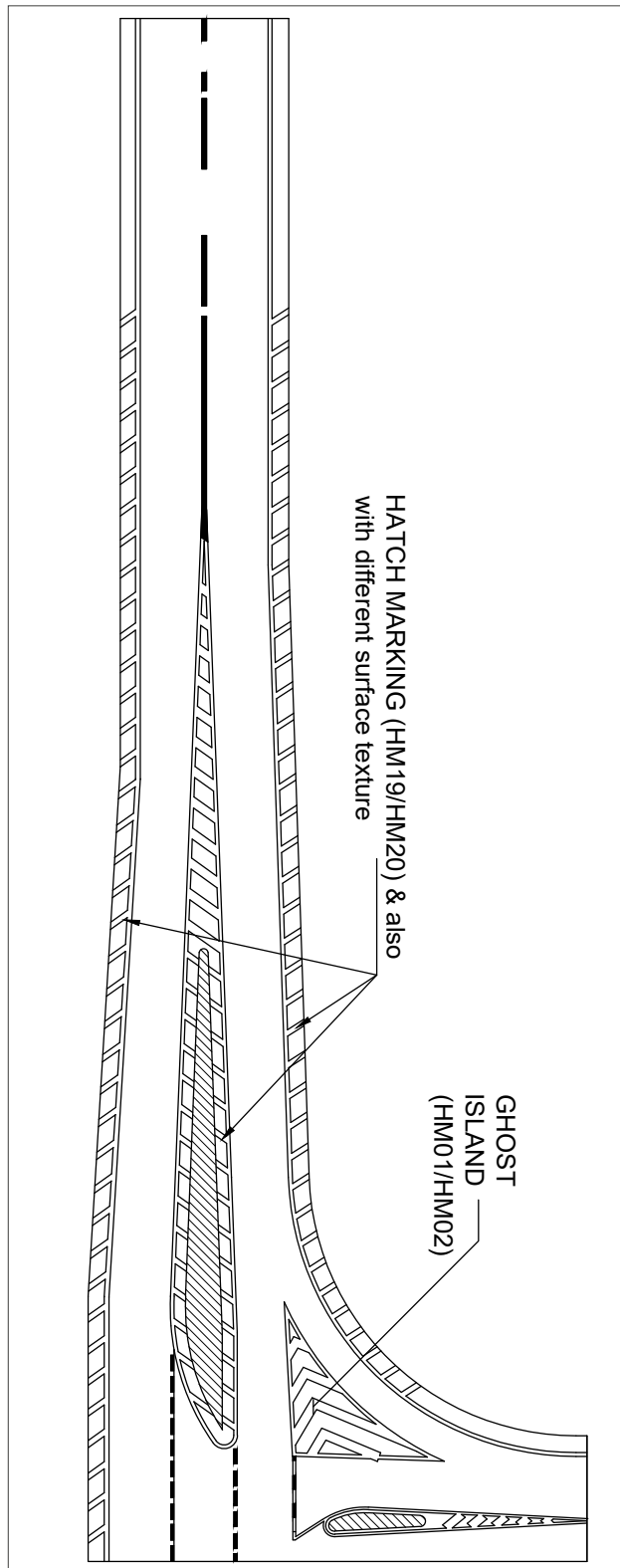
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Markings for At-Grade Intersections

Figure 7.5 Hatch Marking and Ghost Island**Notes:**

- 1 Applicable when roads of Class B, C or D meet with Class A or B road.
- 2 Applicable when class UC road meets class UL road.

7.3 Skew or Y-junction

A skew or Y-junction is a layout where a side road approaches the main road at an oblique angle. A triangular island can be created so that the side road is made to align almost perpendicular to the main road. By this way safety can be enhanced substantially. Figure 7.4 depicts a typical layout.

If the forward visibility is clear enough, the broken centre line can be continued. If the main road has visibility issues due to horizontal or vertical geometry or other external factors, the centre line shall be no-overtaking. The provision of edge lines and the width of the centre line are governed by carriageway width.

A Stop line marking, along with Stop sign, shall be provided. If the visibility funnel is clear, Give Way marking with Give Way sign shall be provided. A barrier line (about 30 m in length) shall be provided for the side road.

7.4 Ghost Island Junction

A Ghost Island Junction is an at-grade junction where an area of the carriageway is marked to direct the traffic movement and to give a shelter lane to the right turning movement.

A wide paved area will generally provoke drivers to move recklessly. Hatch markings direct drivers to confine themselves to the space provided. It also prevents vehicles from hitting the raised islands. Typical hatch marking and ghost island marking is shown in Figure 7.5.

Creating a Ghost Island in the main carriageway makes it easier for the right turning vehicle on the minor road to turn. It also provides shelter to the right turning traffic from through traffic.

The width of through lane should not be more than 3.65 m and not less than 3.0 m. The desirable width of the turning lane is 3.5 m. At urban junctions it is advisable to use a wider turning lane, not exceeding 5 m, so that the right turning vehicle can have some shelter. The width of the ghost island should be between 2.5 and 3.5 m.

Figure 7.6 Junction with Ghost Islands

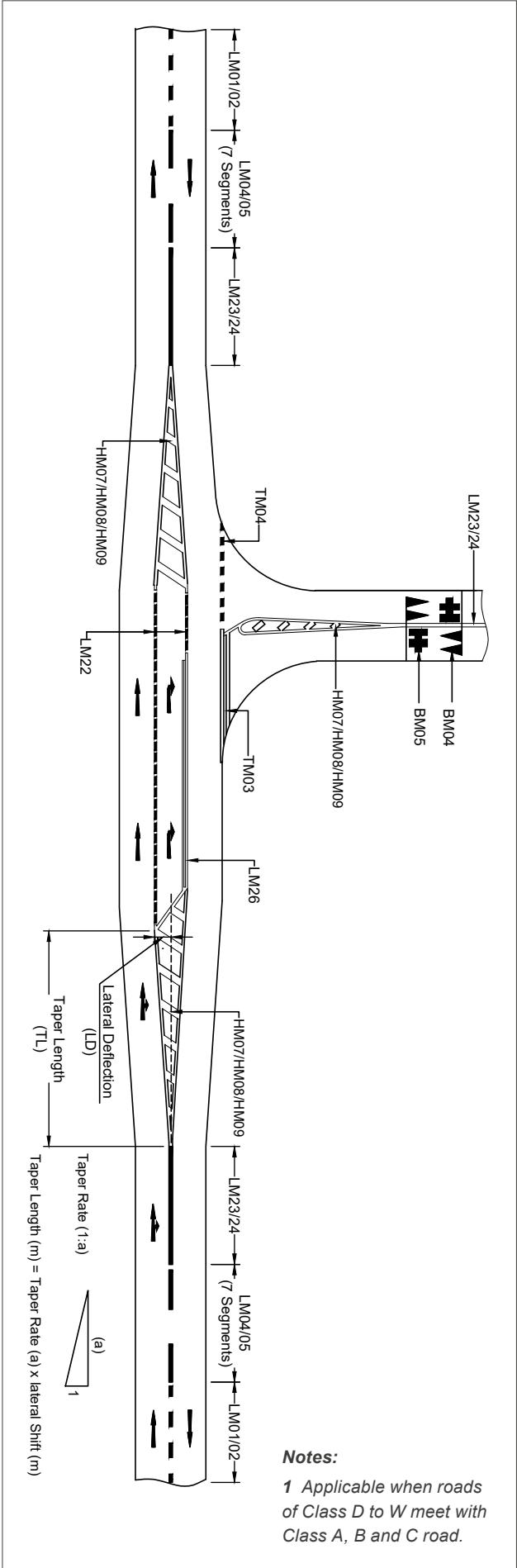
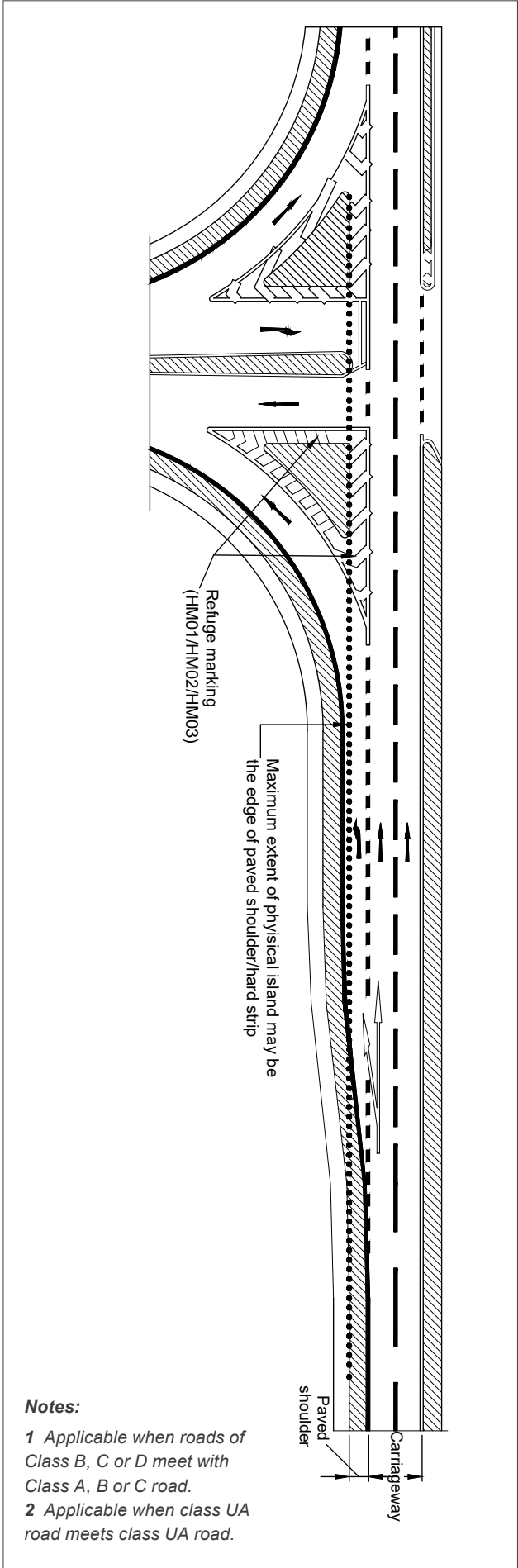


Figure 7.7 Marking for Refuge for Physical Island



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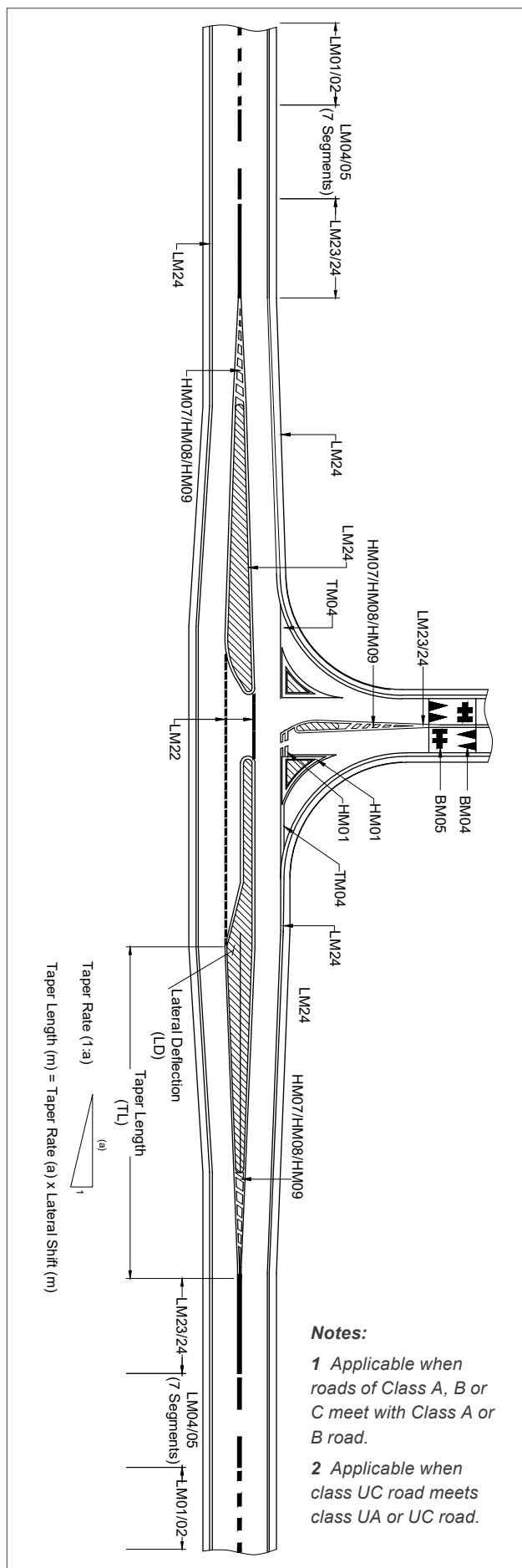
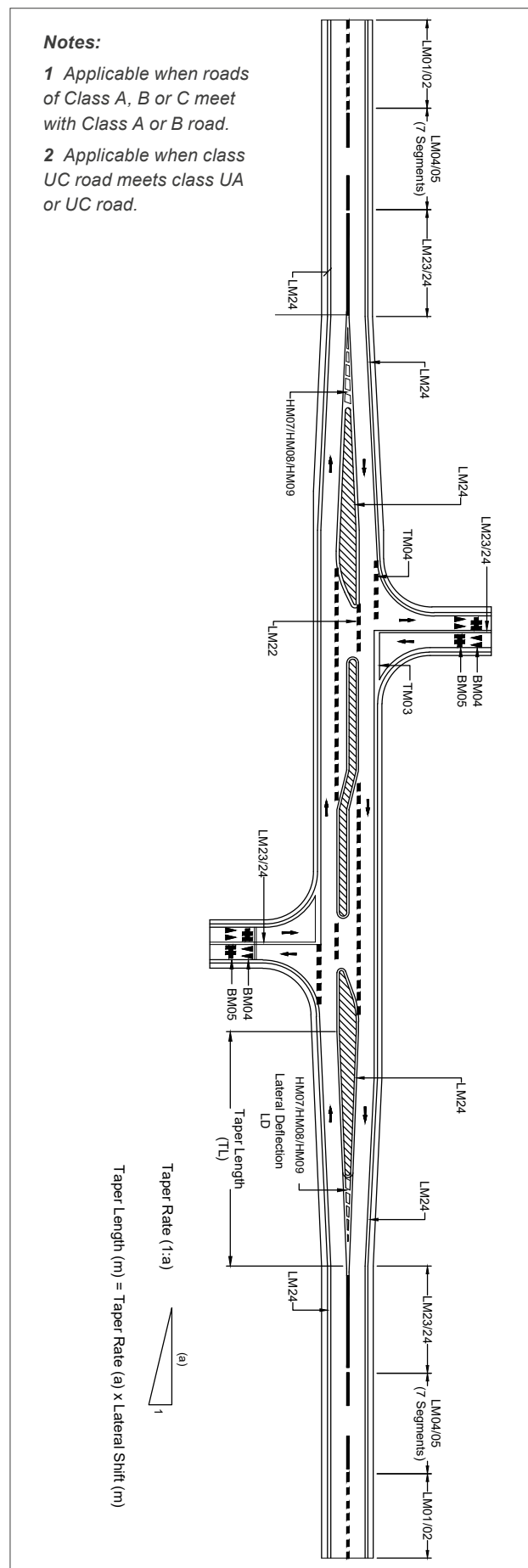
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Markings for At-Grade Intersections

Figure 7.8 Typical Marking T-Junction with Physical Islands**Figure 7.9** Layout for Staggered Intersection (with Right Turn Protected Arrangement)

7.5 Right Turn Protected T- Junction

A Right Turn Protected T-Junction is an at-grade junction where a physical island is created to provide a shelter lane for right turning movement.

The width of through lane should not be more than 5.5 m unlike that in a ghost island. This minimum width shall be catered to while forming the physical island.

In the unlikely event that traffic moves on the paved shoulder and hits the triangular channelising islands, the islands can be protected by refuge island markings instead of a physical island, as shown in [Figure 7.7](#).

The right turn protected bay shall be 3.5 m wide. If sufficient width is not available on the carriageway, the width can be reduced to 3 m. The width of the raised physical island shall be a minimum of 1.2 m to act as a refuge island and also to ensure that signs placed on the islands are not hit by traffic.

In order to create the right turning lane, necessary tapering shall be done as per the taper rate.

A typical layout for a right turn protected T-Junction is given in [Figure 7.8](#). When the carriageway is wide enough and if there is provision for a paved shoulder, shoulder side edge line shall be provided. If sufficient turning radius cannot be provided on a physical triangular island, the islands can be made as a Ghost Island.

Typical marking at a T-Junction with physical islands is shown in [Figure 7.8](#).

7.6 Staggered Junction

At a staggered junction, several minor roads meet a major road at a slight distance apart. [Figure 7.9](#) gives a typical layout for a staggered intersection.

At a staggered junction the major road is continuous. It can accommodate a shelter lane for right turning traffic. Two minor T-junctions are eccentrically placed so that there is enough space to provide a right turning lane. Wherever there is sufficient room, it is desirable to convert crossroads into a staggered junction. This enhances safety.

7.7 Dual Carriageway Junctions

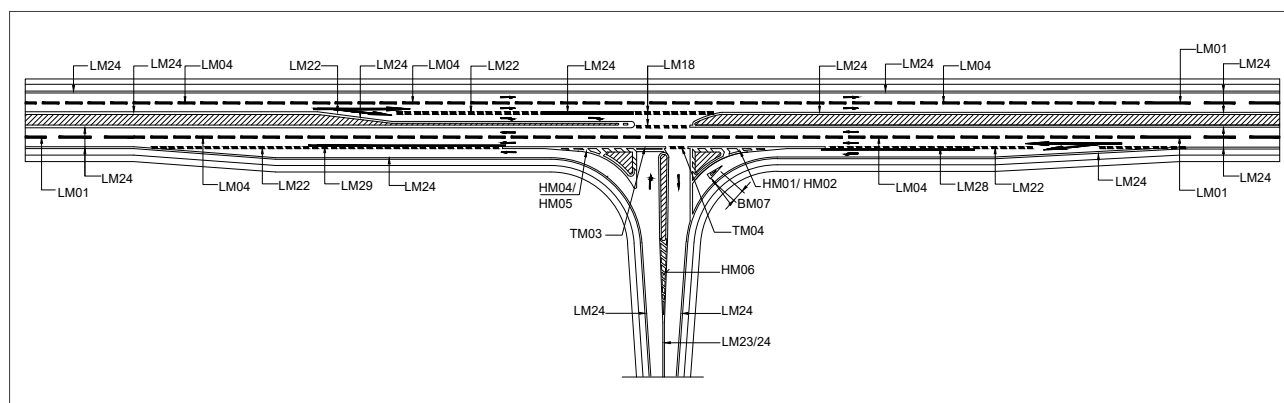
The standard layout for right turns on a dual carriageway is shown in [Figure 7.10](#). The through lanes should maintain their width throughout the carriageway and be separated from the right turn lane with broken markings.

7.8 Signalised Intersection

At signal-controlled junctions, the Stop lines should be placed as near to the intersection as possible, to address the need of pedestrians. Care should be taken that the driver can have an uninterrupted view of at least one signal. Lane lines should also be marked so that the maximum available space of the carriageway is used. Near the intersection and approaches, appropriate arrows should be marked to show the turning lane.

The Stop line should be positioned 1.5 m in advance of the nearside primary signal. The markings should be perpendicular to the centre line of the carriageway. [Figure 7.11](#) shows the basic principles in signal control markings.

The marking layout for a four-armed junction controlled by signals is shown in [Figure 7.12](#). Dedicated arrow markings shall be provided according to signal traffic phase, based on traffic turning movements. Pedestrian markings shall be provided and shall be user friendly. The height of raised kerbs shall be lowered and provided with different surface textures to make them distinct. The width

Figure 7.10 Typical Marking for Junction with a Divided Carriageway Islands**Notes:**

- 1 Applicable when roads of Class A, B or C meet with Class A or B road.
- 2 Applicable when class UC road meets class UA or UC road.

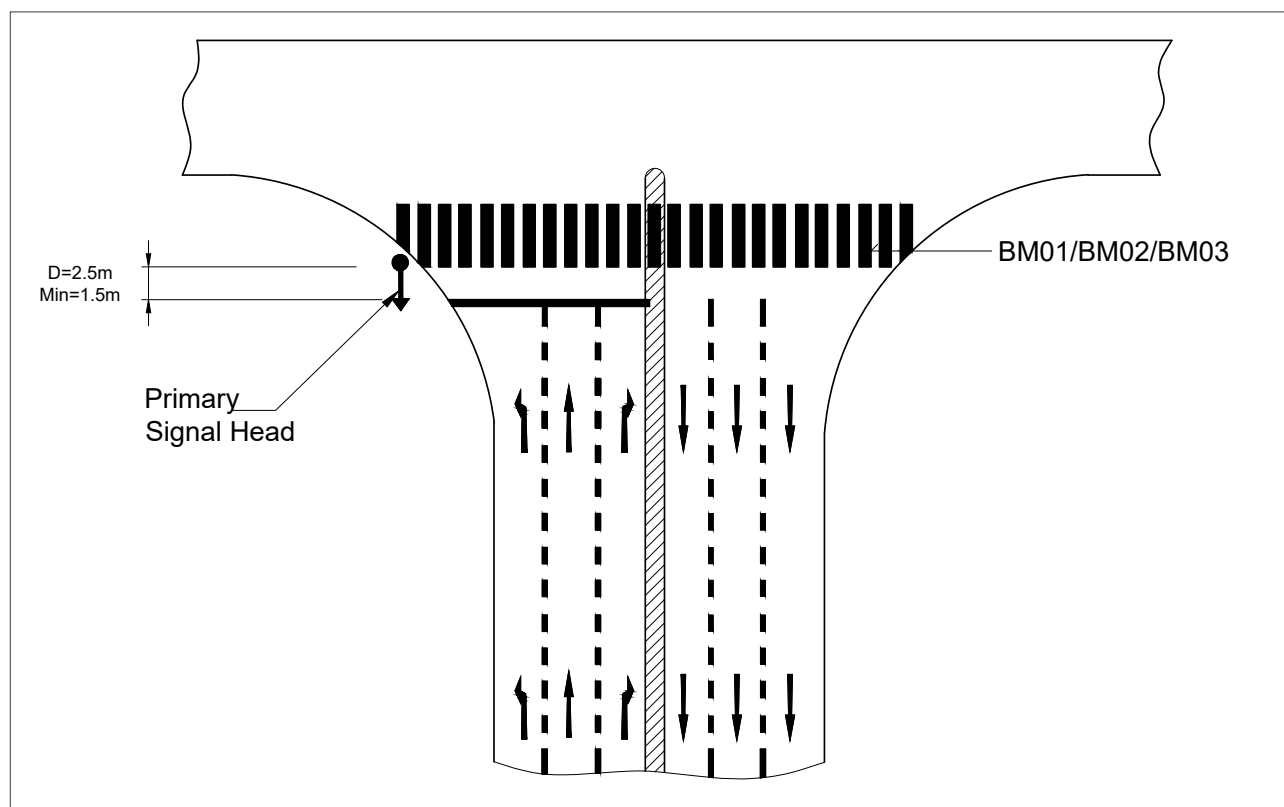
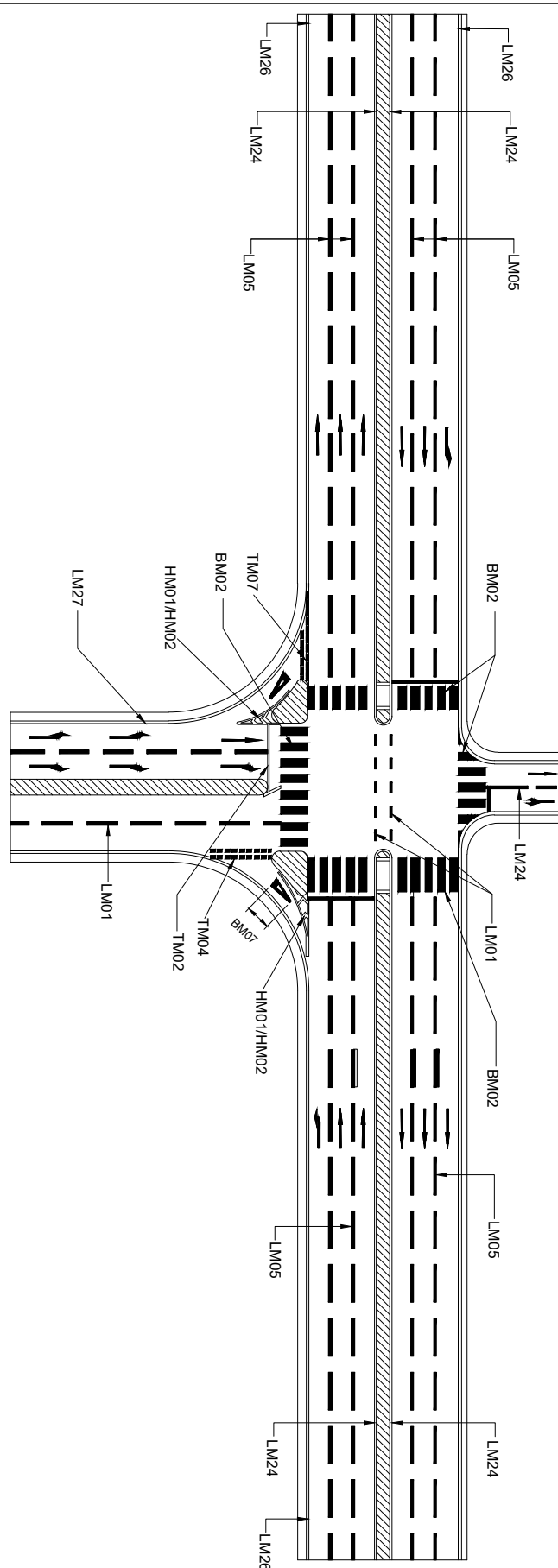
Figure 7.11 Marking in Signal Controlled Junction

Figure 7.12 Typical Marking for Signalized Junction

**Notes:**

- 1 Applicable when roads of Class A, B or C meet with Class A or B road.
- 2 Applicable when class UA or UC road meets class UA or UC road.

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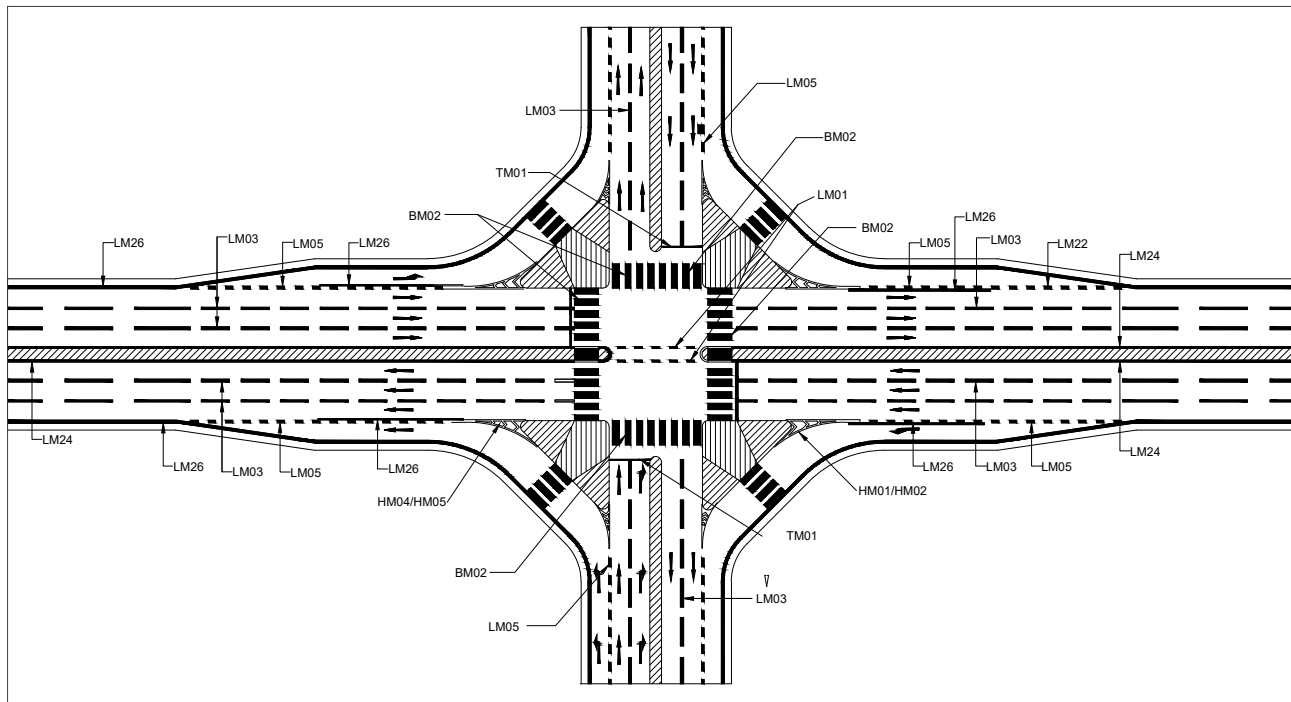
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Markings for At-Grade Intersections

Figure 7.13 Typical Marking for Signalized Junction (with free left turn)

Note 1: Applicable when roads of Class A, B or C meet with Class A or B road

Note 2: Applicable when class UA or UC road meets class UA or UC road.

of the pedestrian crossing shall be based on pedestrian crossing volume and shall be provided with road studs.

A signalised intersection facilitating free left turn movement is shown in [Figure 7.13](#). If turning movement through the free slip road is high, the pedestrian crossing on the slip road shall warrant signal control.

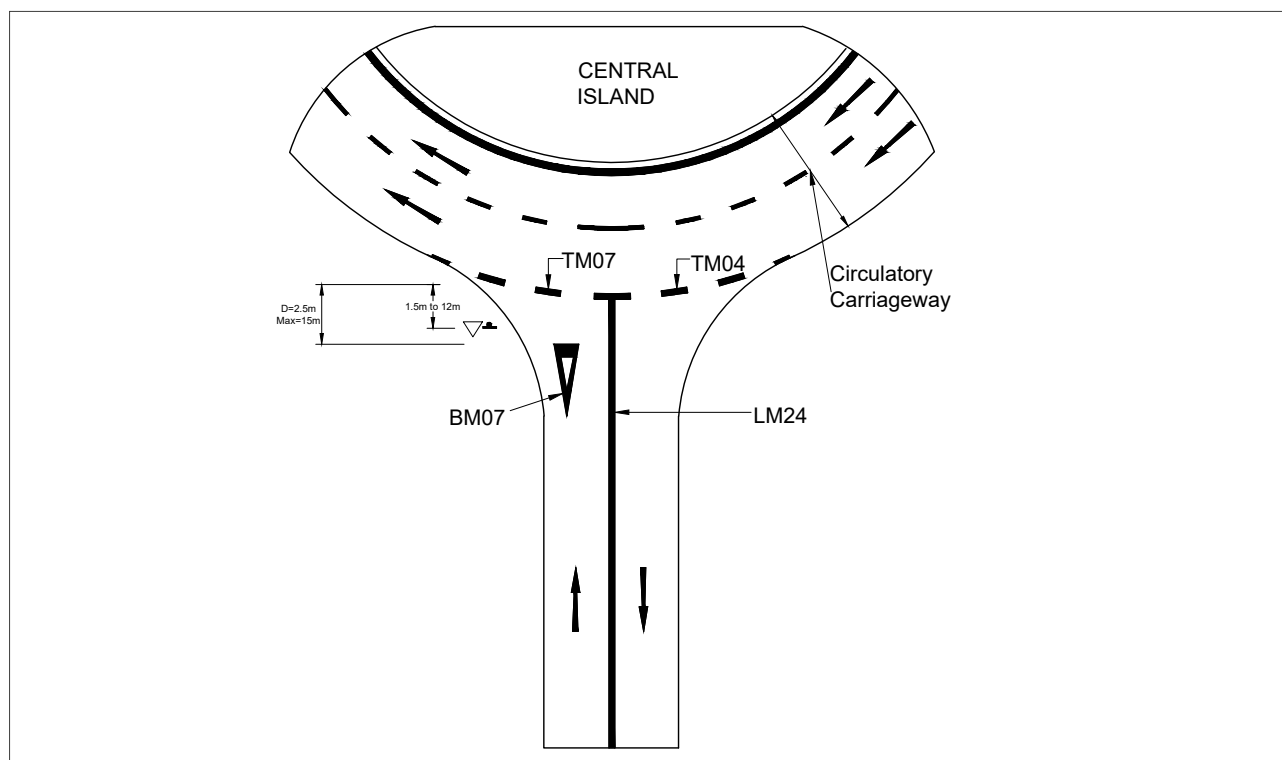
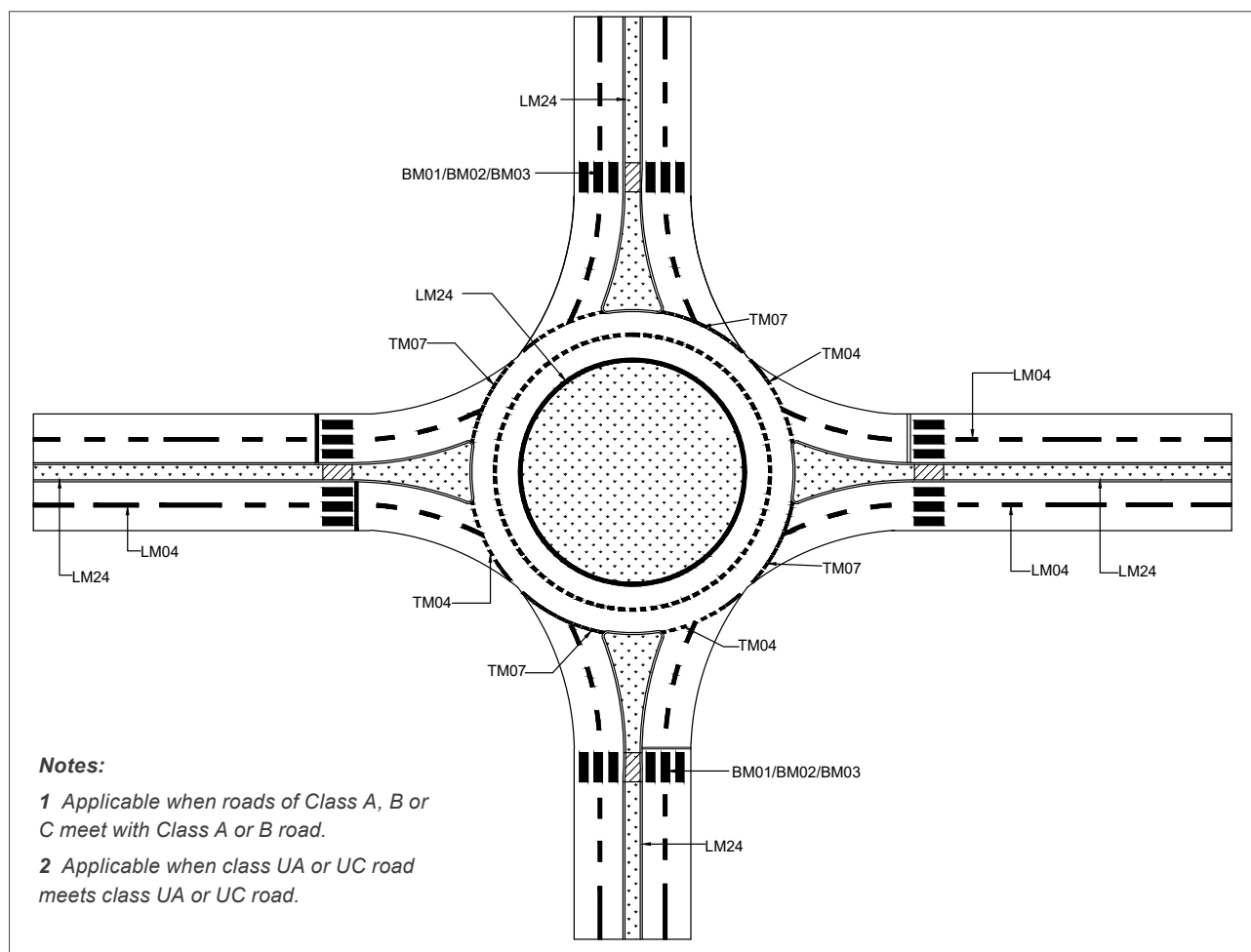
7.9 Roundabouts

The general rule governing traffic movement at roundabouts is that drivers should give way to traffic to their immediate right. Conventional roundabouts always operate based on the 'priority from right' rule. Traffic from an approach road entering the roundabout shall ensure the priority is given to the vehicles already in the roundabout. Give Way marking shall be provided at the approaches as shown in [Figure 7.14](#).

Exit from the roundabouts should be indicated by signs and directional arrows placed at the edges of the central island and also at the exit. Pedestrian and cyclist crossings should be provided with suitable pavement markings.

The marking for a conventional roundabout is given in [Figure 7.15](#) wherein approach roads of more or less same category intersect at the roundabout.

The [Figure 7.16](#) shows the layout of a relatively small roundabout with and without triangular splitter islands. The triangular island can take the form of a physical island or a ghost island.

Figure 7.14 Give Way Marking for Priority in Roundabout Approach**Figure 7.15** Typical Marking for Roundabout

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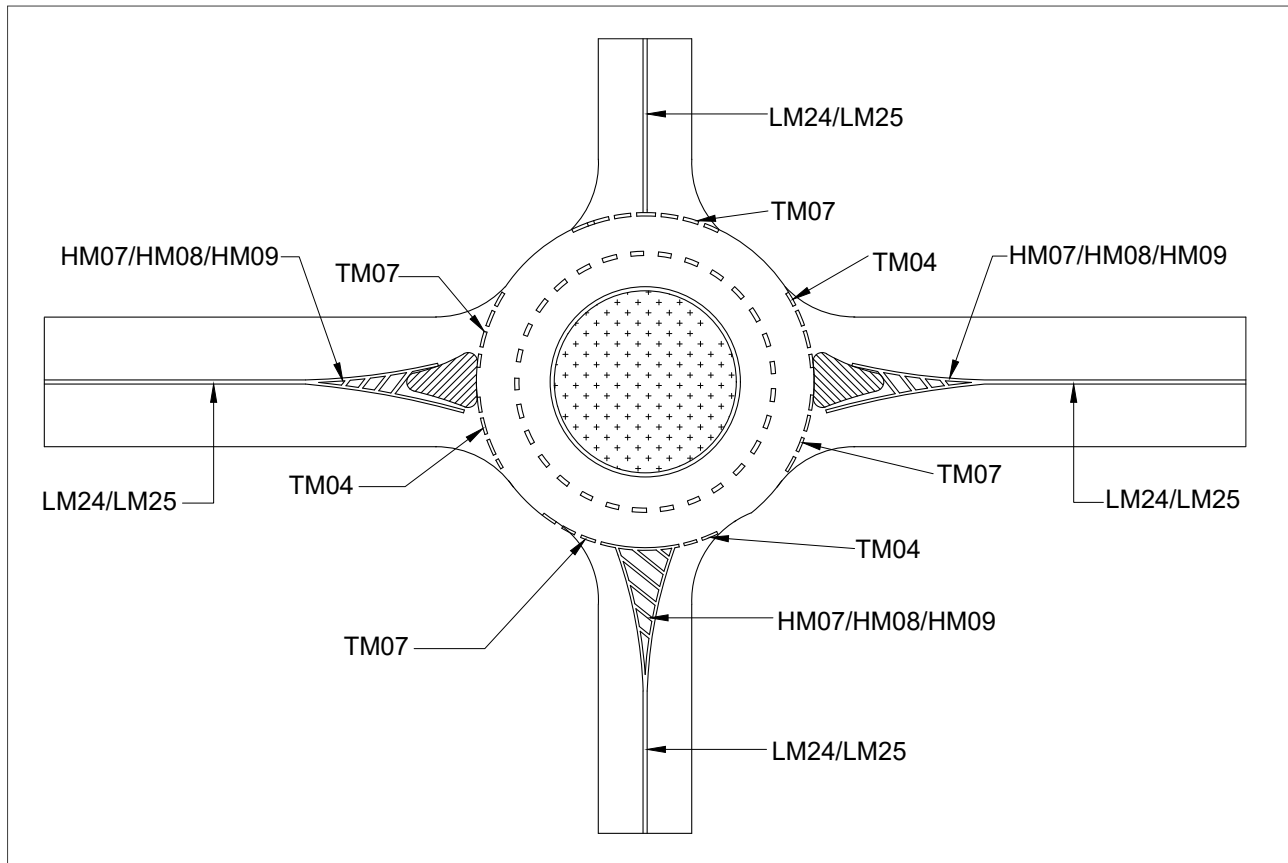
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Markings for At-Grade Intersections

Figure 7.16 Typical Marking for Roundabout (Approach Roads of Different Category)

7.9.1 Signalised Controlled Roundabouts

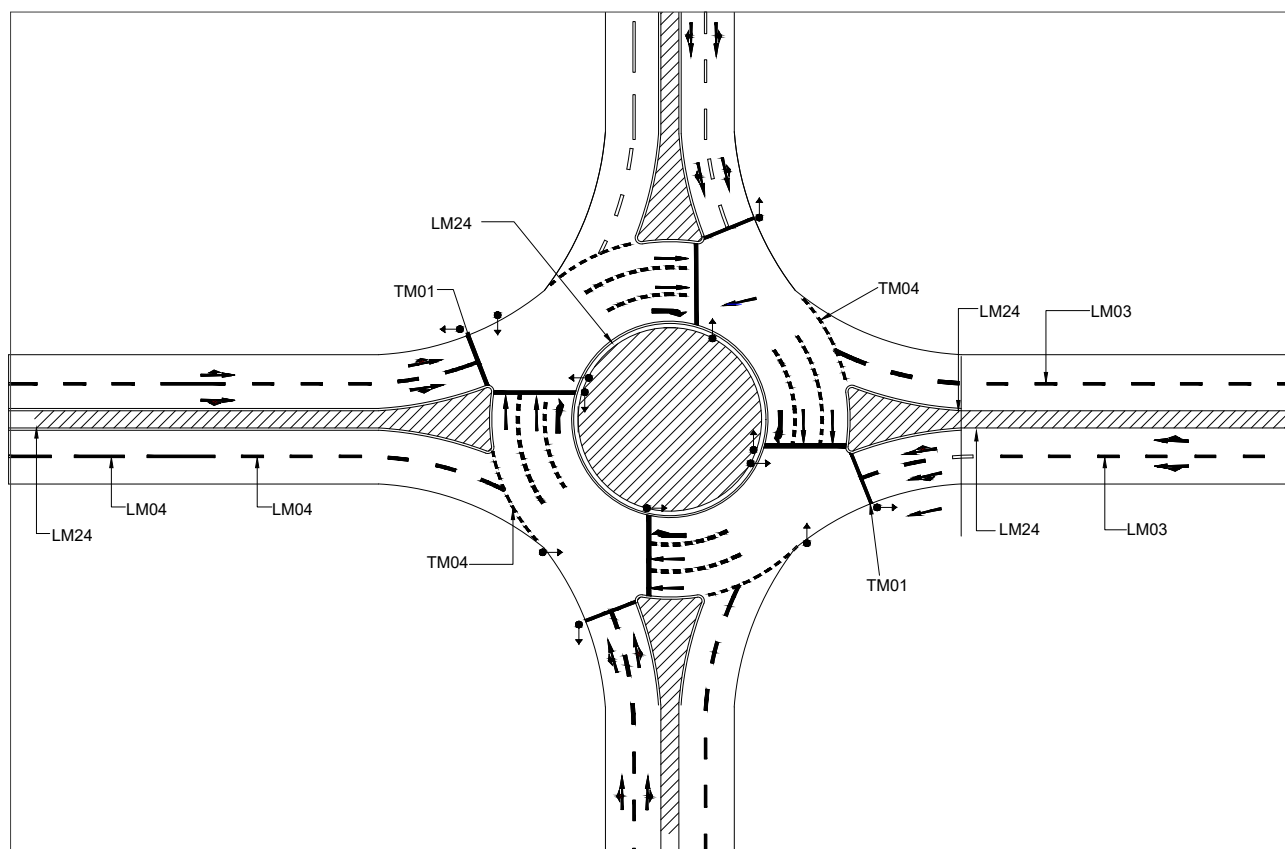
Signal-controlled roundabouts have traffic signals on one or more approach arms. The Stop line should be laid at right angles to the carriageway edge. Lane markings should be provided on the circulatory path of the roundabout.

The common configurations of road markings which are used on circulatory carriageways are concentric circles, which help drivers to know the extent of lane width in which they are circulating. Lane arrows and destination arrows are usually repeated on roundabout circulatory carriageways. At the entries of roundabouts these arrows help drivers to identify the correct lane for the specific exit on the roundabout.

Pedestrians should not be directed to cross the roundabout. Pedestrian crossing should be provided with safer crossing points elsewhere.

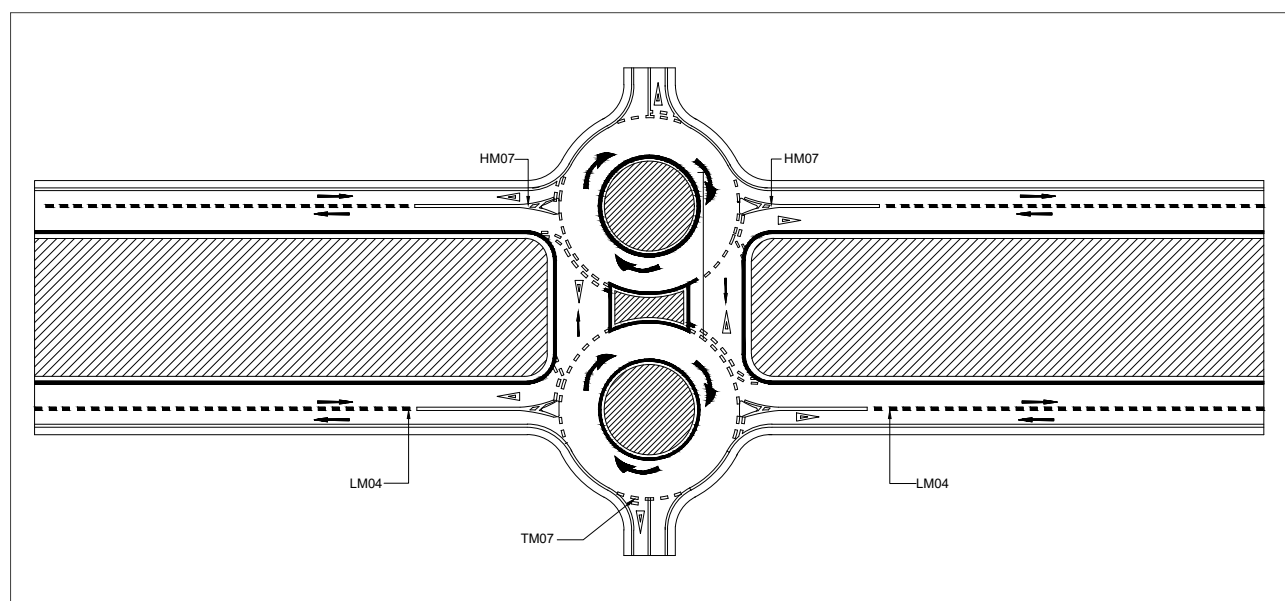
Yellow box markings at un-signalised roundabouts are not allowed but could be used on signal-controlled roundabouts.

The marking for a signal-controlled roundabout is given in [Figure 7.17](#).

Figure 7.17 Signal Controlled Roundabout

7.10 Double Roundabout

Double roundabouts are required in situations where a number of roads meet, and it is not possible to handle them in one large roundabout. In a double roundabout, as shown in [Figure 7.18](#), the two roundabouts function as standalone roundabouts.

Figure 7.18 Typical Marking for Double Roundabout

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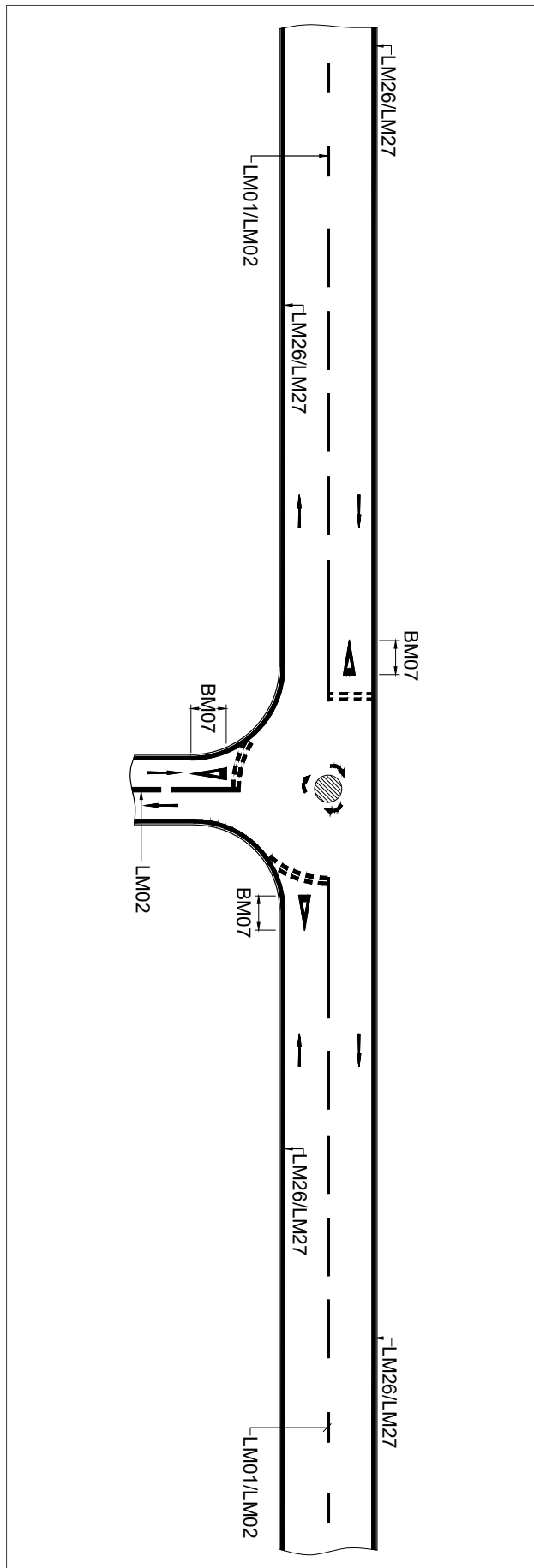
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Markings for At-Grade Intersections

Figure 7.19 Mini Roundabout (Traffic Control by Give Way)

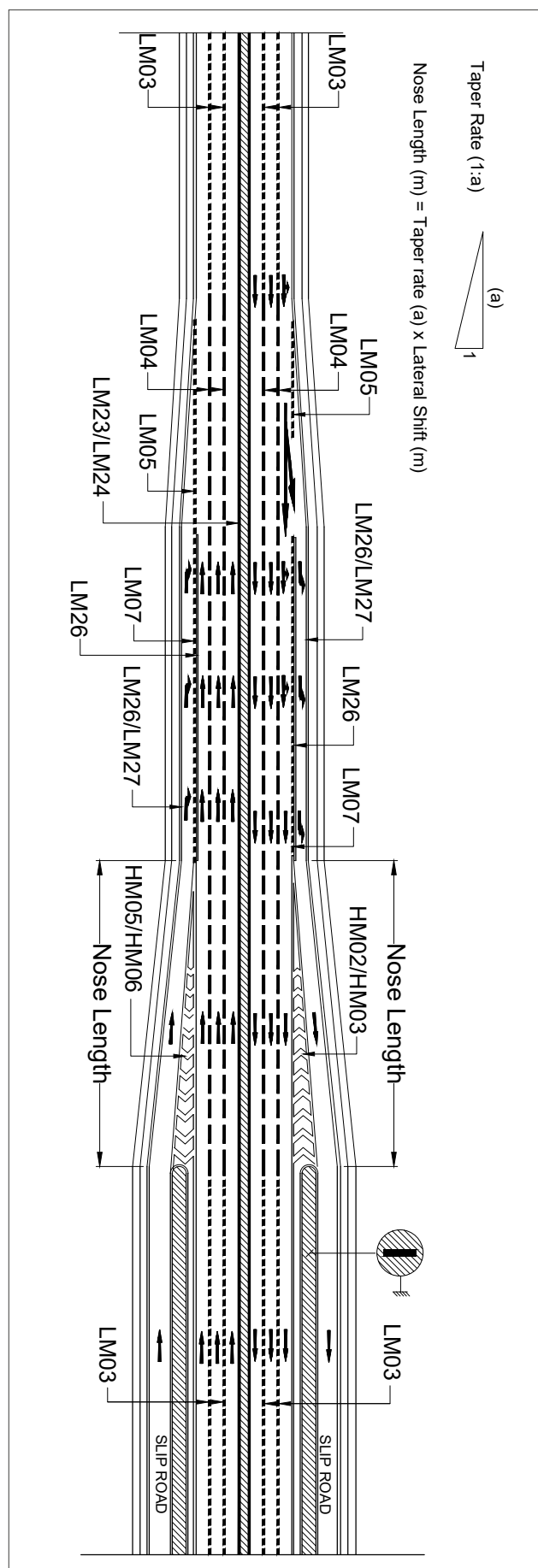
7.11 Mini Roundabout

Mini roundabouts are effective in improving safety at junctions prone to side road delays in urban areas. It can be installed with minimal alterations to kerbs. Mini roundabouts have a one-way circulatory carriageway around a slightly raised central disc, with or without flared approaches. The layout of mini roundabout is shown in Figure 7.19.

The layout (including signs and markings) should be designed in such a way that drivers are aware in good time that they are approaching a roundabout. Mini roundabouts should be used when all approaches are subject to a speed limit of 30 km/h or less. The mini roundabout is not recommended when the speed limit on any of the roads is above 30 km/h as markings will not be conspicuous at higher speeds and the space on the mini roundabout will not be sufficient for deflection of the vehicle. The choice of size of mini roundabout will depend on the available space. The standard Give Way marking should be used at mini roundabouts accompanied by a sign indicating a roundabout, sited at about 1.5 m advance of the Give Way markings. The central islands of mini roundabouts should be kept free of all types of furniture.

8 Grade Separated Intersections

Figure 8.1 Merging/Diverging (Without Lane Gain/Lane Drop)



8.1 General

In the case of grade separated intersections, vehicles change lanes from the main carriageway to the auxiliary lanes at high speed. It is important to guide the driver to the proper lane.

Merging and diverging marking can be provided with lane gain /drop or without lane gain / drop depending upon the volume of traffic diverging / merging traffic.

8.2 Merging/Diverging without Lane Gain or Lane Drop

Merging and diverging movements near a flyover or a Vehicle Under Pass (VUP) are generally carried out without any lane gain or drop. It is shown in Figure 8.1.

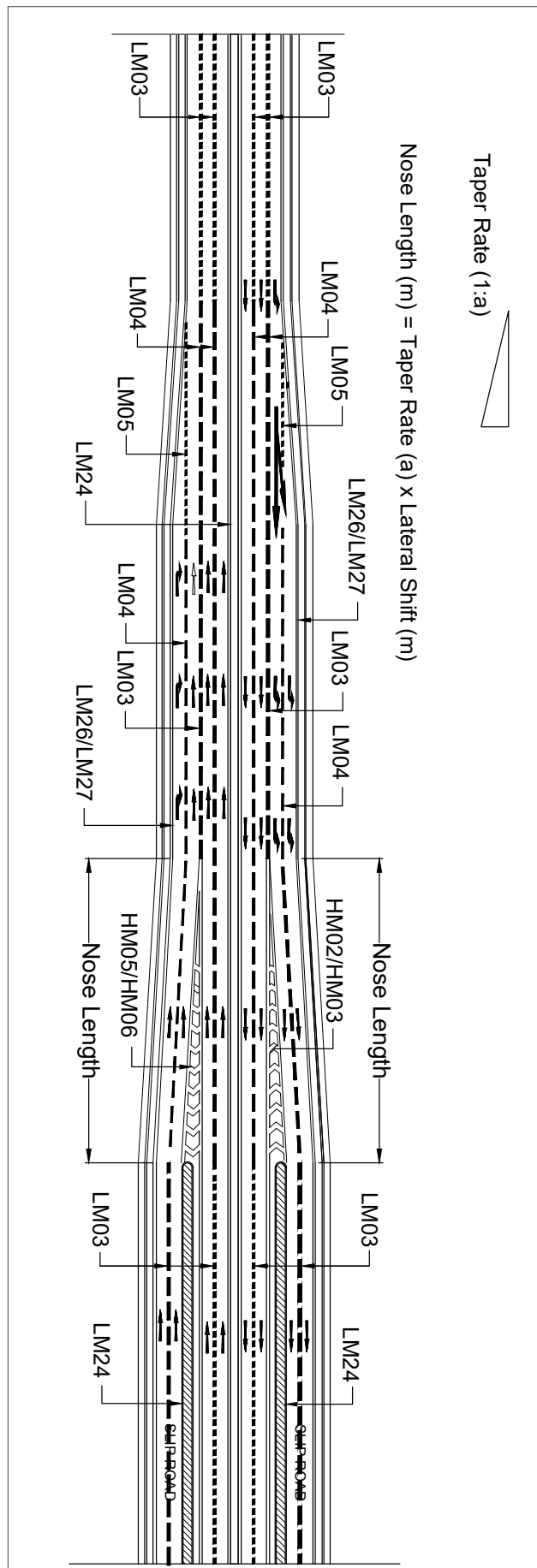
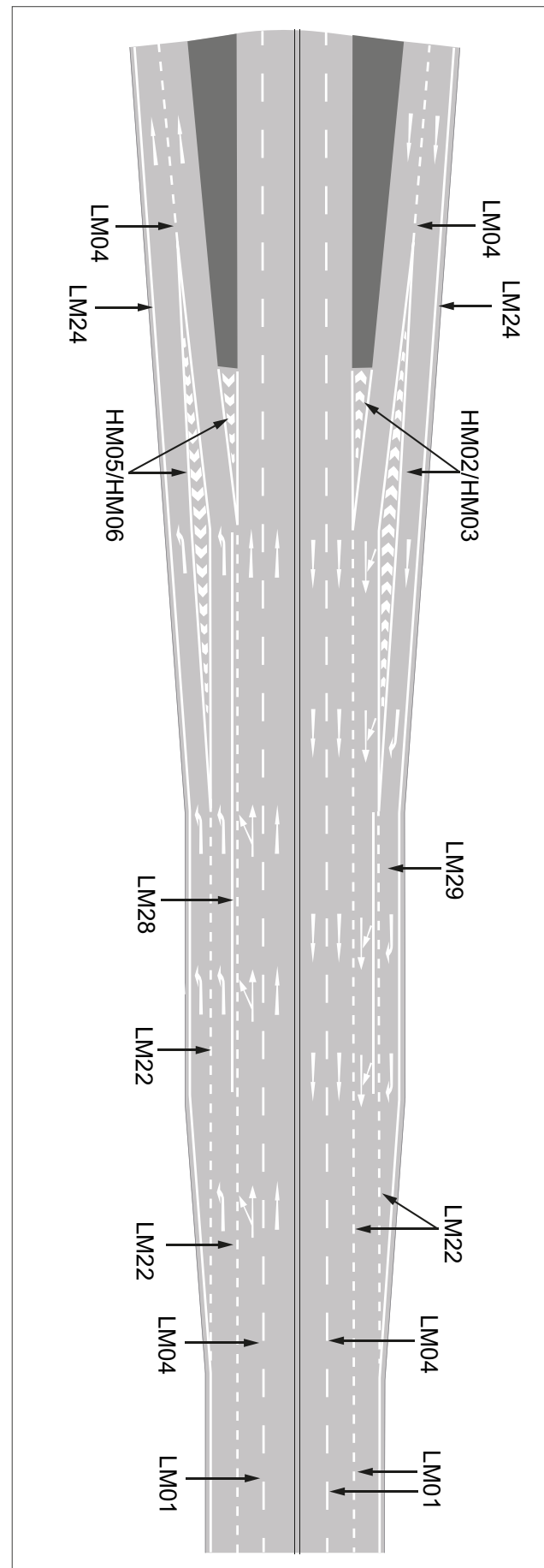
8.3 Merging/Diverging with Lane Gains and Lane Drop

If there is a substantial volume of traffic either diverging or merging, these operations are to be facilitated by the provision of lane gain or lane drop, as shown in Figure 8.2.

The nose length shall be determined based on the speed and actual lateral shift of traffic. The length of auxiliary lane and taper shall be as per the design.

For diverging and merging situations with two lane slip roads, a safer practice of lane gain and lane drop is to provide ghost marking, as shown in Figure 8.3.

The abbreviations LM01, LM02 etc, HM03, HM04 etc refer to type of longitudinal markings and hazard markings to be used. The markings are detailed in Table A.1 to A.7 in Appendix A of this Manual.

Figure 8.2 Merging/Diverging (With Lane Gain/Lane Drop)**Figure 8.3** Merging/ Diverging with Ghost Island (With Lane Gain/ Drop)

9 Markings for Pedestrians and Cyclists

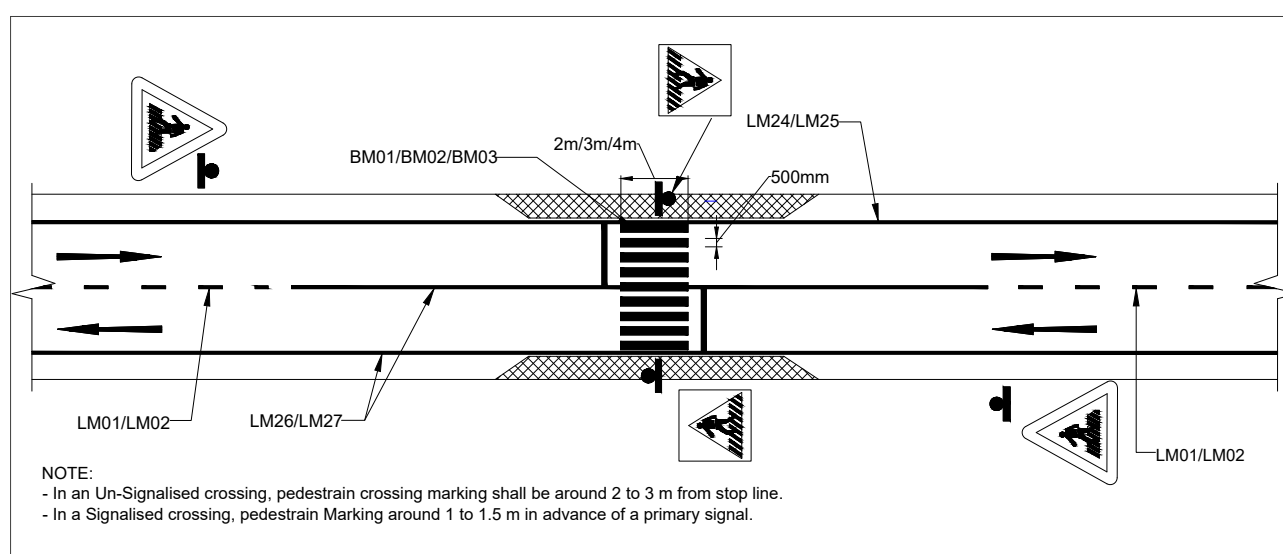
9.1 General

This chapter details the markings for pedestrian crossing at mid-block section, T-junctions, and hazardous areas. The chapter also deals with cycle marking on roads and railway level crossings.

9.2 Pedestrian Crossings

Pedestrian crossings are provided at important intersections. It is essential where conflict exists between vehicles and pedestrians. The success of a pedestrian crossings depends on where and how it is marked. The site should be so selected that the pedestrians are subjected to minimum inconvenience and have enough visibility to see the oncoming vehicles. At the same time the vehicular traffic should not be interrupted too often. Figure 9.1 shows markings for pedestrian crossing at a mid-block section.

Figure 9.1 Pedestrian Crossing



At intersections, the pedestrian crossings should be preceded by a stop line. In an unsignalised pedestrian crossing, the stop line shall be set back 2 to 3 m from the start of pedestrian crossing markings. In a signalised intersection, the pedestrian crossing shall be 1 m ahead of the primary signal and the stop line shall be placed another 1 m ahead of the start of pedestrian crossing.

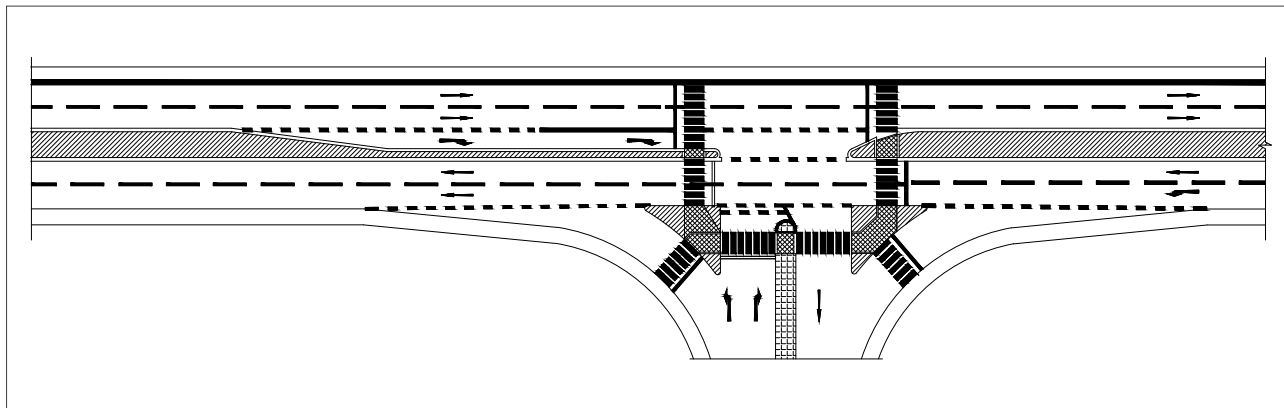
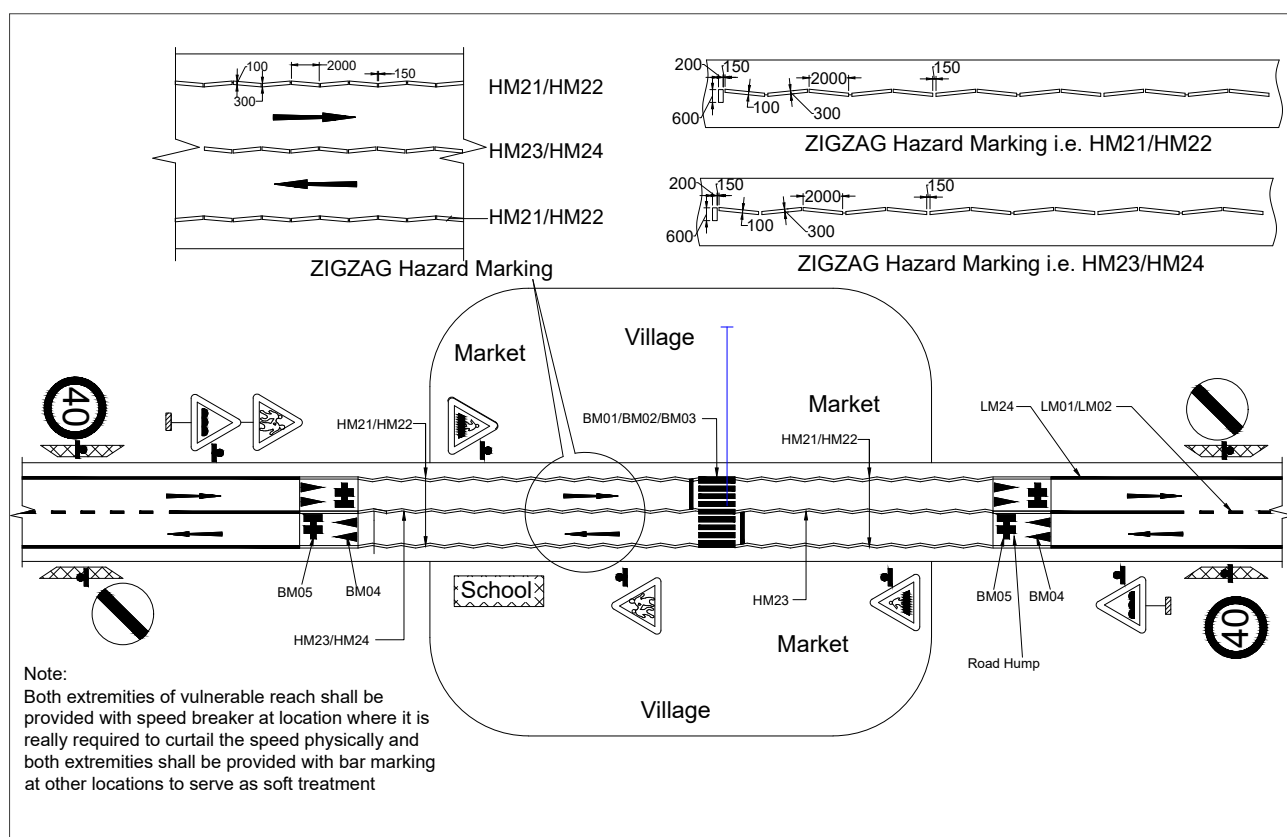
If the pedestrian crossing length is more than 10.5 m at one go, there shall be a refuge island of at least 1.2 m width to serve as a shelter place. Road studs shall be provided to make the pedestrian crossing visible to users during night-time. All kerbs to be mounted by pedestrians shall not be more than 50 mm high and shall be user friendly for wheelchair usage.

Pedestrians shall have sufficient space on the walkway to wait. Obstructions such as trees, signposts, lamp posts etc. shall be cleared in the path of pedestrians at either end of the pedestrian crossing.

The width of the pedestrian crossing is governed by the pedestrian volume crossing and by local requirement. It should not be less than the width of the walkway, subject to minimum of 2.0 m. The width of a crossing is generally between 2 and 4 m. In urban locations, it may be advantageous to install flashing signals 3 to 5 m in advance to warn drivers of the presence of the crossings.

It is desirable to provide a raised kerb, at least for a small length, so that pedestrians feel safe while waiting for an opportunity to cross the road.

Pedestrian crossing marking at priority T junction is shown in Figure 9.2.

Figure 9.2 Pedestrian Crossing Marking on Priority Junction**Figure 9.3** Pedestrian Crossing in Vulnerable Reach (Special Treatment)

9.3 Marking for Vulnerable Road Section

A road passing through an area with human activities for a distance shall be marked separately as shown in Figure 9.3.

The centre line and edge line marking for the entire vulnerable section shall be with zigzag hazard marking to alert the driver. Both extremities of the vulnerable section shall be provided with speed breakers. Both extremities shall be provided with bar marking.

9.4 Cycle Lane Marking

Cycle lane marking should be provided when a portion of the carriageway is marked for exclusive use of cyclists.

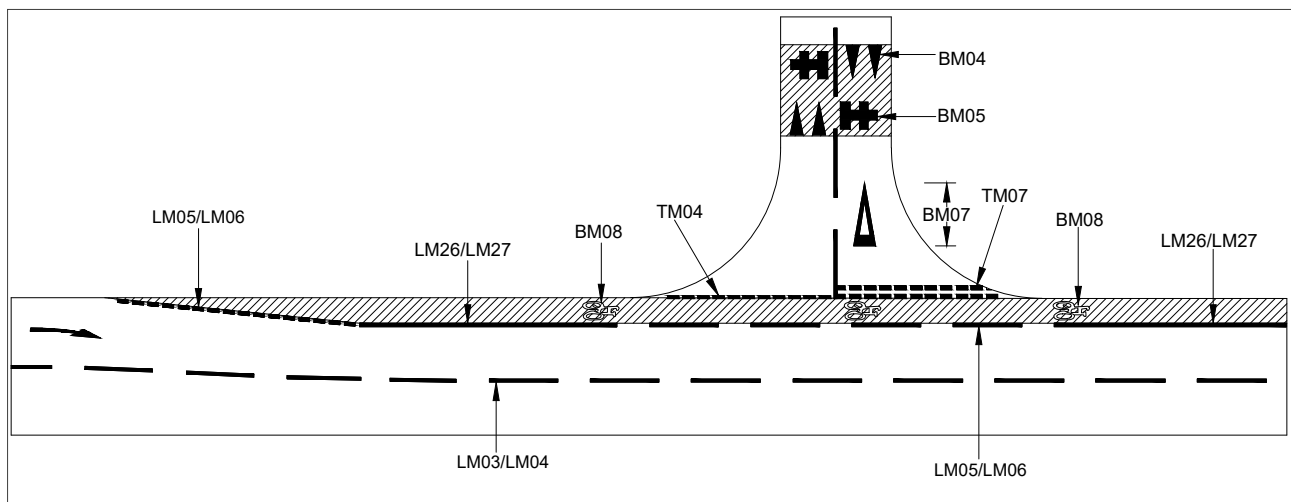
The markings shall consist of 150 mm wide solid white boundary line. The minimum width of the cycle path shall be 1.5 m. It shall be 2 m where the cycle traffic is heavy.

Cycle symbol shall be marked on cycle lanes. The termination of a prescribed cycle lane should be indicated by an END marking on the carriageway. This must always be used in conjunction with a cycle symbol. Cycle lane markings shall be supplemented with cycle lane signs.

The continuity of the white solid line shall be maintained at bus stops, junctions and across driveways. A broken taper line (600 mm mark and 300 mm gap) should precede cycle lanes. It can have an inclination 30 to 45 degrees but not sharper than 1:10, as shown in Figure 9.4.

Advisory cycle lane markings shall be placed at locations where vehicles enter from a side road may enter the cycle lane. These locations shall be marked with warning lines on either side.

Figure 9.4 Cycle Lane Marking



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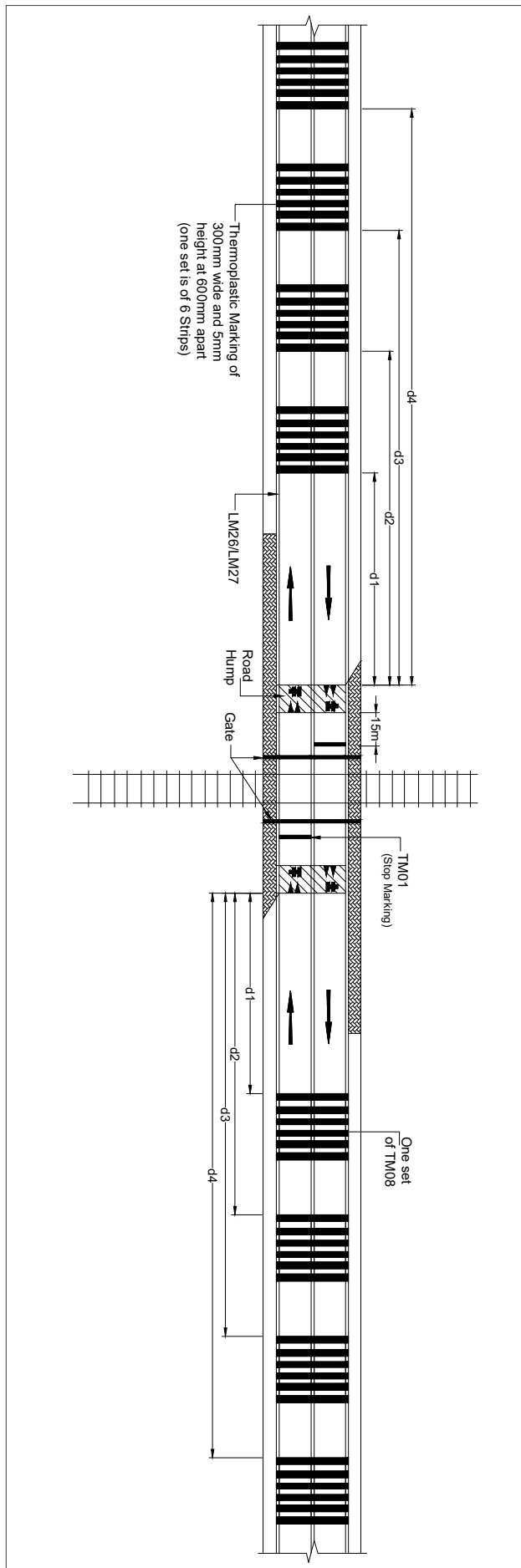
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Markings for Pedestrians and Cyclists

Figure 9.5 Rail Crossing



9.5 Railway Level Crossing

Markings at road-rail level crossings shall comprise of the following:

1. No-overtaking zone marking for roads with two or more lanes (longitudinal).
2. A stop line (transverse).
3. A series of bar marking depending on the approach speed.
4. A speed breaker marking.

The markings shall be placed at all road-rail level crossings with significant vehicle/rail conflict. At unimportant crossings or in urban areas, where other devices are provided with suitable control, these markings may be omitted. Continuous carriageway edge lines shall be marked at rail crossings. The type of centre line marking used generally depends on the width of the carriageway available at the rail crossing. Where the width of the carriageway over the rail crossing is less than 5 m, centre line markings will not normally be provided. If the carriageway width is between 5 and 5.5 m, then longitudinal warning lines shall be provided on either side of the crossing. If the carriageway width is greater than 5.5 m, then the double solid lines at the crossing shall be provided. Arrow markings shall be provided at approach arms to reinforce the solid lines at crossings.

Transverse Marking

If a traffic signal is installed, then a Stop line of 300 mm width shall be placed at 1 to 2 m from the signal. At open controlled crossings, a Give Way line shall be provided at more than 2 m from the near approach rail, preceded by Give Way markings and signs.

Figure 9.5 depicts a typical provision of marking on approach arms by delineating with a series of bar markings, speed breakers and stop line markings.

10 Markings for Buses, Truck Laybys and Toll Plaza

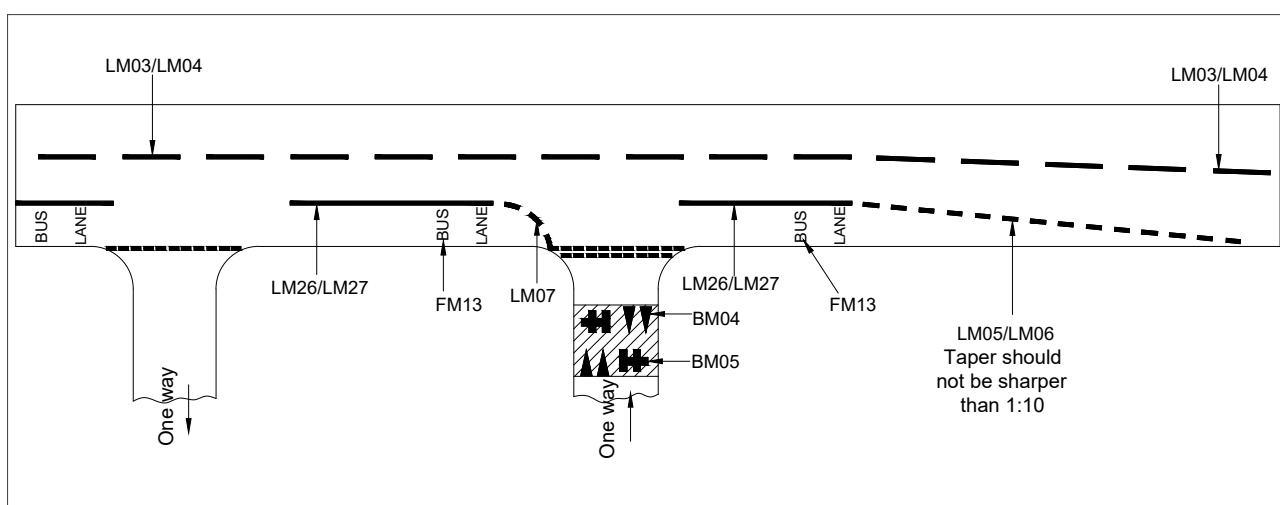
10.1 General

This chapter presents the markings required for bus lanes, bus bays, truck laybys and toll plaza.

10.2 Bus Lane

Lanes reserved for buses without physical separation shall be provided with white lines as bus-lane markings. It is shown in Figure 10.1. Generally, width of 3 m is required for a bus lane. The distance is measured from the edge of the kerb to the centre of the continuous white line. A gap in the white line should be provided adjacent to each side road.

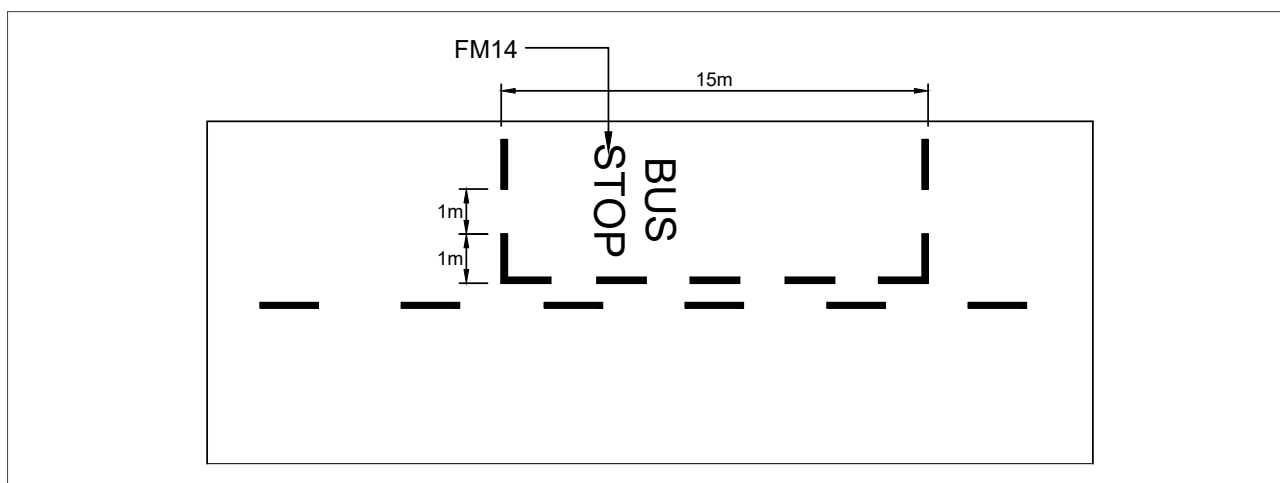
Figure 10.1 Bus Lane Marking



The legend BUS LANE should be marked on the carriageway across the lane at its commencement and repeated after each junction. Where junctions are more than 300 m apart, this legend should be repeated between junctions at approximately 150 m intervals.

Where a bus lane commences just beyond an intersection, adequate length should be left for the taper to commence at the intersection so that the inclined line does not extend across the intersection mouth. The taper shall be 1:10. The continuous bus lane should end ahead of any intersection with major left-turning flow.

Figure 10.2 Bus Stop Marking



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Markings for Buses, Truck Lay-Bys and Toll Plaza

10.3 Bus Stop

Pavement markings at bus stops should be provided with the word BUS STOP written prominently on the pavement. The length of the bay for bus stops shall be 15 m as a minimum, as shown in Figure 10.2. It may be increased in stages of 30 m up to a maximum of 45 m. The word message BUS STOP shall be repeated in each box of 15 m length. The line marking for the bay shall be white in colour, 100 mm wide.

Pedestrian crossings should be marked slightly behind the standing position of buses in order to avoid conflicts. The kerbs should be marked with a continuous yellow line to indicate no parking.

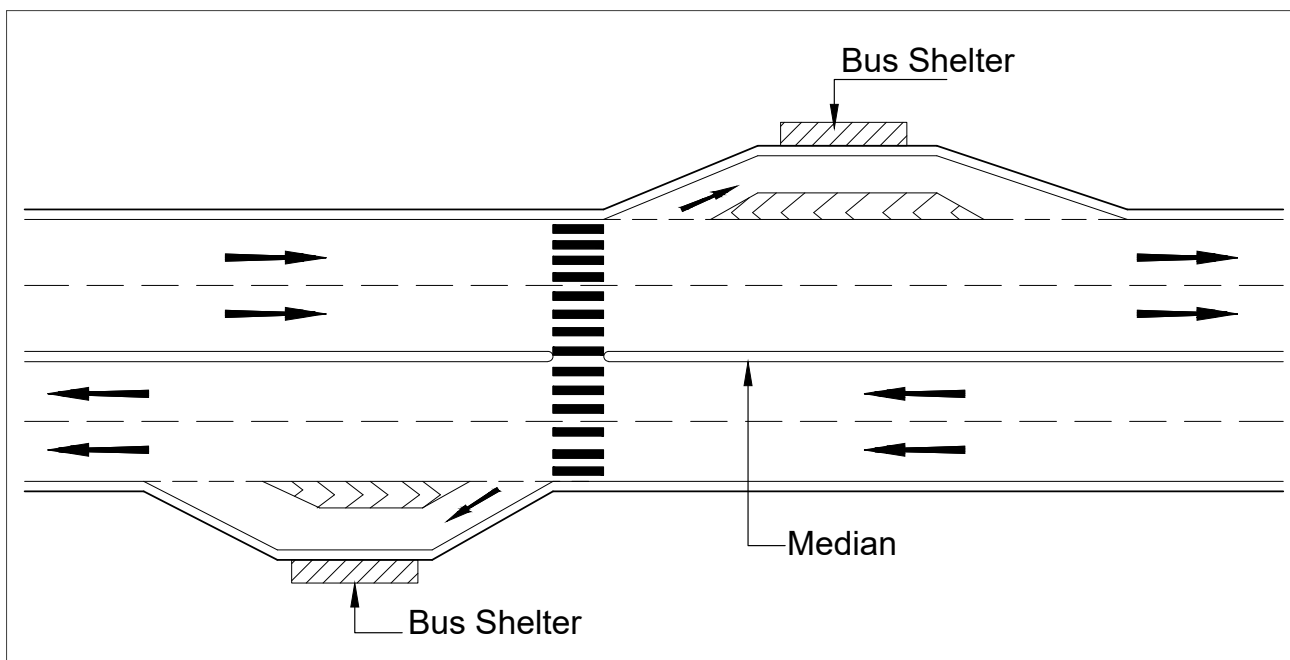
Bus Stop markings shall be supplemented with Bus Bay signs at the start of a transition approach to bus stop.

10.4 Bus Bay Marking

Bus bay marking should be provided with proper chevron marking and can be either with physical island or with ghost island.

Generally, a physical island shall be aligned in such a way that a vehicle moving on a paved shoulder should not collide with the separating island. Figure 10.3 illustrates the marking requirement for bus bay with ghost Islands.

Figure 10.3 Bus Bay Markings



10.5 Truck Layby Marking

Truck bay marking should be provided with proper chevron marking and with advance informatory signs.

Generally, the physical island shall be aligned in such a way that a vehicle moving on a paved shoulder not hit the separating island, as shown in Figure 10.4. The left lane shall be provided throughout, clearly defining the through lanes and diverging and merging lanes.

Figure 10.4 Truck Lay-by Markings

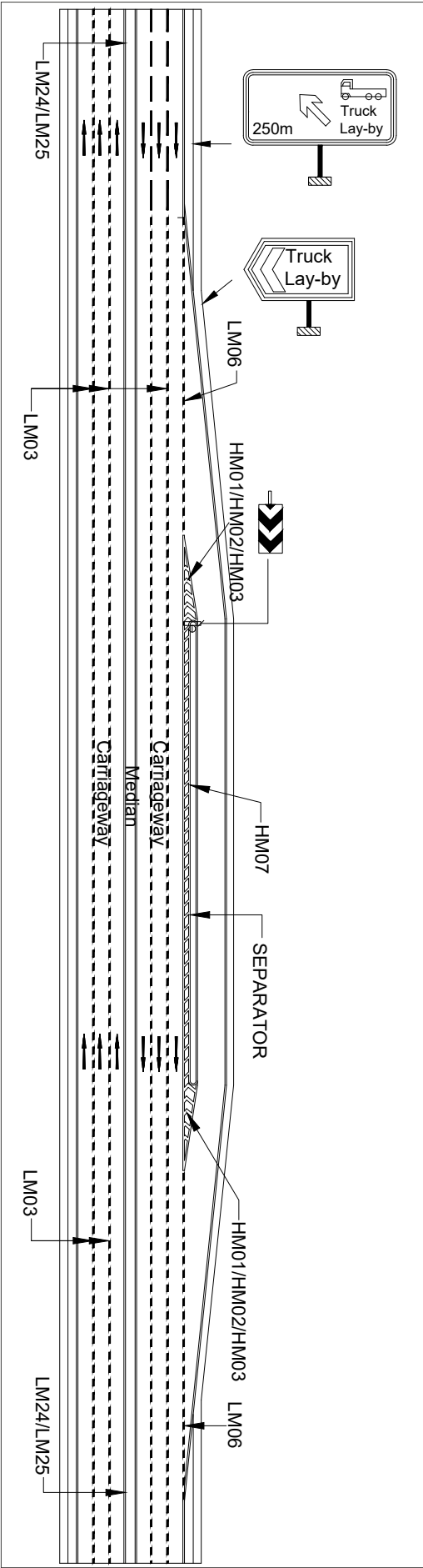
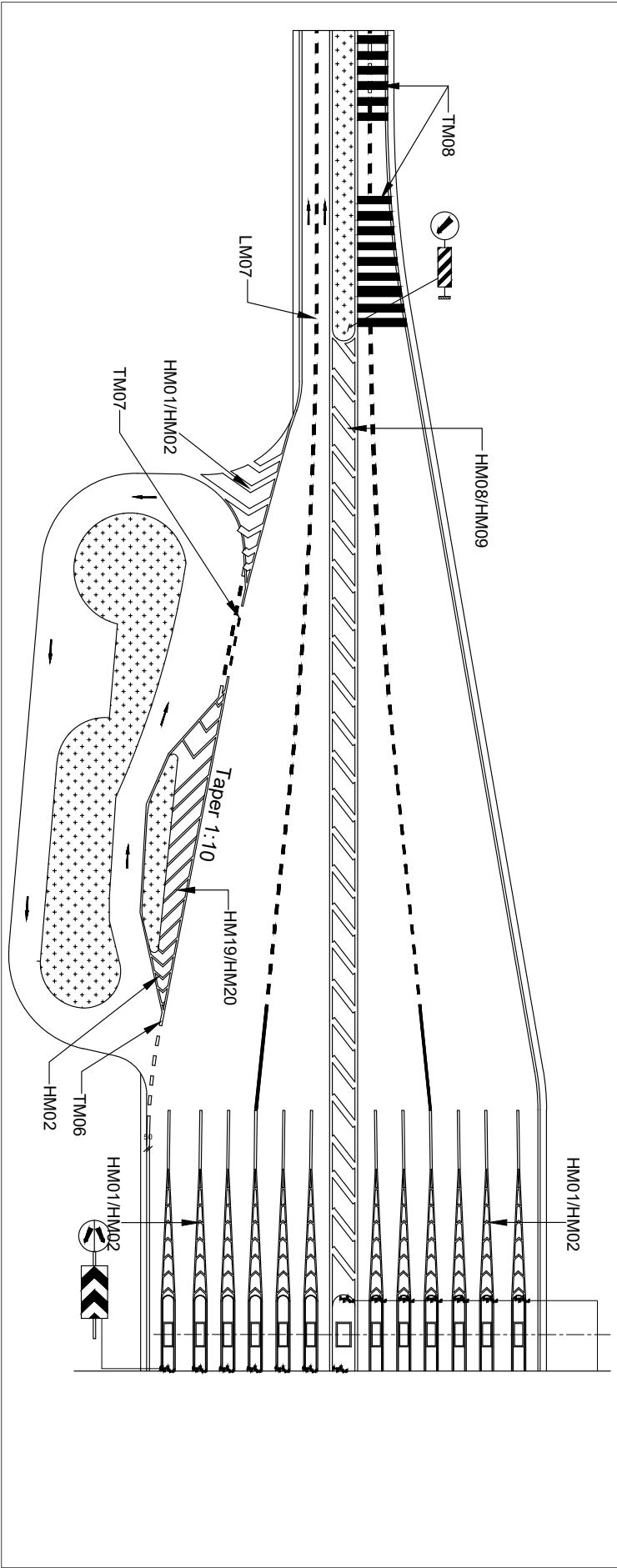


Figure 10.5 Toll Plaza Markings



10.6 Toll Plaza Marking

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A toll plaza area shall be equipped with proper chevron marking and with advance signs.

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The edge line on the median and shoulder side shall be continuous. The traffic lane line shall also be continued in such a way that the traffic emerging from the through lane can align uniformly to a toll booth, as shown in [Figure 10.5](#). The lanes designated for electronic tolling shall be marked accordingly. Toll booths shall be provided with chevron marking and continuous lines.

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In [Figure 10.5](#), abbreviations LM are for longitudinal marking, HM for hazard marking and TM for Thermoplastic marking. These abbreviations with details are shown in [Appendix A](#) of this Manual.

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Markings for Buses, Truck Lay-Bys and Toll Plaza

11 Parking and Restrictions

11.1 General

This chapter presents the marking details for parking spaces of vehicles. Typical on-street and off-street parking layout are presented. Markings for restricting parking in the school zone and other sensitive areas are also presented.

11.2 Parking Spaces

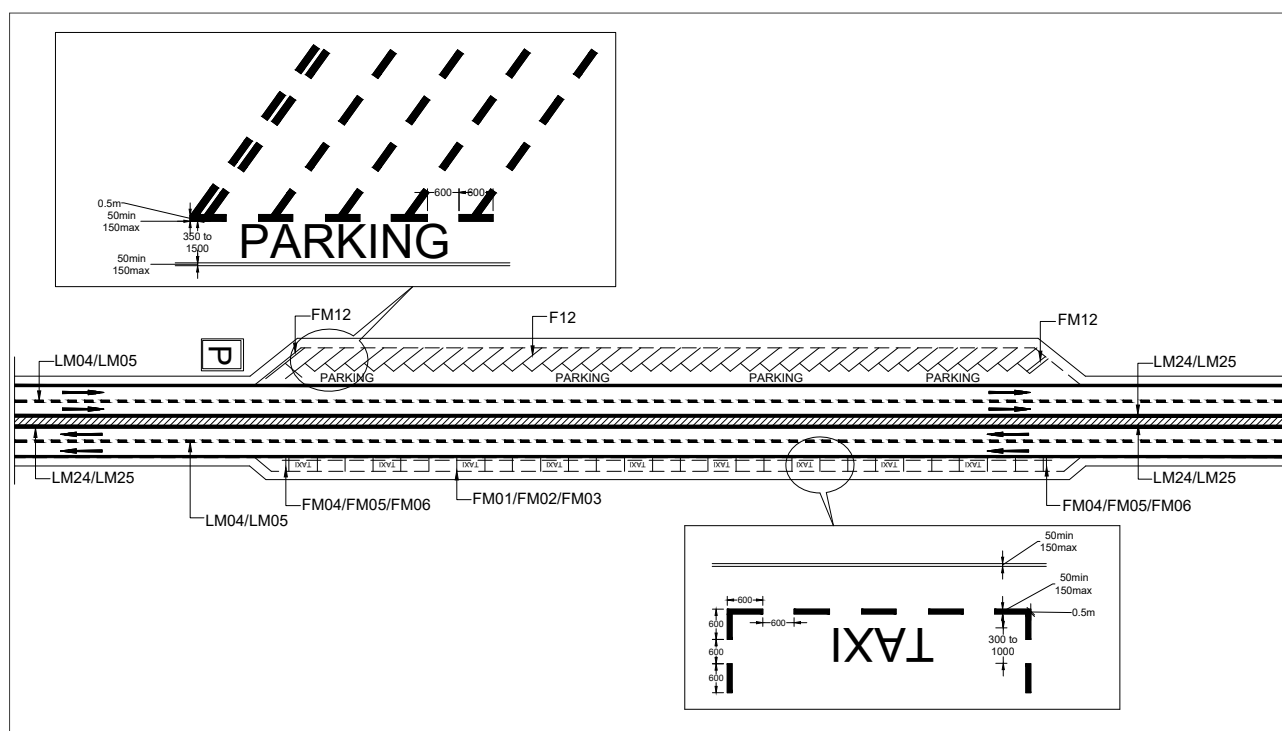
The limitation in parking space on urban roads warrants efficient use of parking spaces. Proper markings avoid the tendency of encroachment on fire hydrant zones, bus stops, loading / unloading zones, and other such locations where parking of vehicles is undesirable. The markings shall be solid white lines 100 mm wide.

The limits of the designated parking places should also be indicated by informative parking signs mounted on the kerb side in accordance with RDM Vol 6 Part 2 Traffic Signs Manual.

The word TAXI, CARS, MOTORCYCLE etc., should be written if the parking area is specific for any particular type of vehicle. These words should also be indicated on the supplementary plate. Typical details for off street parking are given in Figure 11.1.

Parking can be on or off-street and a typical detail for off-street parking is given in Figure 11.2

Figure 11.1 On Street Parking



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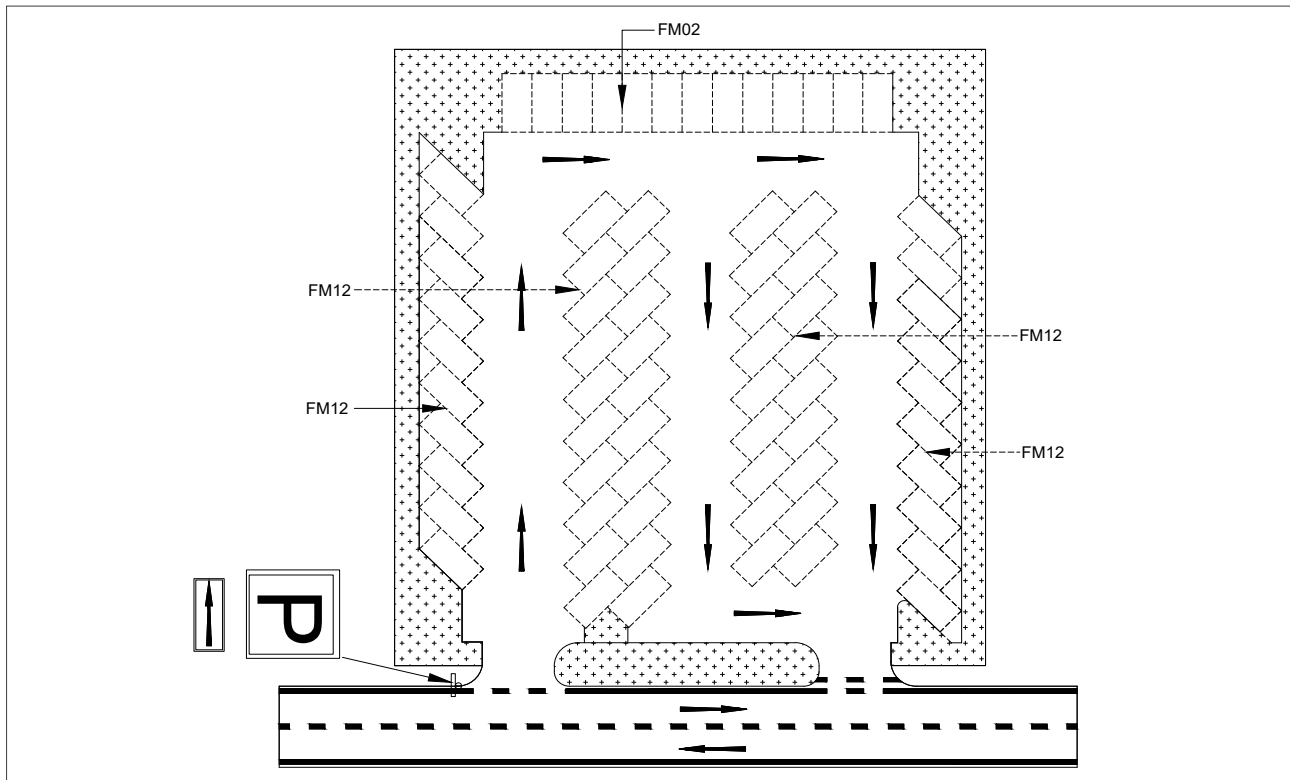
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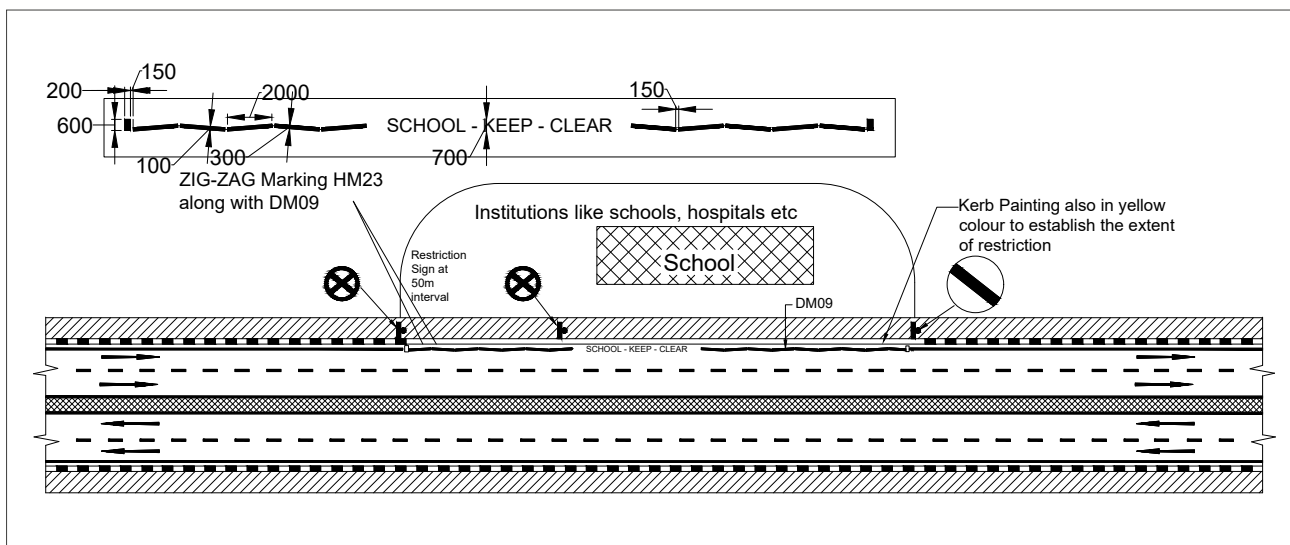
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Parking and Restrictions

Figure 11.2 Off Street Parking

11.3 Restrictions

The stretch where parking is to be restricted has to be delineated with yellow kerb painting coupled with the provision of yellow edge line. Restrictions like Keep Clear marking can be placed on road stretches located in front of schools and hospitals by providing zigzag marking. This may be further supplemented by painting the kerb yellow for the length intended to be kept clear of traffic, along with proper signs, as shown in Figure 11.3.

Figure 11.3 Keep Clear Markings

A stretch of road where parking is prohibited for certain hours can be distinguished with yellow single solid edge, as shown in **Figure 11.4**.

A stretch where parking or waiting is prohibited at all times has to be distinguished by providing yellow double solid edge lines, as shown in **Figure 11.5**.

The zigzag markings, as shown in **Figure 11.6**, can be used to delineate the No-Parking or No-waiting areas for stricter compliance.

In order to restrict loading or unloading of goods, for certain hours of the day, restrictions can be shown by road marking as given in **Figure 11.4** or **Figure 11.5**. The marking is to be supplemented by an appropriate sign board with messages displaying day and time restrictions, such as 'No Loading - Mon-Sat 8.30 am to 6.30 pm' or 'No Loading', as the case may be.

Figure 11.4 No Parking for Certain Hour

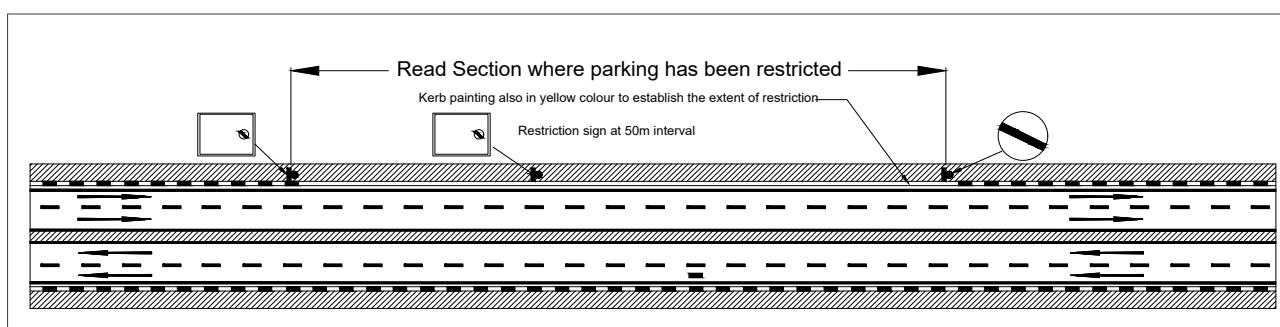
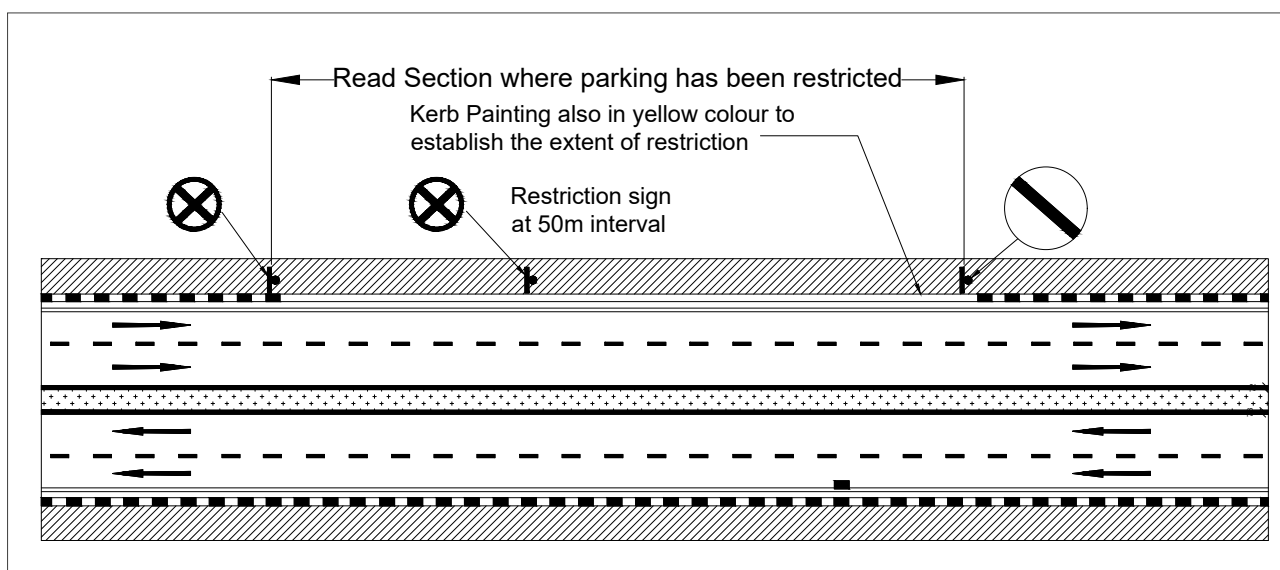


Figure 11.5 No Parking



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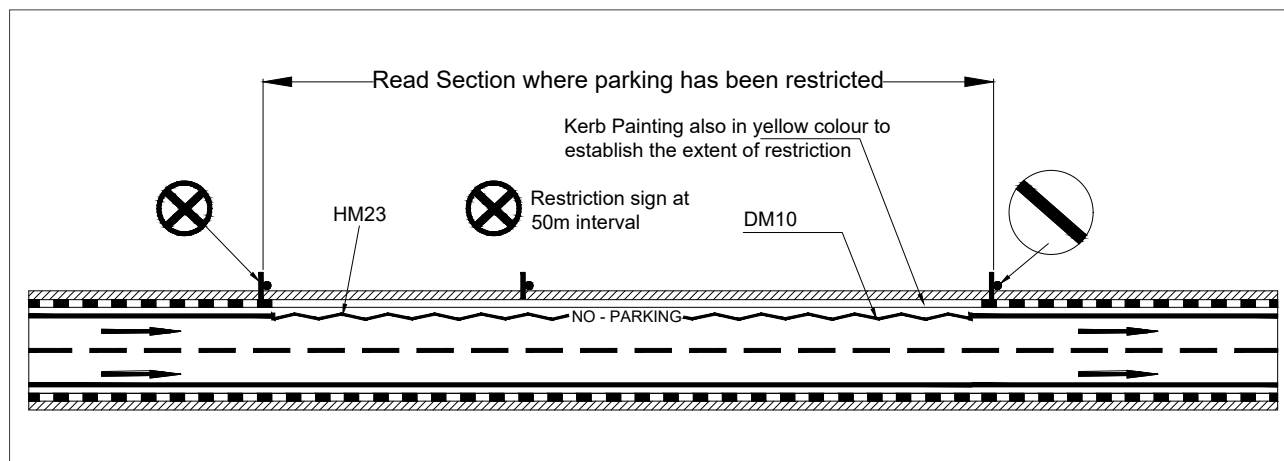
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Parking and Restrictions

Figure 11.6 No Parking for Stricter Compliance

12 Object Marking

12.1 General

Objects within / adjacent to the carriageway are physical obstructions which constitute a serious hazard to the traffic flow. These shall be adequately marked. The objective of delineating such objects is to make them more visible during the night-time, to reduce the likelihood of a vehicle colliding with them.

Physical obstructions can be classified as:

1. Objects within carriageway.
2. Objects adjacent to the carriageway.

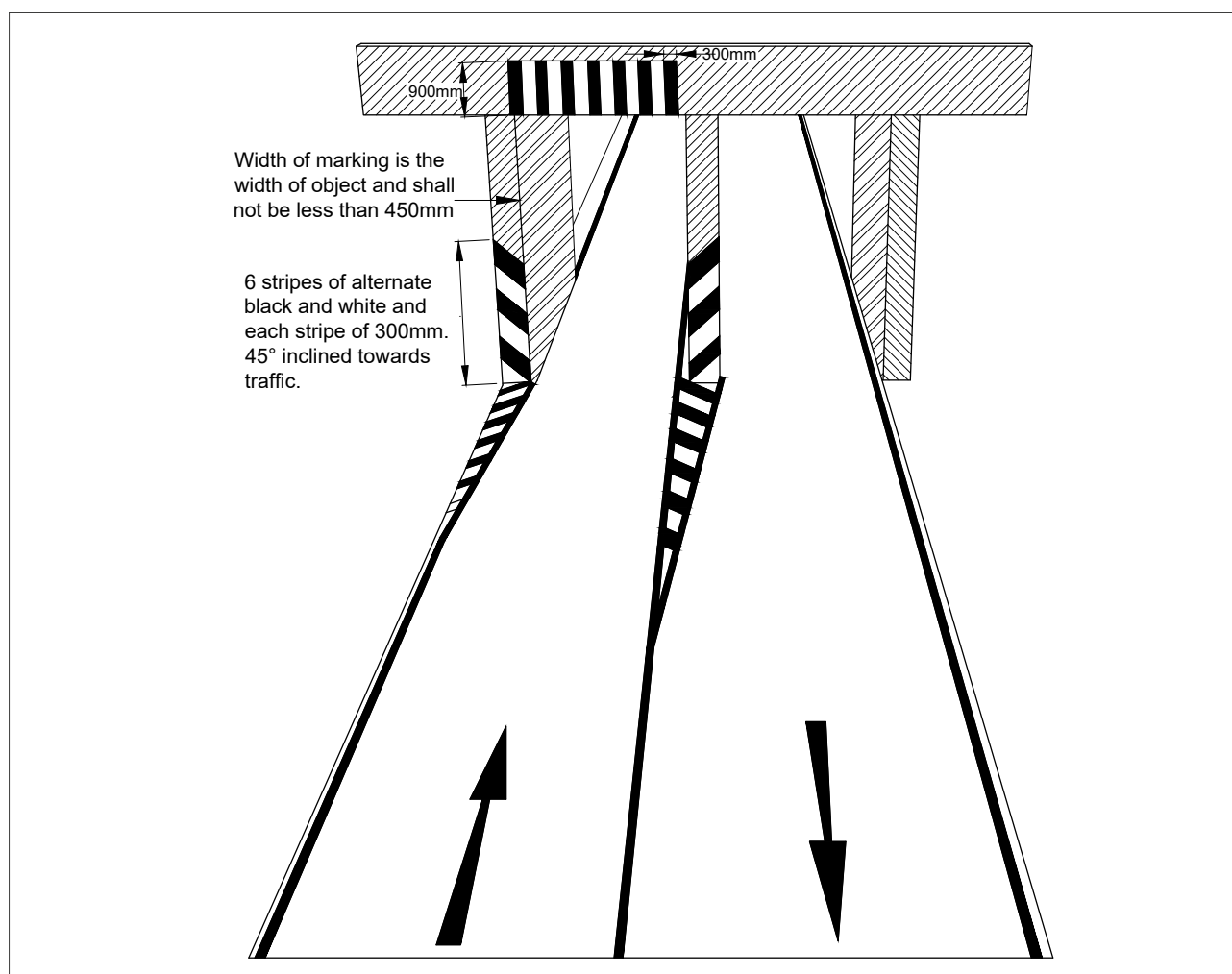
Typical obstructions within the carriageway are monuments, raised channelising islands, central island and central median, underpass structures such as piers and abutments.

Typical obstructions adjacent to the carriageway are continuous raised, culvert head walls, subway abutments and piers, electric poles, signposts, signal supports, trees, guard rails and stones.

12.2 Markings for Objects within the Carriageway

The face of obstructions located within the carriageway shall be marked by at least six alternate black and yellow stripes sloping downward at an angle of 45 degrees towards traffic. The diagonal stripes shall be uniform, and their width shall not be less than 300 mm, as shown in Figure 12.1.

Figure 12.1 Object and Diagonal Marking



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Object Marking

On structures with restricted vertical clearance, markings of vertical black and yellow stripes with a width of at least 300 mm shall be used. If the object does not easily lend itself to markings, the markings may be placed on an independent surface mounted immediately in front of the object. The width of the road marking shall be equal to the width of the obstruction protruding into the carriageway, and shall not be less than 450 mm.

In addition to the object markings, warning lines on the pavement shall be placed. Diagonal marking shall be placed in front of object markings as shown in [Figure 12.1](#).

Kerbs of raised channelising islands, central island and central medians shall be marked with alternate vertical black and white strips of 500 mm width to improve visibility of the islands.

12.3 Markings for Objects adjacent to Carriageway

Subway piers, abutments and culvert head walls adjacent to the carriageway shall be marked with at least six alternate black and white stripes sloping down at an angle of 45 degrees towards the traffic as shown in [Figure 12.2](#).

Electric poles close to the carriageway shall be marked with alternate horizontal black and white stripes up to a height of 1.25 m above the road level. The stripes shall be uniform and not less than 100 mm wide.

Guard rails, guard stones or drums and trees shall be painted solid white. In the case of trees, the marking shall be for a height of 1.25 m above the road level with a 300 mm band of black paint in the middle.

All objects located within 2.4 m from shoulders/kerbs shall be painted. Object markers shall be placed in front of objects to enhance visibility. The height of object markers shall be at least 1.2 m above the nearest traffic lane.

Kerbs of all islands located in the line of traffic flow shall be painted with vertical black and white stripes of 500 mm width as illustrated in [Figure 12.3](#).

Kerbs on bridges and flyover shall also be painted with vertical black and white stripes of 500 mm width as shown in [Figure 12.3](#).

Where restrictions such as No-Parking or No Stopping are required, the kerb shall be painted with yellow colour as shown in [Figure 12.4](#).

Figure 12.2 Object Marking Adjacent to Carriageway

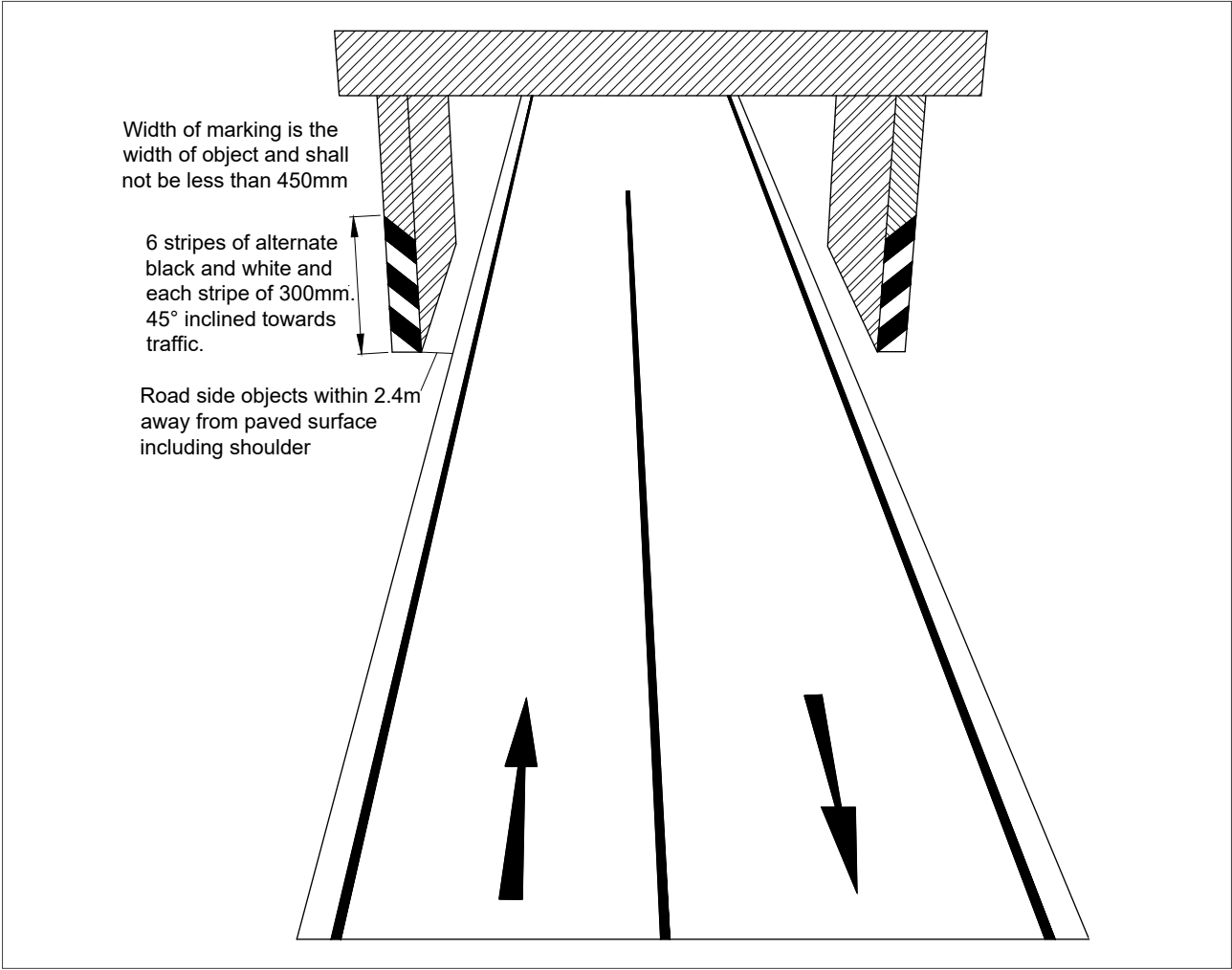
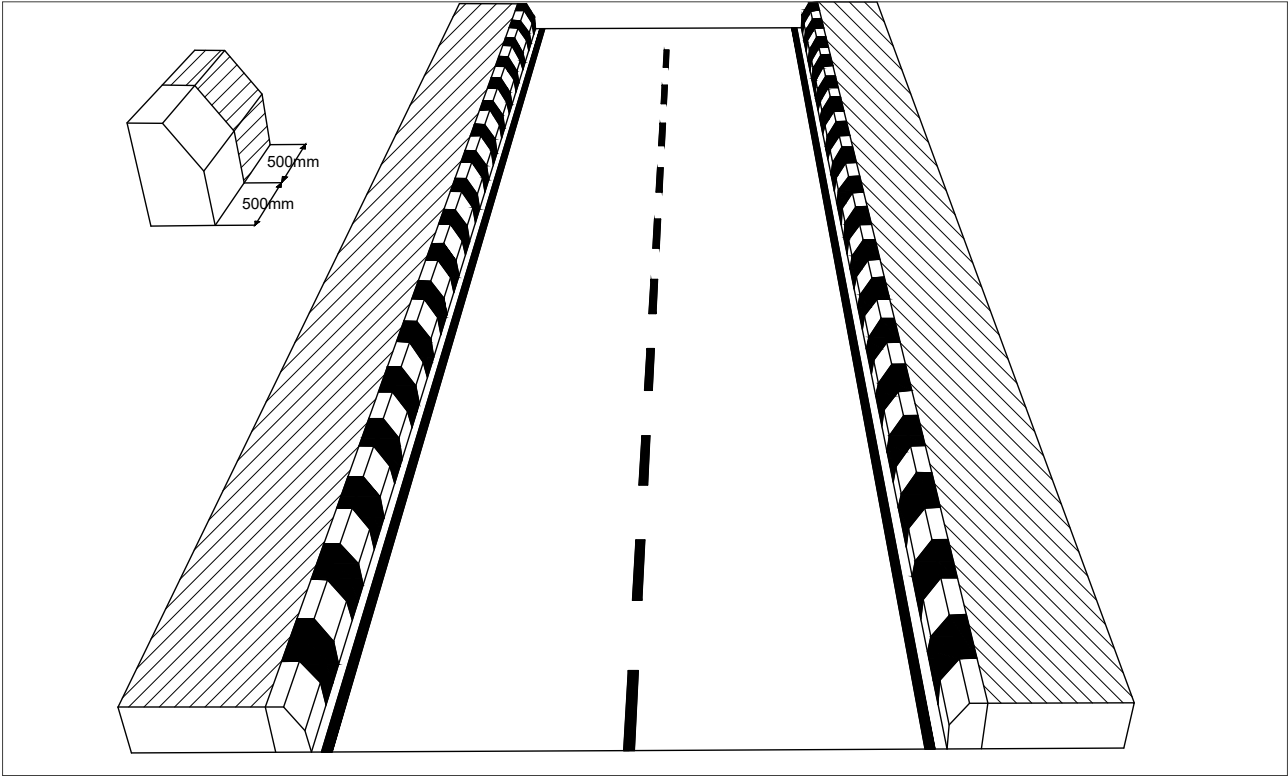


Figure 12.3 Kerb Marking



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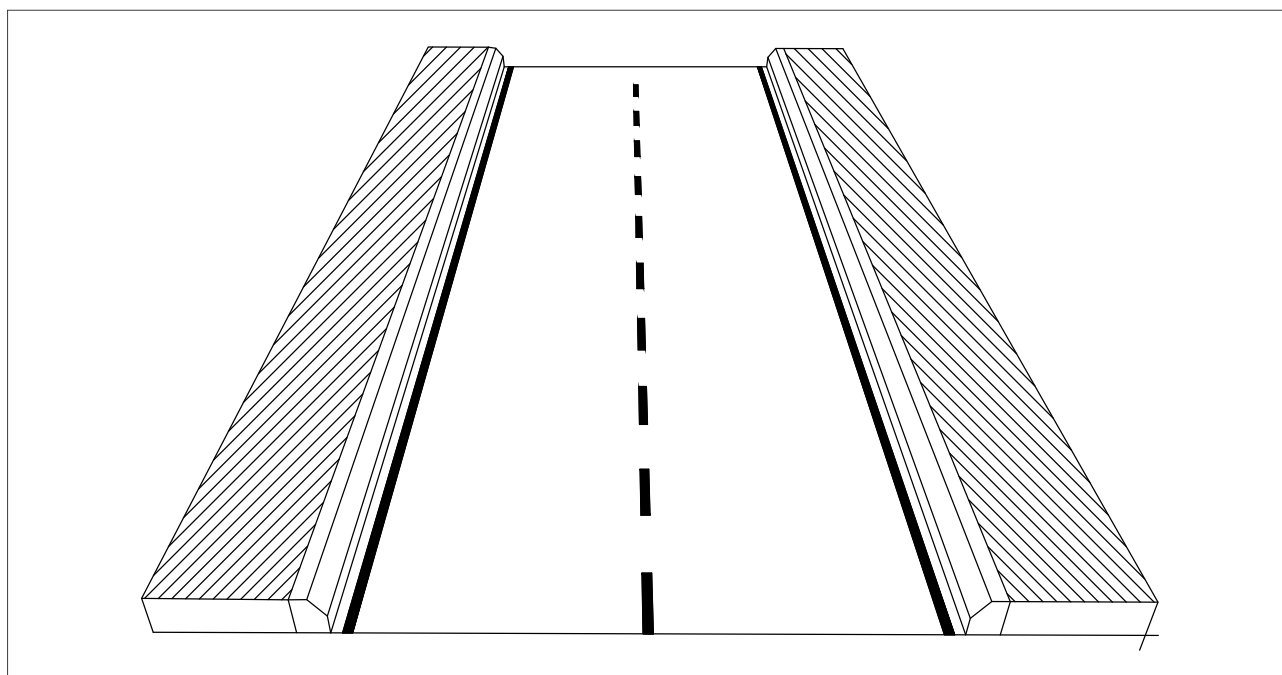
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Object Marking

Figure 12.4 Kerb Marking Where Restrictions Apply



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Object Marking

13 Road Studs

13.1 General

Retro-reflective road studs are used to supplement longitudinal/transverse reflectorised road marking. They improve visibility at night-time and in adverse weather conditions. Road studs are also used across the carriageway to serve as Speed Arrestors by giving a warning through the creation of a rumbling sensation to the user. A series of such road reflector studs are laid in advance of junctions/crossings/the ends of the flyovers, where road crashes are more prevalent.

The technical specifications of road studs shall be as per EN 1463 Part 1 for performance requirements of temporary and permanent studs. Testing methods for both permanent and temporary studs shall be as per EN 1463 Part 2. Refer to Appendix H for list of standards.

13.2 Types of Road Studs

Road studs can be classified based on substrate materials (e.g., metallic, or non-metallic); reflector type (e.g., beads, retroreflective sheeting, or solar); anchorage (i.e., with or without anchor).

Studs shall be made with reflectors/light emitters for use on roads. These reflectors/light emitters should have abrasion resistant protective coating. Studs with anchorage or without anchorage can be bonded to the road surface with suitable adhesive as recommended by the manufacturer. Bitumen and epoxy adhesives are available for attaching road studs to the pavement surface. In addition, some road studs are fitted with pressure sensitive adhesive pads. To use stud with adhesive pad, a liner is to be removed from the pad at the bottom of the stud before placing on the road surface.

Uni-directional Road studs shall be used on one directional carriageways of undivided roads, one-way streets or slip roads. Bi-directional road studs shall be used for two directional roads to delineate the centre line and edge line.

Epoxy adhesives shall be used for fixing road studs, as per EN 1463.

13.3 Siting and Placing of Road Studs

Road studs are placed to enhance safety. It gives better visibility to the markings in adverse weather conditions. Studs are advised for use on high-speed roads which handle high volume of traffic. It is advisable to use road studs on Class A roads, for centre line, lane lines and edge lines. It is desirable to use studs at vulnerable sections on Class B roads, especially at curve sections. Longitudinal road studs should be used on expressways, for median side edge line, lane lines and shoulder edge lines.

The spacing of road studs on rural and urban roads depending on approach speed is given in Table 13.1. For colour of studs and spacing on divided and undivided roads refer to Table 13.2.

Table 13.1 Road Studs Spacing Based on Approach Speed

Marking	Stud Spacing (m)	Use (m)
Rural Highways (Type A and B) for Approach Speed 60 km/h and above		
Lane lines (white)	18	Separation of lanes
Lane lines (white)	9	Next to auxiliary lane for ramp
Centre line (yellow)	12	Approach to junctions
Centre line (yellow)	9	Two-lane undivided carriageway Four-lane undivided carriageway
Edge line (white)	18	Edge of carriageway
Rural Highways (Type A, B and C) for Approach Speed Less Than 60 km/h		
Lane lines (white)	12	Separation of lanes
Lane lines (white)	9	Approach to junctions
Centre line (yellow)	12	Two-lane undivided carriageway
Centre line (yellow)	9	Four-lane undivided carriageway
Edge line (white)	12	Edge of carriageway

1

Road stud placement is mandatory at sharp curves, no overtaking sections, approach to junctions, adjacent to auxiliary lanes for exit/entry and on expressways.

2

Road studs should be placed on broken longitudinal markings at the centre of the gap. It should not be placed on the line segment or by the side of the line segment.

3

Where road studs are to be placed on a carriageway with a paved shoulder, they shall be placed outside the shoulder side edge line. They shall be set back by 50 mm from the edge line. When road studs are to be placed the edge of a road with no paved shoulder, they can be placed on the edge line due to space constraints.

4

Roads studs on the median side edge shall not be placed on the median line marking. They should be in the hard strip or kerb shyness width, with a 50 mm set back distance from the median side edge line. A minimum 100 mm clearance must be maintained from the vertical face of the raised kerb.

5

In the case of a flushed or depressed median, the minimum requirement of 100 mm clearance does not hold. Studs can be placed 50 mm from the edge line.

6

If the width of the hard strip or shyness is not adequate to accommodate the required set back distance, the studs can be placed adjacent to the edge line and even directly on the edge lines.

7

Studs with anchorage are preferable for edge lines. Studs with anchorages are not recommended for concrete roads.

8

Application shall conform to AASHTO M237 Standard Specification for Epoxy Resin Adhesives for Bonding Traffic Markers to Hardened Portland Cement and Asphalt Concrete.

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13.4 Colour of Road Studs

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Studs with different coloured reflectors, such as white, red, yellow, and green, are used for highways. Their usage is as follows:

White Colour: white road studs are to be used to indicate traffic lane line.

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Red Colour: to be used to indicate a line which should not be crossed and mainly to delineate the left-hand edge of the running carriageway i.e., red studs are to be used on shoulder side edge lines. Red studs shall be omitted, or can be replaced with green studs, at locations where traffic exits from the main carriageway, such as entry to a Truck Layby / Bus Bay, 'Start of Service Road', etc.

12

Yellow Colour: Yellow Road studs should be used on yellow markings. It is to be used on centre lines on undivided highways. It is to be used on the right edge of the carriageway in the case of the multi-lane divided carriageways, i.e., on the median side edge line.

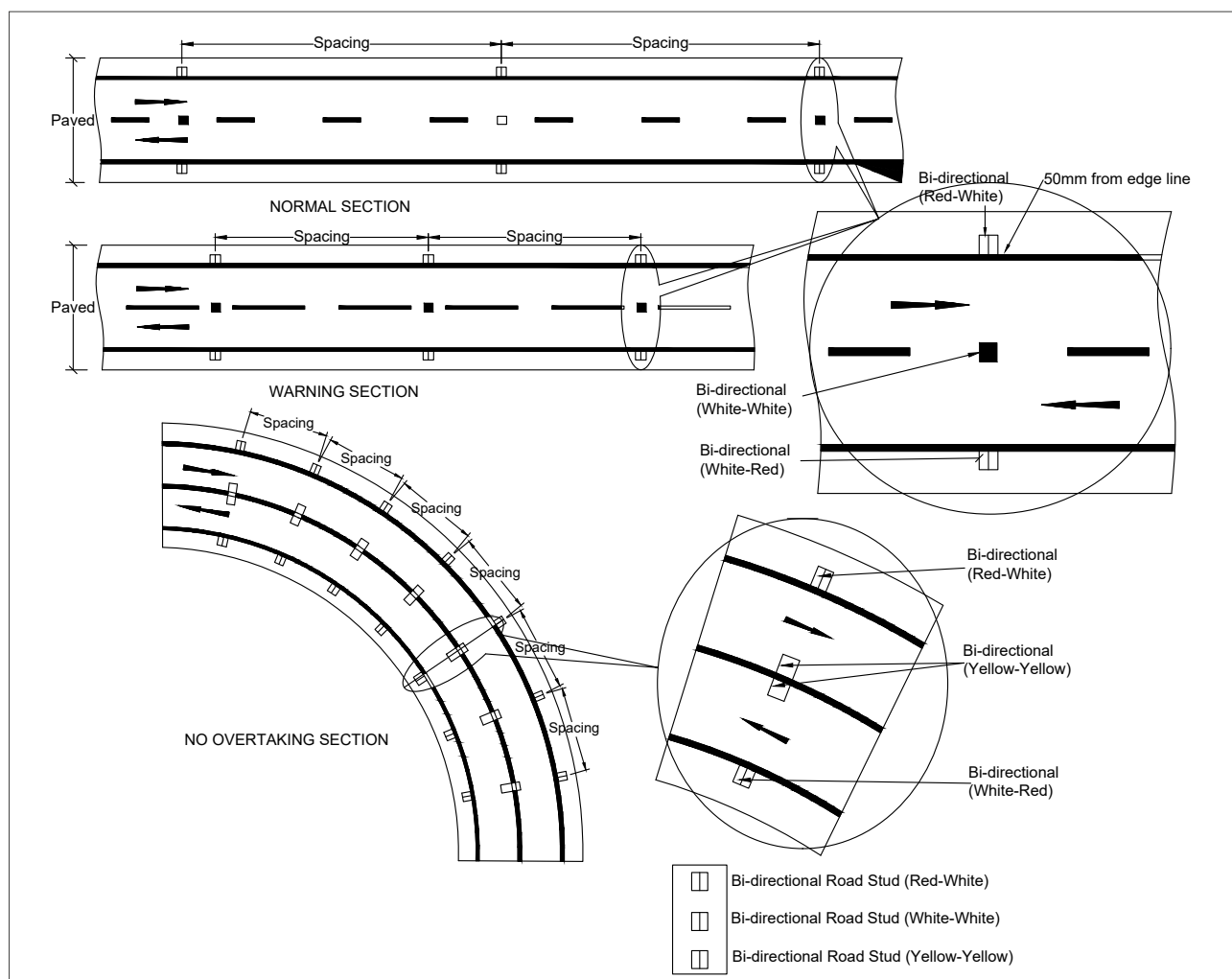
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Green Colour: to be used to indicate crossable edge lines such as laybys and to show the boundary of acceleration or deceleration lanes on the left-hand side of the carriageway in the case of the multi-lane divided carriageways.

Uni-directional carriageways shall have one coloured face with the applicable colour. Bi-directional carriageways shall have two coloured faces which can be a combination of two colours such as red-white, or same colour on two faces such as white-white or yellow-yellow.

13.5 Spacing of Road Studs

The colour and spacing pattern to be followed in an undivided carriageway are given in [Table 13.2](#). [Figure 12.1](#) shows the road studs for a two directional road for normal, and no-overtaking sections.

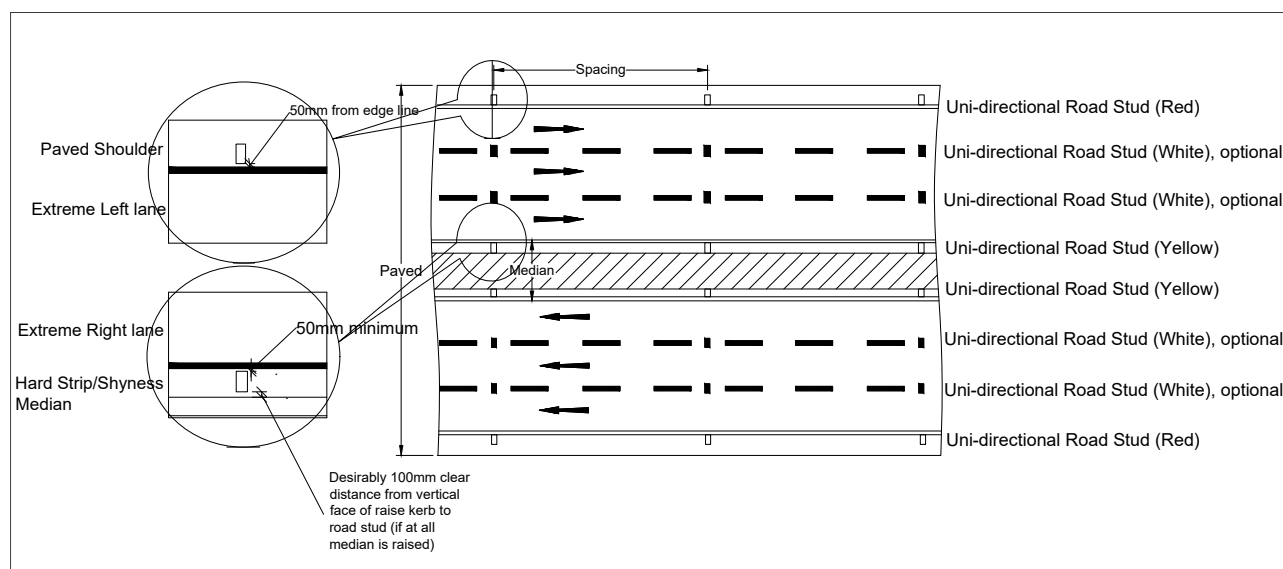
Figure 13.1 Road Studs for Two Directional Road**Table 13.2** Road Studs Spacing for Undivided Roads

Description	Road Category	Traffic Movement	Carriageway	Normal Section			No-overtaking Section		
				Centre Line	Edge Line	Traffic Lane Line	Centre Line	Edge Line	Traffic Lane Line
Single/Intermediate Lane Road	Two-way	< 5.5 m		NA	12 m interval	NA	NA	9 m interval	NA
Two Lane Road	Two-way	5.5 m to 7 m		12 m interval	12 m interval	NA	9 m interval	9 m interval	NA
Two Lane Road with Paved Shoulder	Two-way	>7 m		12 m interval	12 m interval	NA	9 m interval	9 m interval	NA
Three Lane Undivided Road	Two-way	>11 m		12 m interval	12 m interval	Not Required	9 m interval	9 m interval	Not required
Four Lane Undivided Road	Two-way	>14 m		12 m interval	12 m interval	12 m interval	9 m interval	9 m interval	9 m interval

Table 13.3 Road Studs Spacing for Divided Carriageways

Road Category	One side Carriageway Width (in m)	Normal Section			No-overtaking Section		
		Traffic Lane Line	Shoulder side Edge Line	Median side Edge Line	Traffic Lane Line	Shoulder side Edge Line	Median side Edge Line
Four Lane Divided Carriageways	>7.3 m	12 m interval	12 m interval	12 m interval	9 m interval	9 m interval	9 m interval
Six Lane Divided Carriageways	>10.8 m	12 m interval	12 m interval	12 m interval	9 m interval	9 m interval	9 m interval
Four Lane Divided Expressway	>9.5 m	18 m interval	18 m interval	18 m interval	12 m interval	12 m interval	12 m interval
Six Lane Divided Expressway	>11 m	18 m interval	18 m interval	18 m interval	12 m interval	12 m interval	12 m interval

Source: Indian Roads Congress IRC 35: 2015

Figure 13.2 Road Studs for a Divided Carriageway

The colour and spacing pattern to be followed in a divided carriageway is given in Figure 13.3.

Figure 13.2 Shows the colour pattern and lateral placement of road studs in a divided carriageway road, and the placement of road studs on a one-way ramp or slip road or street is given in Table 13.4.

For chevron/diagonal markings on gore areas, red colour road studs shall be provided and the spacing shall be the same as that of the diagonal and chevrons. The studs shall be placed to mark the continuous line of diagonal and chevron marking. Road studs shall be placed inside the continuous line marking, at the midpoint of diagonal or chevron as shown in Figure 13.3. For crossable continuous lines (such as at acceleration/deceleration lanes where lane changing is involved) green colour road studs shall be provided at 8 m spacing as shown in Figure 13.4.

Table 13.4 Road Studs for Ramp / Slip Road/One Way Street

Road Category	Traffic Movement	Carriageway Width	Normal Section		No-overtaking Section	
			Traffic Lane Line	Edge Lines	Traffic Lane Line	Edge Lines
One Lane Ramp / Slip road	One-way	<5.5 m	NA	12 m interval	NA	9 m interval
One Lane Ramp / Slip Road	One-way	5.5 to 7.5 m	NA	12 m interval	NA	9 m interval
Two Lane Ramp / Slip Road	One-way	>7.5 m	18 m interval	12 m interval	9 m interval	9 m interval
Three Lane Ramp / Slip Road	One-way	>11 m	18 m interval	12 m interval	9 m interval	9 m interval

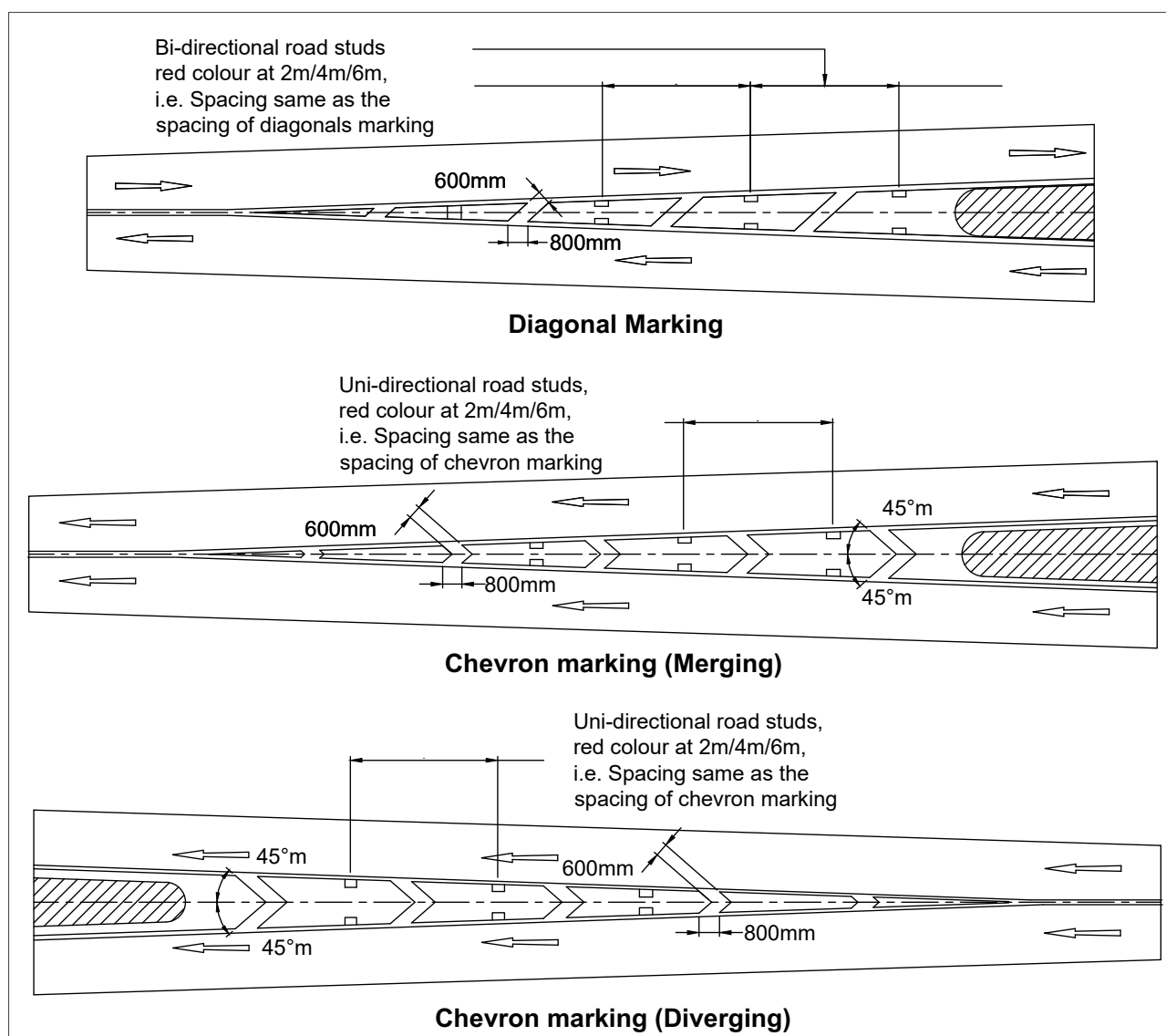
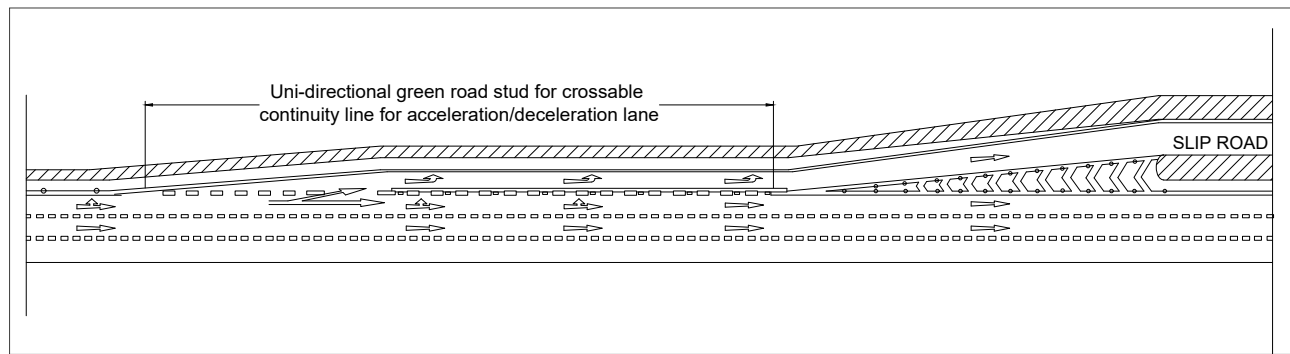
Figure 13.3 Road Studs for Diagonal and Chevron Markings

Figure 13.4 Road Studs for Crossable Continuity Lines

Two rows of yellow road studs shall be provided for pedestrian crossings at 0.5 m spacing across the carriageway; this can be achieved by installing road studs at the four corners of each block of marking. Also, on the approaches to any hazardous situation, two arrays of road studs shall be provided at 0.5 m gap across the carriageway and the two arrays of road studs shall be placed 0.5 m apart.

13.6 Solar Studs

Solar powered road markers can be used at locations at hazardous locations such as: crash prone areas, at bridges, toll plazas, sharp curves, pedestrian crossings. The functions of solar road do not rely on vehicle head light. The intense brightness of the Light Emitting Diodes (LEDs) makes them visible at distances of more than 800 m under favourable conditions. The solar studs are also more distinctly visible during rainy and foggy conditions where the road markings could be ineffective.

The spacing for solar studs is the same as in conventional road studs.

13.7 Studs on Concrete Pavement

The road studs with anchorages are not recommended for concrete roads. The spacing and colour of the studs are same as for flexible pavement. The placement of studs on joints should be avoided. At joints, tolerance of plus or minus 100 mm (before/after the joint) can be used.

13.8 Studs on Guard Rails

Reflective road studs can be placed on metal beam and concrete guard rails to make the carriageway prominent at night-time. The studs are to be placed at curve sections. The spacing shall be as per the spacing of the studs on the carriageway. The colour of studs shall be as per [Section 13.4](#). The reflector sheeting shall meet ASTM D4956 Type IX or Type XI for luminance level. Minimum reflector dimensions should be 102 mm × 76 mm.

14 Performance Assessment and Monitoring

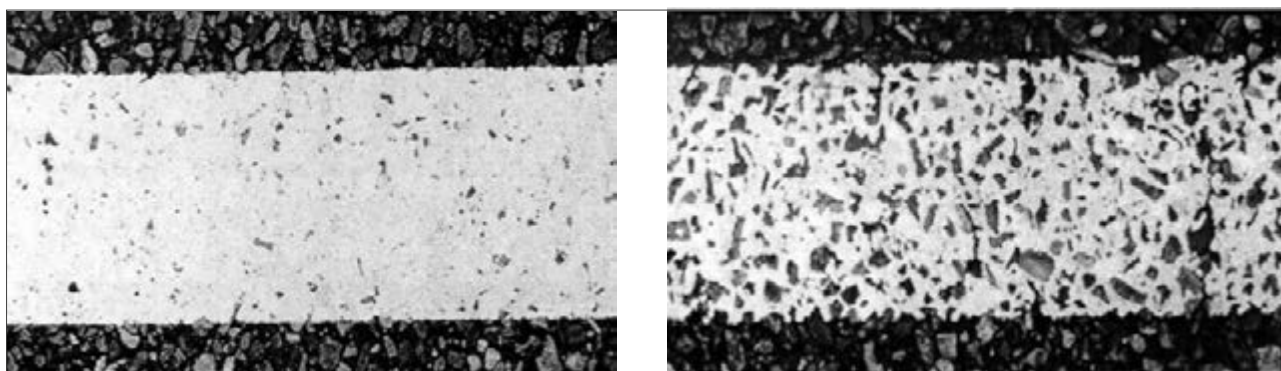
14.1 General

This chapter discusses the level of performance required to maintain the quality of the road marking. It also describes the assessment mechanism for various performance characteristics. The performance of road markings is based on wear durability, day and night-time visibility, and wet reflectivity.

14.2 Wear Durability

Assessment of the degree of wear is measured as a percentage of the pavement marking area remaining intact. Pavement marking systems demonstrate different wear profiles as shown in Figure 14.1. Some pavement marking systems wear out faster. This depends on its wearing resistance. As wear may be more apparent on left-hand edge lines or side road entries, the overall performance of the marking system needs to be taken into account.

Figure 14.1 Above 95% of Area Intact (left) and Below 70% of Area Intact (right)



Source: Indian Roads Congress IRC 35: 2015

14.3 Day Time Visibility

Reflection in day light or under road lighting is measured by luminance coefficient according to EN 1436. It is expressed in milli candela per square meter per lux ($\text{mcd}/\text{m}^2/\text{lux}$). Day time luminance is measured using the method described in Appendix D of this manual.

14.4 Night Visibility

14.4.1 Dry Retro-reflection

Night-time retro-reflectivity results from vehicle headlamps being reflected back to the driver. The measurement of the retro-reflected light is the coefficient of retro reflected luminance, expressed as $\text{mcd}/\text{m}^2/\text{lux}$.

Night-time pavement marking visibility is a function of both the line dimension and retro-reflectivity. Dry retro-reflectivity is measured using the method described in Appendix D of this manual.

Another important parameter is the quality and the quantity of Titanium Dioxide used in the paint. This chemical is the white pigment used to render brightness to the pavement marking.

14.4.2 Wet Retro-reflection

A film of water can coat the glass beads in wet conditions and reduce the retro-reflective performance of the pavement markings. Wet retro-reflectivity is measured using the method described in Appendix D of this manual.

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Pavement markings may change the micro texture and the macro texture of the pavement surface. For cyclists and motorcyclists, this may have a destabilising effect. Improved skid resistance is brought about by the inclusion of surface applied angular particles to the marking.

14.5 Performance Criteria

The performance criteria for daytime and nighttime visibility are detailed in sections 14.5.1 and 14.5.2.

14.5.1 Day-time Visibility

The luminance coefficient under diffuse illumination for road marking:

- minimum of 130 mcd/m²/lux on cement concrete road.
- minimum of 100 mcd/m²/lux on asphalt road.

Testing method shall be according to Appendix D of this manual.

14.5.2 Night-time Visibility

Dry Retro-reflection shall be as per Table 14.1 and measured by the method described in Appendix D of this manual.

Table 14.1 Initial and Threshold Reflectivity (Dry Retro Reflectivity During Night-time)

S No.	Design Speed (km/h)	Retro-reflectivity (mcd/m ² /lx)	
		Initial (7 days)	Minimum Threshold Level after 2 years
1	Up to 65	200	80
2	65 - 100	250	110
3	Above 100	350	150

Source: Indian Roads Congress IRC 35: 2015

Wet Retro-reflection shall be as per Table 14.2 and measured by the method described in Appendix D of this manual, irrespective of the design speed.

Table 14.2 Initial and Minimum Performance for Night Visibility in Wet Conditions

S No.	Retro-reflectivity (mcd/m ² /lx)	
	Initial (7 days)	Minimum Threshold Level after 2 years
1	100	50

14.6 Skid Resistance

The skid resistance parameter is considered for urban/city traffic conditions, for pedestrian crossing, cycle path, intersection delineation etc. Skid resistance is measured by British Pendulum Number (BPN). Table 14.3 shows the acceptable skid resistance values.

Skid resistance measurement method is described in Appendix G of this manual.

Table 14.3 Initial and Minimum Performance for Skid Resistance

Type of Road	Skid Resistance (BPN)	
	Initial (7 days)	Minimum Threshold Level
Urban/City	55	45

Source: Indian Roads Congress IRC 35: 2015

15 Warranty and Testing Methods

15.1 General

Road markings are laid on the pavement either for temporary period or for a long term. The life of the temporary road markings is limited to the duration of the work zone. For long term road markings, the functional life is considered as two years. The road authority must demand for a warranty period for the pavement marking. During the warranty period, all the performance parameters must remain higher than the minimum threshold value. Performance parameters should be checked every year on a random section of the network, conforming to the procedure presented in [Chapter 14](#).

15.2 Testing Methods

15.2.1 General

Road Markings shall be inspected using the methods and frequency defined in this chapter. The purpose of inspection is to identify the deterioration of road markings. In this regard, it is essential to determine the appropriate timing for maintenance intervention.

15.2.2 Tests for Pavement Marking Material

Road marking shall be tested as performance criteria given in EN1436 European Standards for Road Marking for retro-reflectivity, luminance, chromaticity and skid resistance.

Thermoplastic paints shall be tested as per ASTM D7735. Quality of paints shall be tested by methods given in ASTM E 1710 and ASTM D713-90 and ASTM D 5680.

15.2.3 Characteristics Requiring Maintenance Inspection of Pavement Marking

Road marking shall be subjected to routine inspection, once in six months, for all the characteristics, as per the inspection methods set out in the flow diagrams in [Appendix F](#).

- Retro-reflectivity Dry, Wet Recovery, Wet Continuous during rain.
- Wear.
- Luminance coefficient.
- Skid resistance.

The location of the road markings will determine which of these characteristics require inspection and the methods of inspection to be employed.

15.2.4 Methods of Inspection and Frequency

The road corridor/ network should be surveyed on a routine annual basis. For all longitudinal road markings, annual survey of retro reflectivity shall be carried out by using High Speed Monitor (HSM) instrumentation system or any other appropriate instrumentation system. This assessment would help to understand whether the pavement marking conforms to the criteria shown in the Flow Diagram of [Appendix F](#). The results may be averaged for every 100m interval. The areas of the network that cannot be surveyed by HSMs (like STOP lines, Give Way lines, exit arrows etc.) shall be inspected annually by the alternative inspection methods set out in [Flow Diagram B](#) in [Appendix F](#). These inspections shall include an assessment of retro reflectivity by the use of handheld retro-reflectometer. Measurements shall be carried out as given in [Appendix F](#).

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15.3 An Assessment of Wear

A visual assessment shall be carried out on 50 % of the road markings at each location. An average of the results shall be taken and assessed against the criteria set out in Appendix F, **Flow Diagram B**.

15.4 Road Marking Measurements

Measurements shall be carried out on 50 % of the road markings and the average is to be assessed, using method given in **Appendix F**.

Skid resistance measurements shall be carried out annually on a quarter of the critical areas of the network as detailed in **Appendix G**. Measurements should be taken on the most trafficked areas of the road markings at each location and an average of the results shall be taken.

Road markings for hatched areas or ghost islands shall be inspected and compared with marking in the adjacent area.

15.5 Road Marking Evaluation

All pavement marking should be assessed and evaluated annually. Measurements of performance parameters are to be made in the direction of travel. On the centre line of measurements should be made in both directions.

15.5.1 Longitudinal Pavement Marking Line

For pavement marking lines of 3 km or less, measurements shall be taken on any random section. The sample size shall be at least of 20 numbers. The average of the measurements shall be used to determine retro-reflectivity.

For pavement marking lines of 3 to 16 km, measurements shall be taken on at least three randomly selected sections. The evaluation sections shall not overlap. Measurements in each section shall be of 20 samples. The average retro-reflectivity for each section shall be calculated. The grand average of all sections shall be used to evaluate retro reflectivity.

For pavement marking lines of more than 16 km, at least three sections shall be selected and the average retro-reflectivity for each section shall be calculated. The grand average of all sections shall be used to evaluate retro reflectivity.

15.5.2 Legends, Symbols, Pedestrian Crossing, and Other Non-Longitudinal Markings

Legends: Each letter is considered as an evaluation section; three measurements shall be made in each letter in the direction of travel. The average value shall be calculated for each letter and the average value shall be used to compute the grand average. The grand average shall be used to determine compliance with the appropriate specification.

Symbols or Transverse Lines: Each symbol or transverse line is considered a separate evaluation section. Six measurements shall be made on symbols or transverse lines that are 2.4 m tall or wide. Three measurements on symbols or transverse lines smaller than 2.4 m shall be made. The average value shall be calculated for each symbol or transverse marking and the average value shall be used to compute the grand average. The grand average shall be used to determine compliance with the appropriate specification.

Pedestrian Crossings: Three random stripes shall be selected as an evaluation section; six measurements shall be made in each of the selected stripes. The average value of each stripe shall be calculated, and the average value shall be used to compute the grand average. The grand average shall be used to determine compliance with the appropriate specification.

15.6 Category of Defect and Maintenance

Any defects noticed during evaluation shall be categorised as:

15.6.1 Category 1

Defects that require prompt attention because they represent an immediate or imminent hazard, (e.g., A Badly Worn-out STOP or GIVE WAY line, double white lines) or a slippery road marking.

Category 1 defects shall be corrected if it is reasonably practical, in 24 hours. If it is not possible to repair the defects within 24 hours, then the appropriate prescribed sign (e.g., 'NO ROAD MARKINGS FOR X KM') shall be displayed until long term repairs are carried out. These repairs shall be completed within 7 days of the notification of the defect.

15.6.2 Category 2

Defects that require rectification within six months of evaluation.

15.7 Records

An inspection report in a format agreed by the road authority shall be maintained for all inspections and evaluations. All measurements collected and defects identified during inspection shall be reported and recorded, including details of any action taken or required.

An annual report including a summary of inspections, routine maintenance operations and changes in the network shall be supplied to the road authority at the end of the financial year.

15.8 Inventory

An accurate computerised inventory of road marking shall be maintained and made available to the road authority on request. The stock inventory shall be maintained through back-up facilities with agreed security procedures.

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Warranty & Testing Methods

Bibliography

Sl. No.	Standard	Name
1	AASHTO M237	Standard Specification for Epoxy Resin Adhesives for Bonding Traffic Markers to Hardened Portland Cement and Asphalt Concrete
2	AASHTO M247	Standard Specification for Glass Beads Used in Pavement Markings
3	AASHTO M248	Standard Specification for Ready-Mixed White and Yellow Traffic Paints
4	AASHTO M249	Standard Specification for White and Yellow Reflective Thermoplastic Striping Material (Solid Form)
5	ASTM D 3330	Standard Test Method for Peel Adhesion of Pressure-Sensitive Tape
6	ASTM D 4505-17	Standard Specification for Preformed Retroreflective Pavement Marking Tape for Extended Service Life
7	ASTM D 4592-17	Standard Specification for Preformed Retroreflective Pavement Marking Tape for Limited-Service Life
8	ASTM D 4956	Standard Specification for Retroreflective Sheeting for Traffic Control
9	ASTM D 713-90	Standard Practice for Conducting Road Service Tests on Fluid Traffic Marking Materials
10		Standard Specification for Preformed Retroreflective Pavement Marking Tape for Limited Service Life
11	ASTM D 7735	Standard Test Method for Type A Durometer Hardness Testing of Thermoplastic Pavement Marking at Elevated Temperatures
12	ASTM D 903-98	Standard Test Method for Peel or Stripping Strength of Adhesive Bonds
13	ASTM E 1710-18	Standard Test Method for Measurement of Retroreflective Pavement Marking Materials with CEN-Prescribed Geometry Using a Portable Retro Reflectometer
14	ASTM E 303-22	Standard Test Method for Measuring Surface Frictional Properties Using the British Pendulum Test
15	BS EN 1871	Road Marking Materials
16	EN 1436	European Standards for Road Marking for Retro-reflectivity, Luminance, Chromaticity, and Skid Resistance
17	EN 1463 Part 1	Performance Requirements of Temporary and Permanent Studs
18	EN 1463 Part 2	Testing Methods for both Permanent and Temporary Studs
19	EN 1871	Road Marking Materials - Paint, Thermoplastic and Cold Plastic Materials - Physical Properties
20	IRC 35	Indian Roads Congress Code of Practice for Road Marking
21	ISO 10526	CIE standard Illuminants for Colorimetry
22	ISO 4622	Paints and Varnishes
23	ISO 48	Rubber, Vulcanised or Thermoplastic
24	KS 1610	Kenya Standard Specification for Road Marking Material
25	KS 2563	Standard Test Method for Peel Adhesion of Pressure-sensitive Autoclaving Tape
26	KS EAS 927	Road Marking Paints
27	KS EAS 928 – 2	Kenya Standard — Hot Applied Thermoplastic Road Marking Paint — Specification Part 2: Road Performance
28	KS EAS 928-1	Hot applied thermoplastic road marking paint-Specification-Part 1: Constituent Material and Mixtures

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References

Appendix A

Pavement Markings

Table A.1 Longitudinal Markings

Code	Type	Description	Length (mm) a	Gap (mm) b	Width (mm) c	Colour	Distance between lines	Pattern
LM01	Broken	Single line	2000	6000	100	Yellow		
LM02	Broken	Single line	3000	9000	150	Yellow		
LM03	Broken	Single line	2000	6000	100	White		
LM04	Broken	Single line	3000	9000	150	White		
LM05	Broken	Single line	1000	1000	100	White		
LM06	Broken	Single line	1000	1000	150	White		
LM07	Broken	Single line	2000	2000	100	White		
LM08	Broken	Single line	2000	2000	150	White		
LM24	Continuous	Single line	-	-	100	Yellow		
LM25	Continuous	Single line	-	-	150	Yellow		
LM26	Continuous	Single line	-	-	100	White		
LM27	Continuous	Single line	-	-	150	White		
LM28	Combined line	Solid & broken line	2000	6000	100	Yellow	100	
LM29	Combined line	Solid & broken line	3000	9000	150	Yellow	150	
LM30	Combined line	Two solid lines	-	-	100	White	100	
LM31	Combined line	Two solid lines	-	-	150	White	150	
LM32	Combined line	Two solid lines	-	-	100	Yellow	100	
LM33	Combined line	Two solid lines	-	-	150	Yellow	150	

Table A.2 Transverse Marking

Code	Type	Length (mm)	Gap (mm)	Width (mm)	Colour	Pattern
TM01	Continuous	One solid line		400	White	
TM04	Broken	600	300	200	White	
TM07	Broken	600	300	400	White	
TM08	Bar Marking (One set)	Across carriageway	6 strips 5/10 mm high, 600 mm apart		Yellow	

Table A.3 Longitudinal Markings

Code	Type	Width (mm)		Space (mm)	Colour	Pattern
		Longitudinal	Diagonal / Chevron			
HM01	Chevron (diverging)	150	600	2000	White	
HM02	Chevron (diverging)	150	600	4000	White	
HM03	Chevron (diverging)	150	600	6000	White	
HM04	Chevron (converging)	150	600	2000	White	
HM05	Chevron (converging)	150	200	4000	White	
HM06	Chevron (converging)	150	600	6000	White	
HM07	Diagonal	150	600	2000	Yellow	
HM08	Diagonal	150	600	4000	Yellow	
HM09	Diagonal	150	600	6000	Yellow	
HM10	Ladder Hatching	100 / 150	200	4000	White	
HM11	Ladder Hatching	100 / 150	200	4000	Yellow	
HM14	Continuous	One solid line 150 mm		-	Yellow	
HM16	Continuous (two lines)	Each solid line 100 mm separated by 100 / 150 mm		-	Yellow	
HM18	Continuous (two lines)	Each solid line of 200 mm separated by 'x' m, 'x' > 100 mm		-	Yellow	
HM19	Deflecting	150	200	4000	White	
HM20	Deflecting	150	200	6000	White	
HM21	Zig zag	100	-	-	White	
HM22	Zig zag	150	NA	NA	White	
HM23	Zig zag	100 / 150	-	-	Yellow	
HM25	Bar Marking	-	300	As designed	Yellow	
HM26	Raised Profile Marking	-	150 / 200	500	White	

Table A.4 Transverse Marking

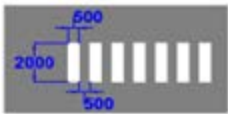
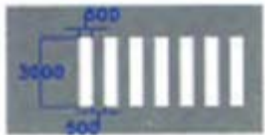
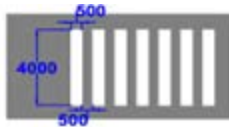
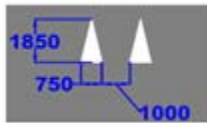
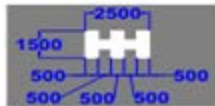
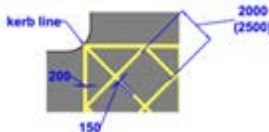
Code	Type	Length (mm)	Breadth (mm)	Space (mm)	Colour	Pattern
BM01	Rectangular	2000	500	500	White	
BM02	Rectangular	3000	500	500	White	
BM03	Rectangular	4000	500	500	White	
BM04	Triangular	750	1850	1000	White	
BM05	Chequer	500	500	500	White	
BM06	Box	2000	2000	–	Yellow	
BM07	Give Way	3750	1100	–	White	
BM08	Cycle	As shown	–	–	White	

Table A.5 Arrow Marking

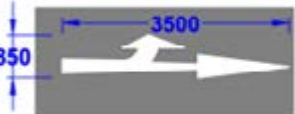
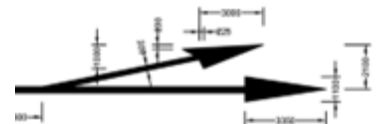
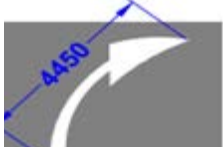
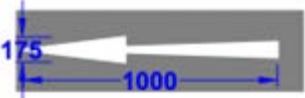
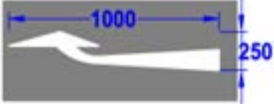
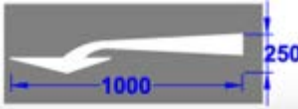
Code	Type	Length (mm)	Width (mm)	Colour	Pattern
AM01	Straight Arrow	3500	500	White	
AM02	Left Arrow	3500	750	White	
AM03	Right Arrow	3500	750	White	
AM04	Straight & Left Arrow	3500	750	White	
AM05	Straight & Right Arrow	3500	850	White	
AM06	Right & Left Arrow	3500	850	White	
AM07	Straight, Right & Left Arrow	3500	750	White	
AM08	Straight Arrow	5000 / 9000	500	White	
AM09	Left Arrow	5000 / 9000	750	White	
AM18	Deflection Arrow	6000	1500	White	
AM20	Bifurcation Arrow	8000	2100	White	
AM21	Bifurcation Arrow	14000	2100	White	
AM24	Roundabout Rotation Arrow	4450	-	White	

Table A.5 Arrow Marking (continued)

Code	Type	Length (mm)	Width (mm)	Colour	Pattern
AM26	Arrow for Cycle Path	-	1000	White	
AM27	Arrow for Cycle Path	-	1000	White	
AM28	Arrow for Cycle Path	-	1000	White	

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Appendices

Table A.6 Directional Marking

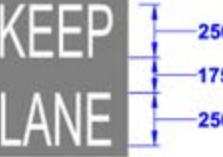
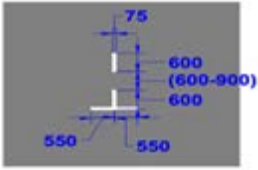
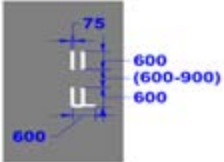
Code	Directional Information	Width (mm)	Height (mm)	Colour	Pattern
DM01	STOP	2050	1600	White	
DM02	STOP	2050	2800	White	
DM03	SLOW	2050	1600	White	
DM04	SLOW	2050	2800	White	
DM05	SPEED LIMIT	1500	1600	White	
DM06	SPEED LIMIT	1500	2800	White	
DM07	KEEP CLEAR	2550	1600	White	
DM08	KEEP CLEAR	2550	2800	White	
DM09	SCHOOL KEEP CLEAR	NA	700	Yellow	
DM10	NO PARKING	NA	700	Yellow	
DM11	NO ENTRY	NA	1600	White	
DM12	NO ENTRY	NA	2800	White	
DM16	TURN LEFT	-	2800	White	
DM18	TURN RIGHT	-	2800	White	
DM22	KEEP LANE	-	2800	White	

Table A.7 Facility Marking

Code	Type	Length (mm)	Width (mm)	Colour	Pattern
FM02	Parking mid block	As shown	75	White	
FM05	Parking last block	As shown	75	White	
FM12	Parking	As shown	100	White	
FM13	Word Message Bus Lane	NA	1250	White	
FM14	Word Message BUS STOP	-	1600	White	
FM17	Word Message TAXI	-	700	White	
FM18	Word Message DISABLED	-	700	White	
FM20	Word Message SLOW	-	1035	White	

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Appendices

Appendix B

Details of Arrow Marking

Figure B.1 Details of Arrow Marking, AM07, AM02 and AM03

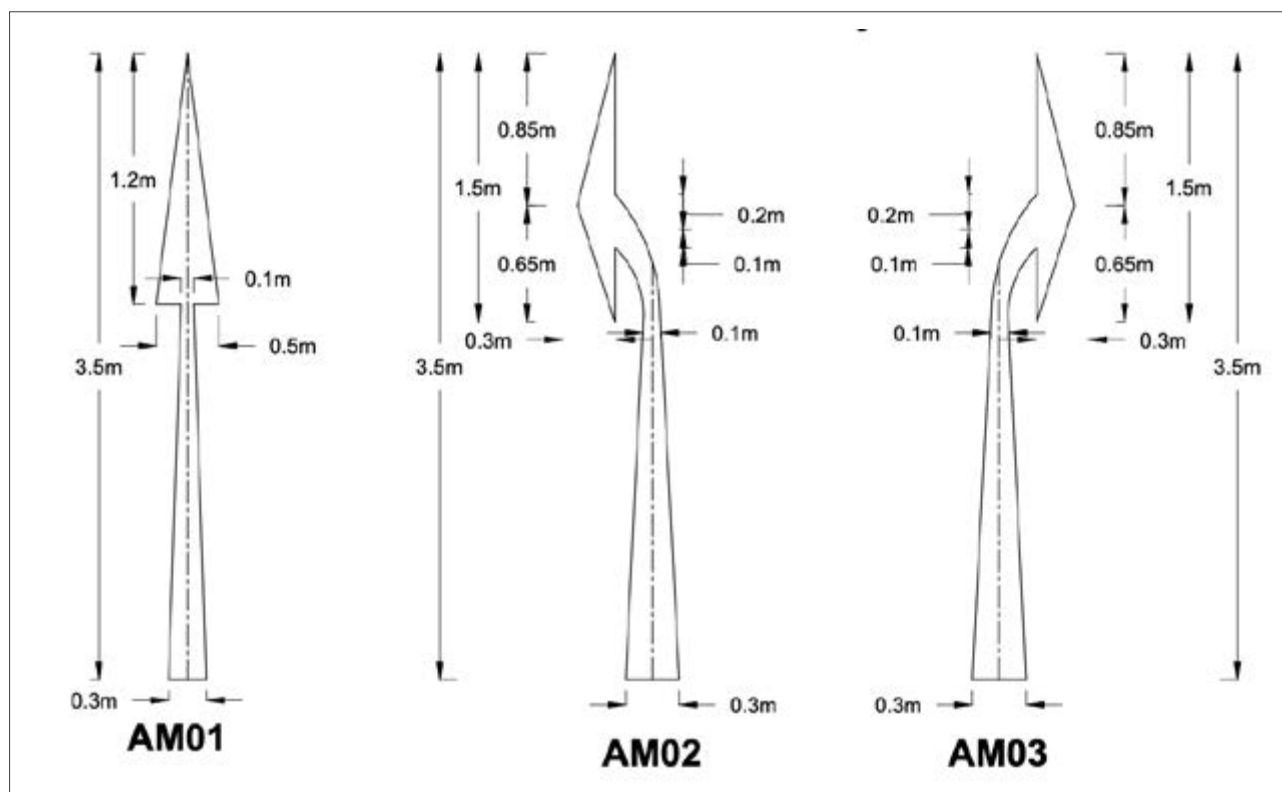


Figure B.2 Details of Arrow Marking, AM04, AM05 and AM06

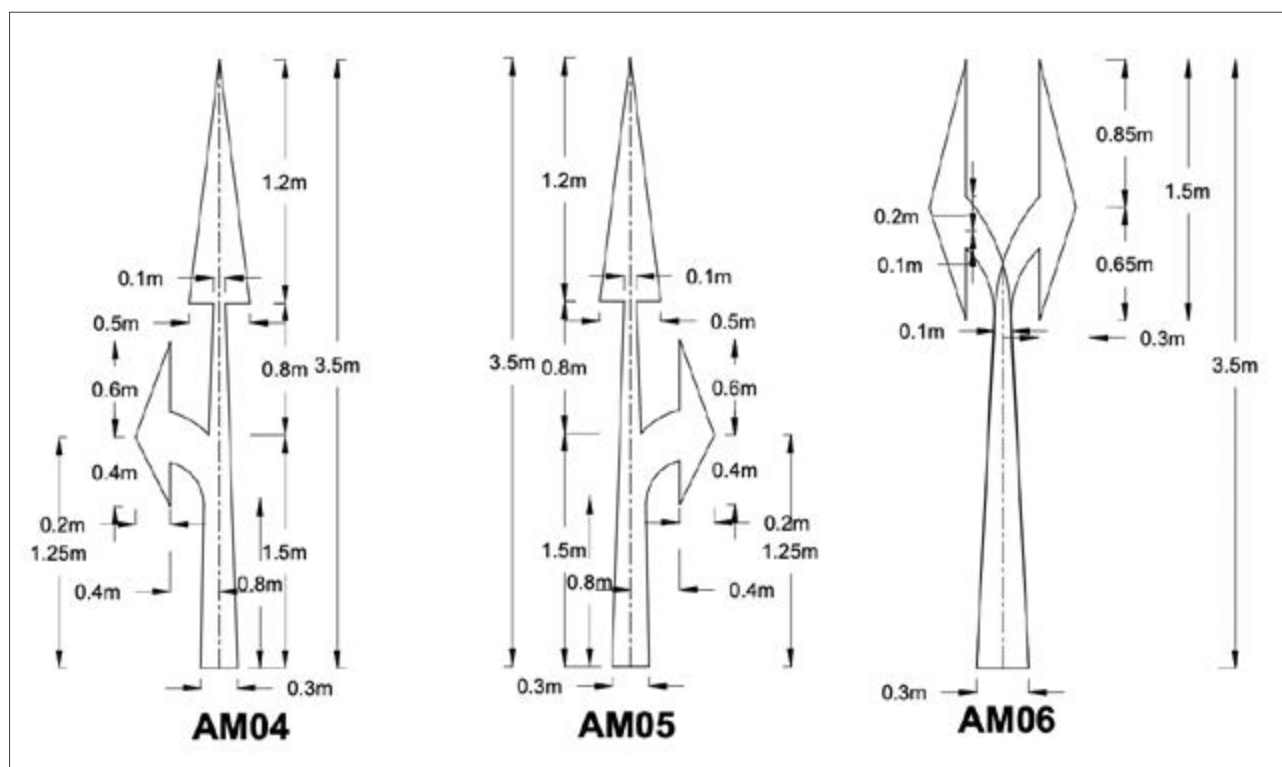
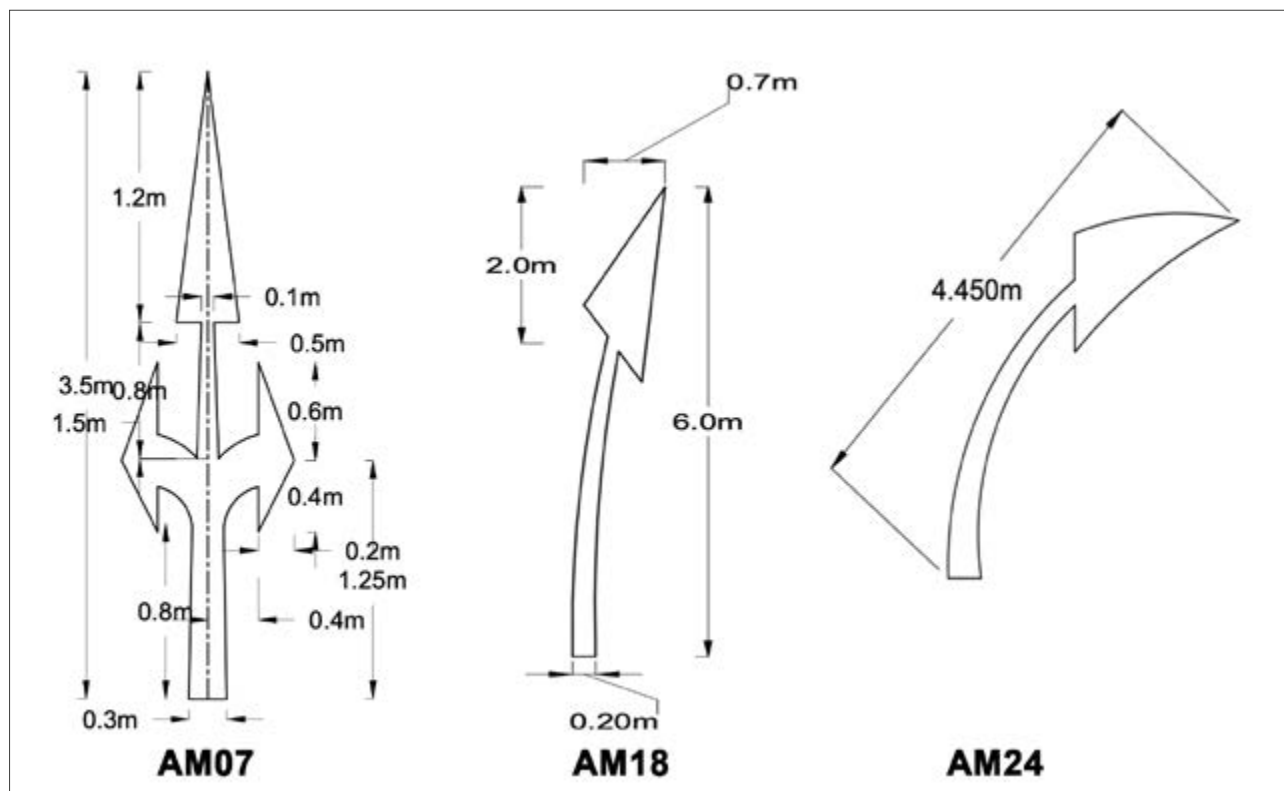
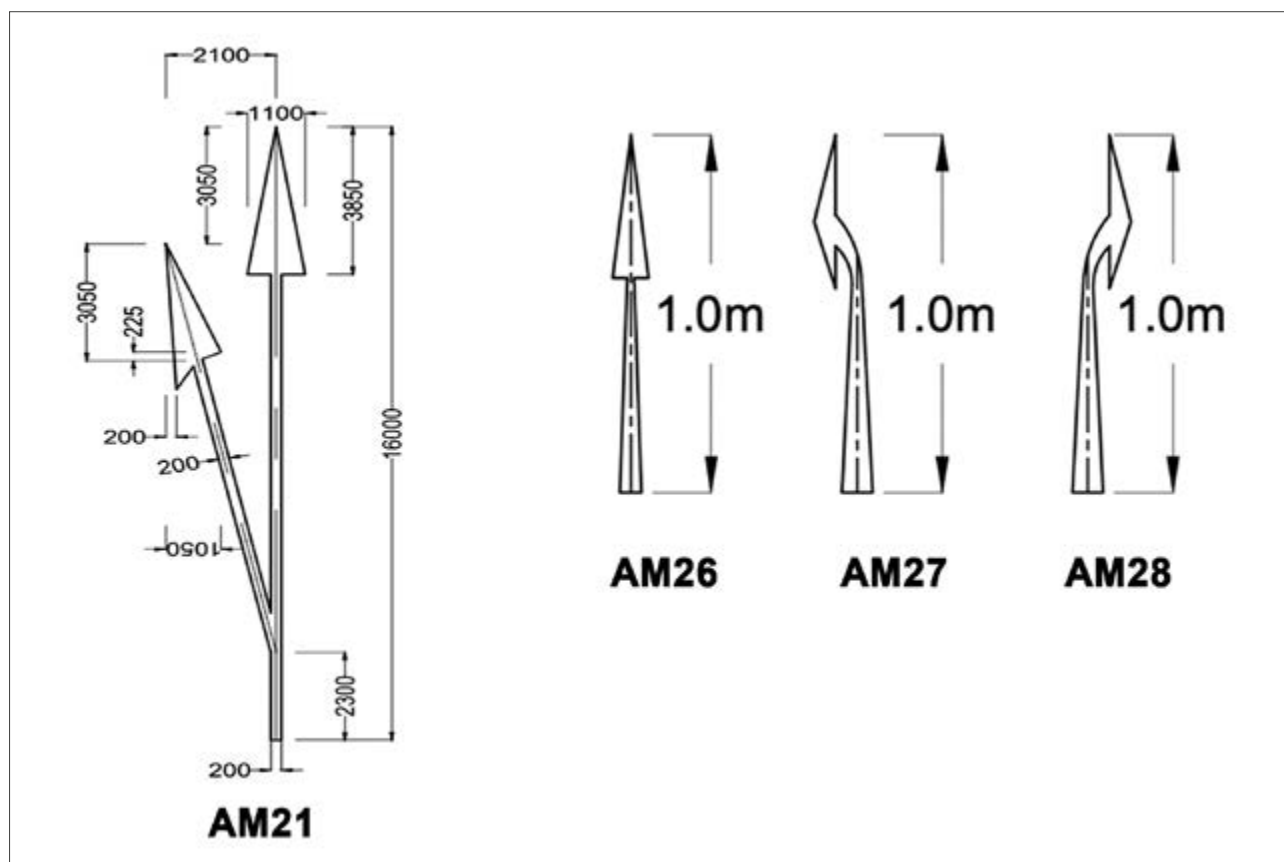


Figure B.3 Details of Arrow Marking, AM07, AM18 and AM24**Figure B.4** Details of Arrow Marking, AM2, AM26, AM27 and AM28

Appendix C

Details of Letter and Numbering

Figure C.1 Details of Letters and Numerals (speed up to 50 mph)

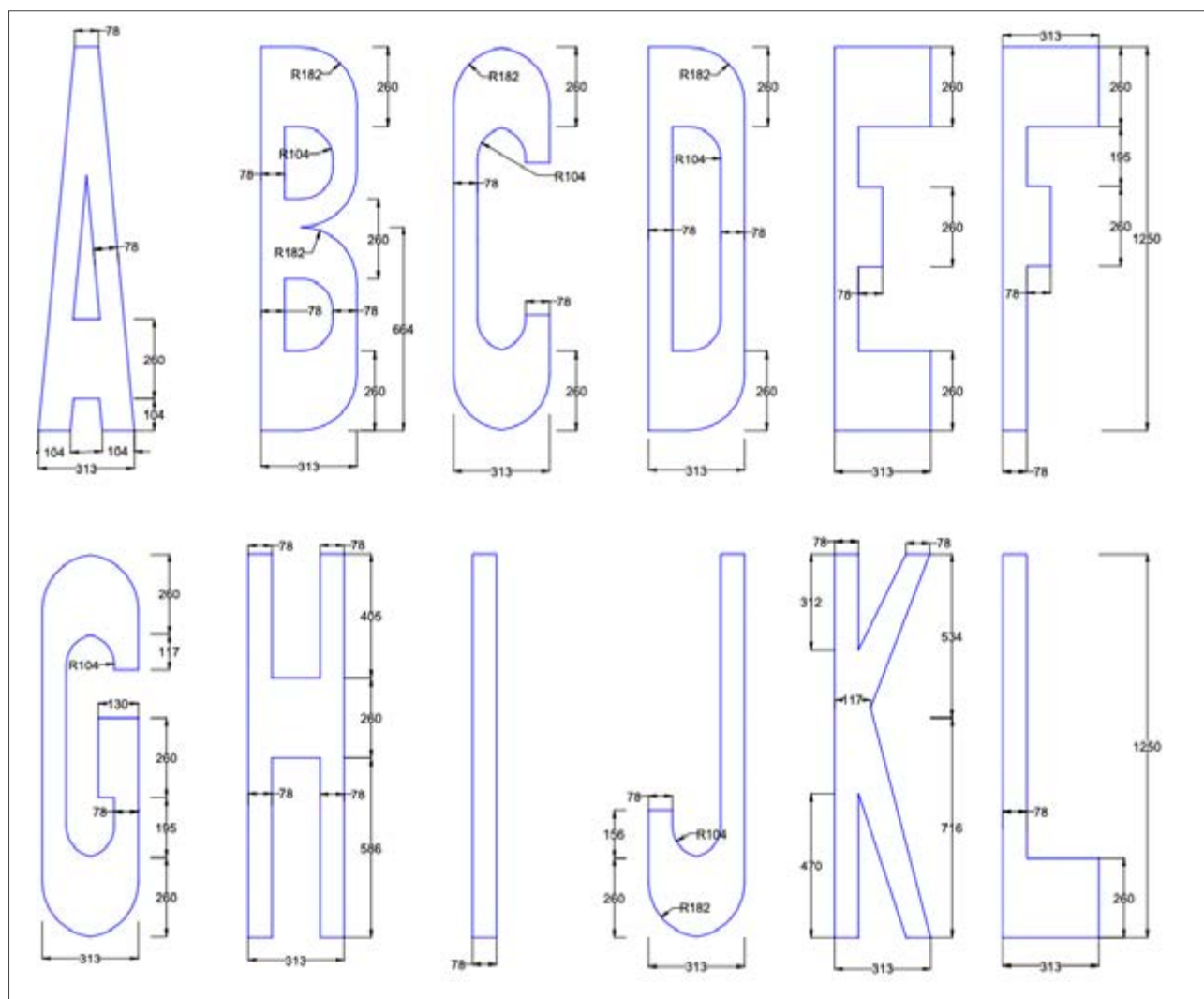


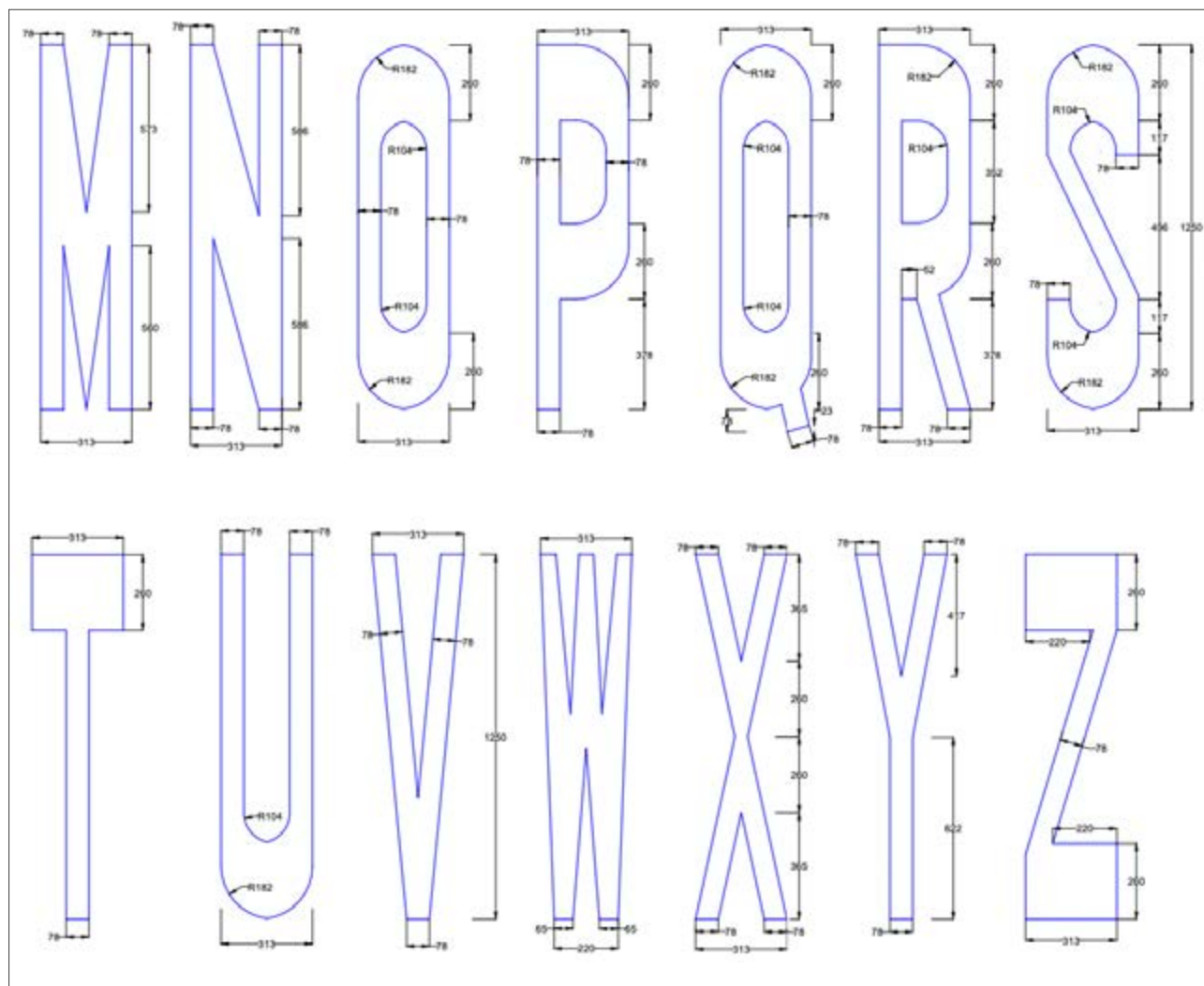
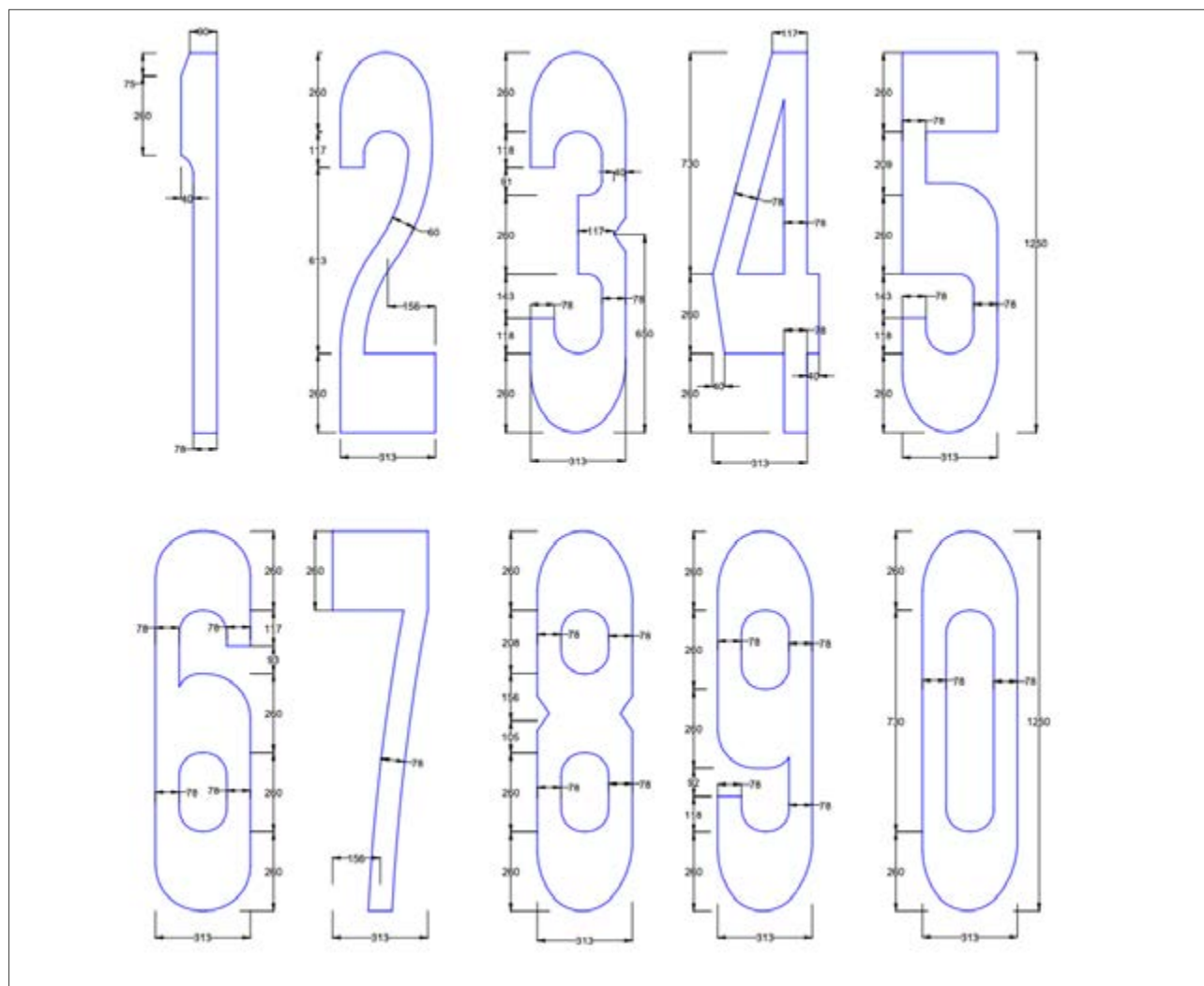
Figure C.1 Details of Letters and Numerals (*speed up to 50 mph*)

Figure C.1 Details of Letters and Numerals (*speed up to 50 mph*)

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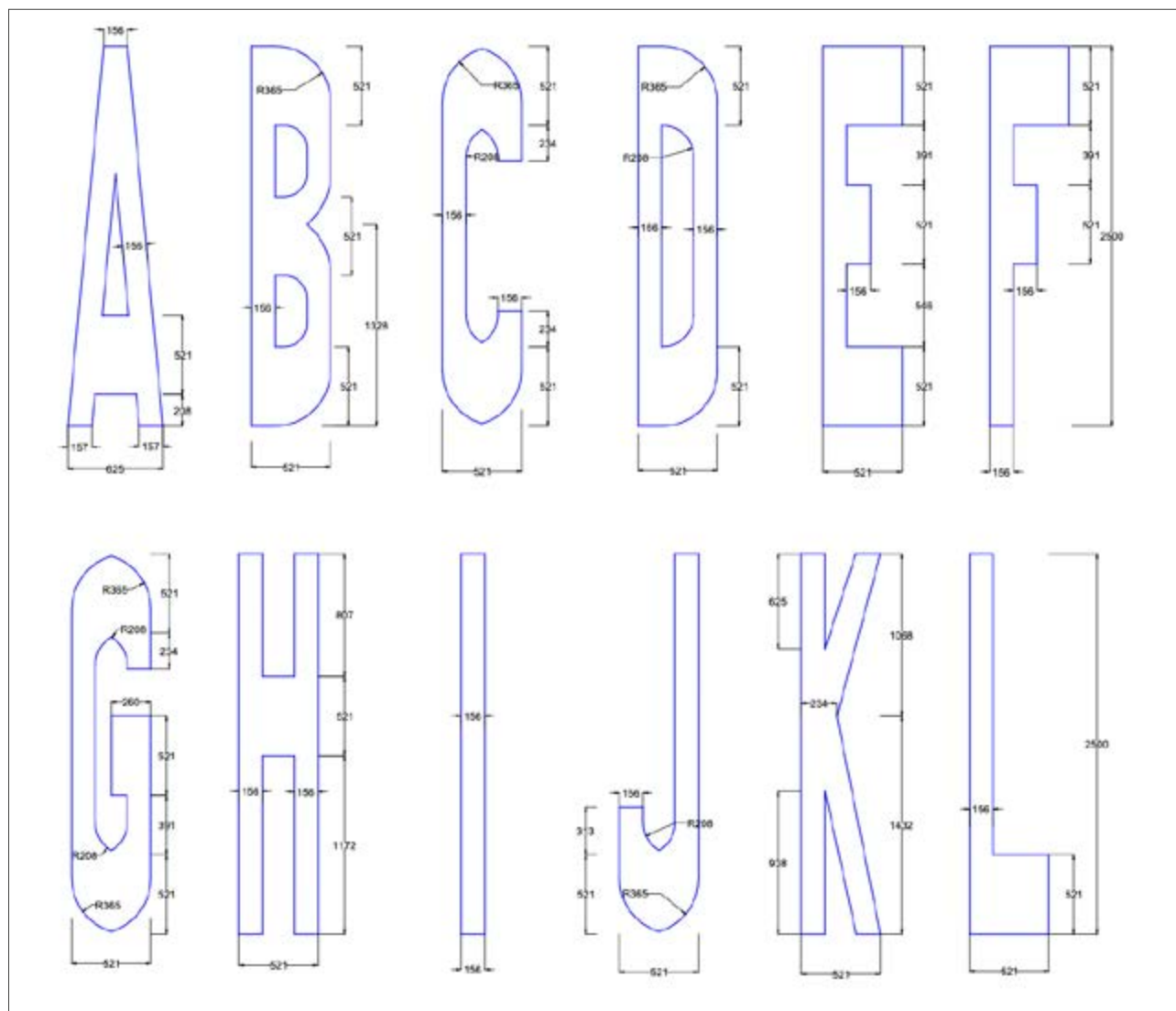
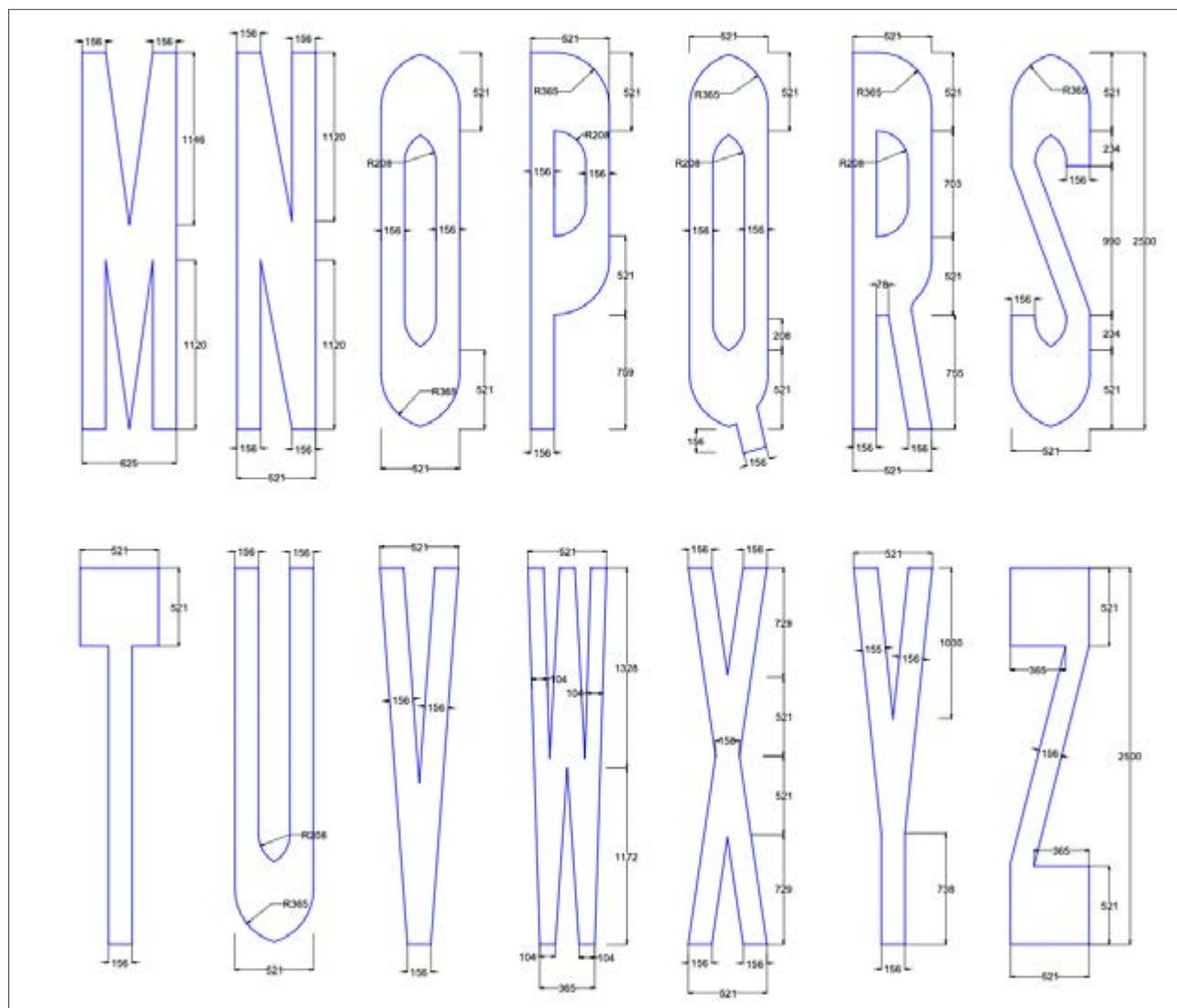
Figure C.2 Details of Letters and Numerals (*speed 51 to 100 mph*)

Figure C.2 Details of Letters and Numerals (speed 51 to 100 mph)



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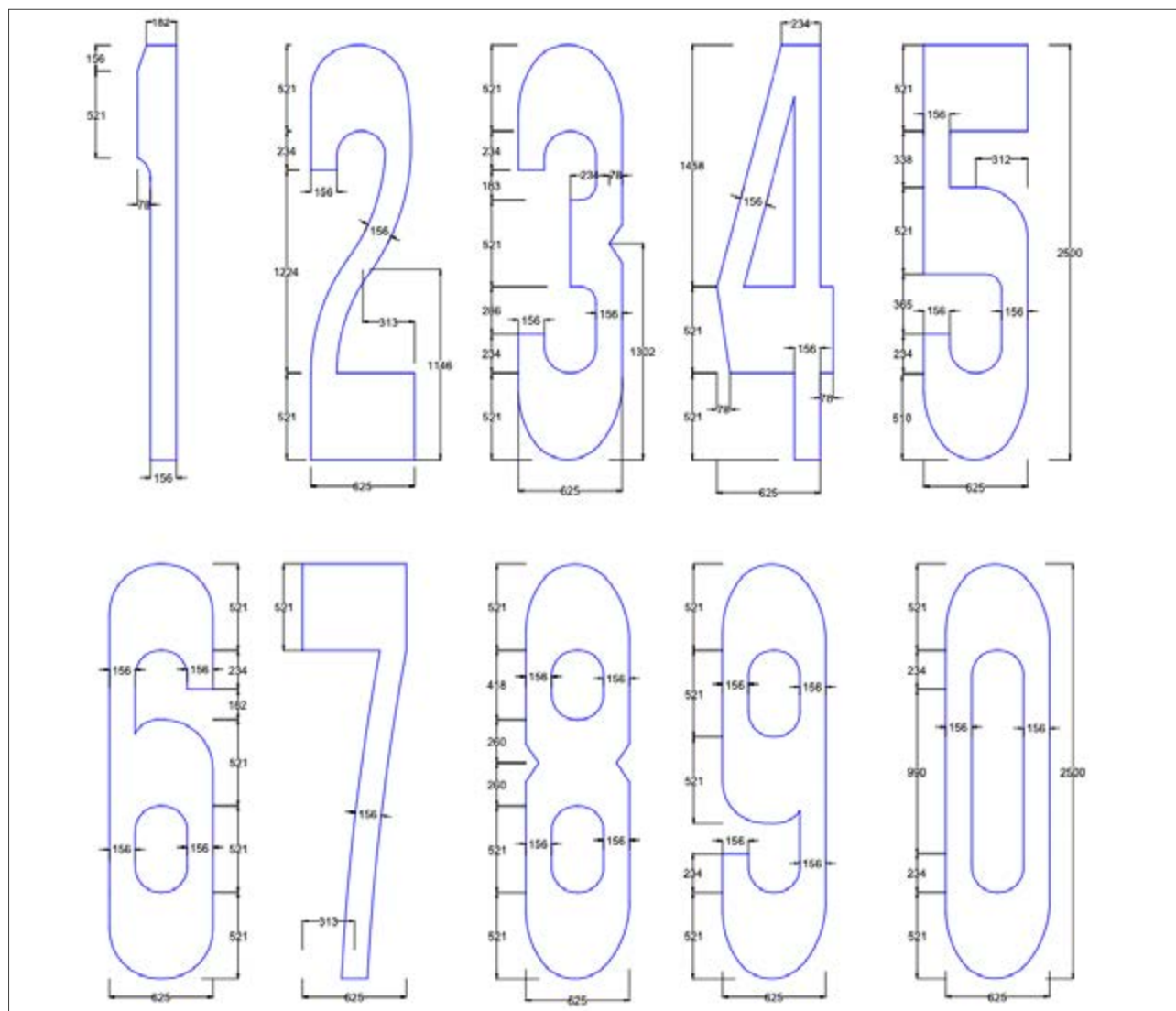
Figure C.2 Details of Letters and Numerals (*speed 51 to 100 mph*)

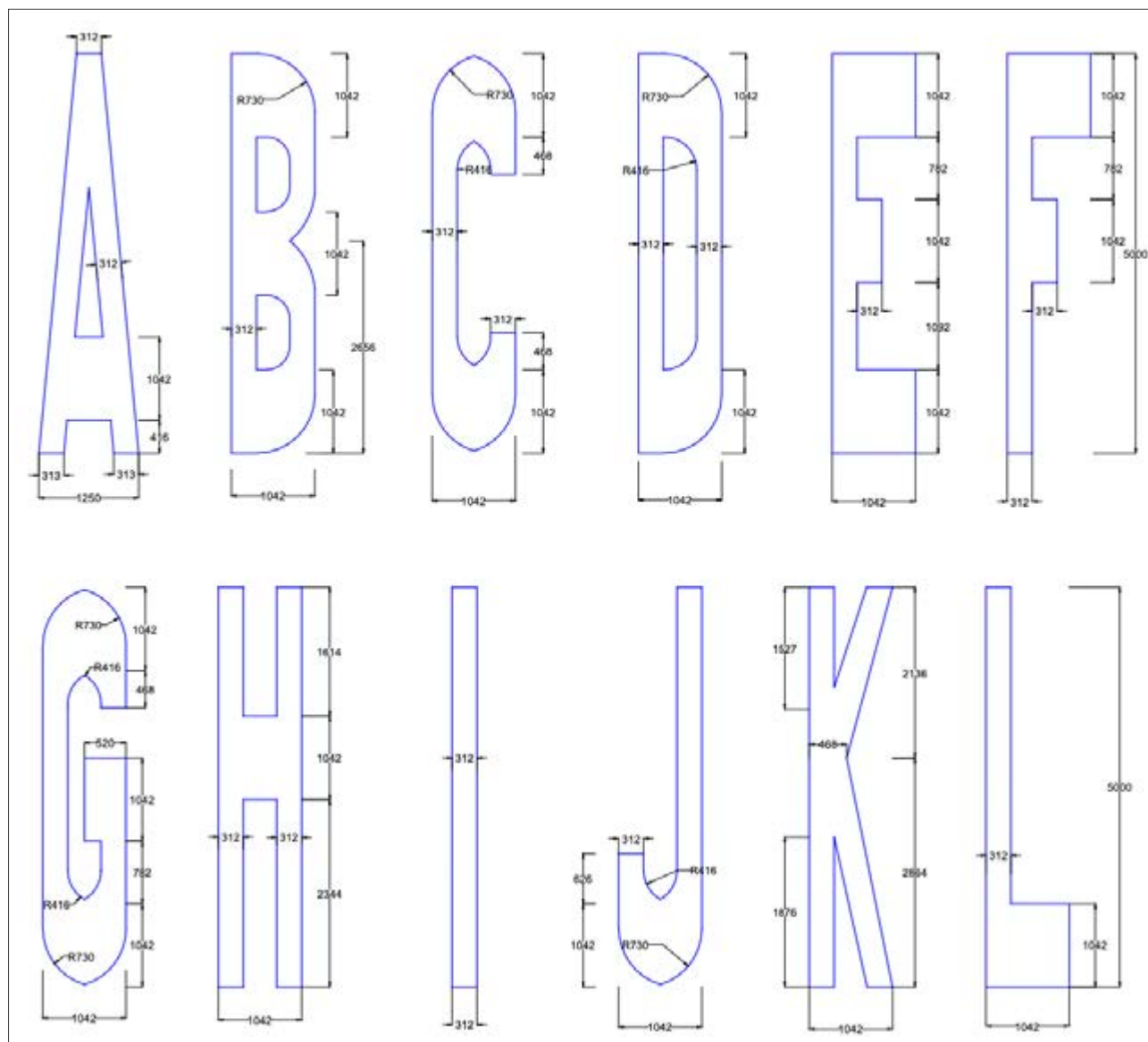
Figure C.3 Details of Letters and Numerals (*speed > 100 mph*)

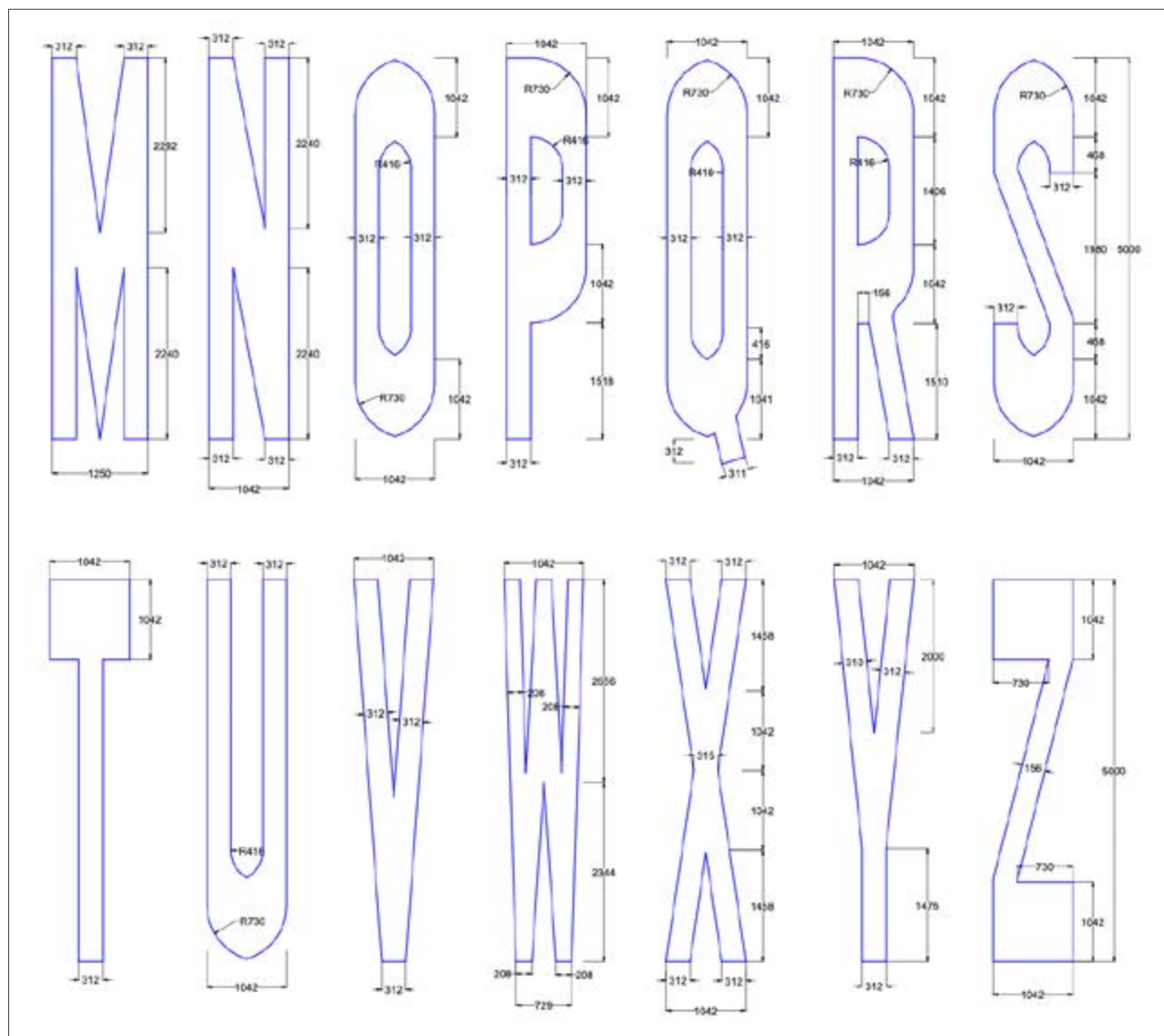
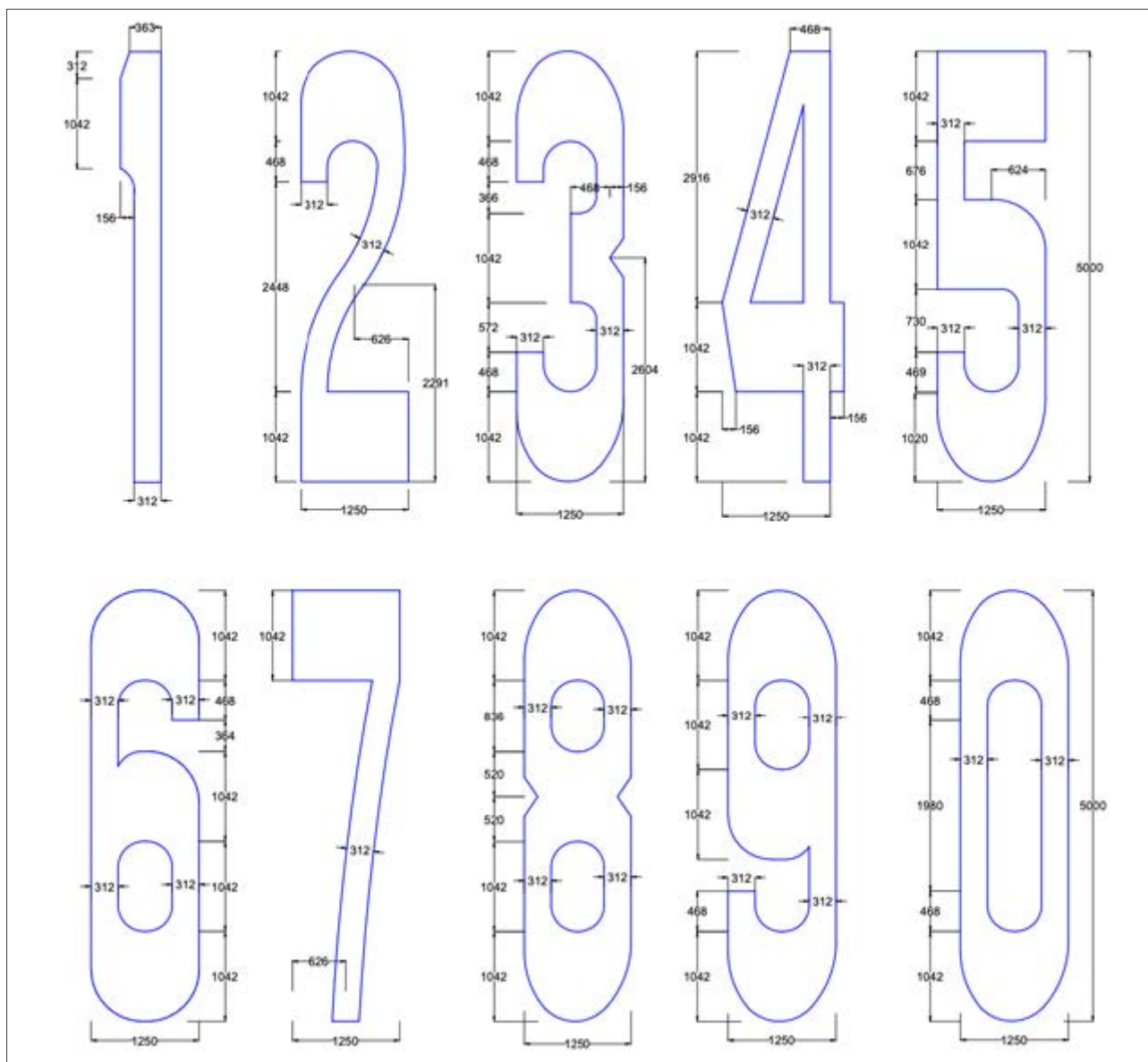
Figure C.3 Details of Letters and Numerals (*speed > 100 mph*)

Figure C.3 Details of Letters and Numerals (*speed > 100 mph*)

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Appendices

Appendix D

Measurement Method for the Luminance Coefficient

1. Introduction

Equipment for measurement of the luminance coefficient are an illumination system, a photometer and means to define a horizontal reference plane with a reference centre. Laboratory measurements are used to establish coefficient values for samples. The reference plane and the reference centre are defined by a sample holder and an alignment procedure.

The diffuse illumination is provided in a photometric sphere. A light source is mounted in the sphere in such a way, that direct illumination falls only on the lower half of the sphere. By reflection and inter-reflection, the upper half of the sphere has a close approximation to a uniform luminance.

The reference plane is the horizontal plane through the centre of the sphere. The reference centre is the centre of the sphere. In some cases, samples can be as short as 200 mm. Samples used to test or calibrate portable instruments need to be at least 40 cm long. Samples used to test or calibrate vehicle mounted equipment need to be longer. The sample can either be laid directly on a substrate, or it can be taken from a road and placed on a substrate.

Portable instruments can be used to measure Q_d values of road marking materials on sample panels before placing them on the road.

A portable instrument integrates the photometer and the illumination system. For a fixed aim instrument, the reference plane and the reference centre are defined by the aiming procedure.

When measuring road marking with a portable instrument, the coefficient value is established as the average of a number of readings taken along the marking, covering one or more spacing of markings.

Vehicle mounted instruments are used to measure coefficient values of road markings while moving at traffic speed. It can be used for longer stretches of road than portable instruments.

Vehicle mounted instruments should comply with the same requirements as for portable instruments. It should be able to cope with the movement of the vehicle and with changing conditions of daylight.

2. Spectral Match

The photometer should have a spectral response and the illumination should have a spectral emission according to standard illuminant D65, as defined in ISO 10526.

Portable instruments should have an overall spectral response that ensures correct measurement of at least white and yellow road markings. The overall spectral response can be tested by means of a white reflection standard, or a white road marking sample, and a set of long pass absorption filters providing colours of light yellow and dark yellow.

The ratio of the coefficient values obtained with a yellow filter inserted in front of the photometer, and without the filter, should be within $\pm 5\%$ of the luminous transmittance of such a filter in standard illuminant D65. The filter shall be inserted at a location, where it does not affect the illumination of the white reflection standard or road marking sample. Filters of other colours may be added to test the ability of an instrument to measure road markings of such colours.

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3. Standard Measuring Condition of Measuring Equipment

The photometer measures a field of the reference plane, which is located about the reference centre. The measured area is of minimum 50 cm².

The observation direction is the central direction of all rays from the measured area to the photometer leading to detection. The observation angle is the angle between the observation direction and the reference plane.

In the standard measuring condition, the observation angle shall be $2.29^\circ \pm 0.05^\circ$ and the total angular spread of the measuring directions shall not exceed 0.33° .

4. Practical Applications of Measuring Equipment

The equipment should be able to measure the expected range of coefficient. The range is typically 1 to 318 mcd/m²/lux.

Offset by stray light in daylight conditions may be tested by measurement in full daylight. Readings obtained without additional cover of an instrument are compared to readings obtained with additional cover by a black cloth or other obstructions about the instrument.

Portable instruments may be tilted and shifted in height because of texture and curve of road markings. For fixed-aim instruments, the sensitivity to tilts and shifts shall be tested by shifting the height position H, of the instrument parallel to a road marking. The movement of the sample is by $H / \sin 2.29^\circ = 25.H$ for method A; the sample is not to be moved for method B. For a non-glossy surface, the measured coefficient value shall not change by more than $\pm 10\%$, when the height position is shifted from '0 mm' to -1 mm, 1 mm, and 2 mm.

Aiming instruments shall meet the same requirements, but verification has to be according to the construction of the instruments. Other height positions may be included to demonstrate the ability of an instrument to measure road markings with extreme texture, in particular structured road markings.

5. Calibration of the Measuring Equipment

For laboratory measurement on road marking panels, direct calibration is performed by mounting the photometer at the reference centre, orienting it towards the upper part of the sphere. Calibration is to be for 318 mcd/m²/lux. The photometer can then be used to measure the Qd values of the road marking panels directly in mcd/m²/lux. Direct calibration is suitable, when the luminance of the upper part of the sphere is perfectly constant. Otherwise, indirect calibration is preferable. It is done on a standard sample with a known coefficient value, measured with direct calibration. The sample may be a curved mirror or another optical system that gives a view to the upper part of the sphere.

A portable instrument shall be calibrated by means of a traceable standard sample with a known coefficient value. A transfer standard may be used for routine tests of the calibration.

6. Uncertainty of Measurement

Uncertainty of measurement involves trueness (or bias) and precision (repeatability or reproducibility).

Trueness can be assured by use of measuring equipment. Precision can be determined by conventional means.

Appendix E

Procedure for the In-Situ Testing of Retroreflection Using Handheld Devices

1. General

This procedure for the use of handheld equipment is described in BS EN 1436. Avoid moving the instrument whenever there are large temperature changes. If necessary, ensure that the instrument is effectively shielded from direct sunlight. For longitudinal lines, position the equipment so that its light source shines in the same direction as a vehicle's headlights would. Measurements on other road markings (such as symbols, letters etc.) should be taken, wherever appropriate, in the general direction from approaching vehicles. All markings to be measured should be free from dirt and completely dry. A suitable brush should be used to remove loose dirt and/or loose glass beads prior to testing. Safety must be given prime consideration during this type of operation. Constant attention must be given to the safety of the operator and of other road users.

2. Selection of the Test Area

Choose an area of marking which appears representative of the total area to be assessed.

3. Number of Readings

For continuous lines, take 15 readings over a 5 m section minimum. If the marking is a centre line, take 15 readings with the equipment facing in each direction. For intermittent lines, take 5 readings per mark for 3 consecutive marks. Note readings should not be taken at the very beginning or end of the line. For markings wider than 150 mm, take readings down the central axis of the line but including some 'off-centre' also. Take care to ensure that any such 'off-centre' measurements are still made within the confines of the marking. For other markings, such as symbols, letters, transverse marking lines, reading should be taken at 5 approximately equidistant points on the surface of the marking. If a particular reading appears inconsistent another reading should be taken.

4. Recording and Interpretation of Results

A survey report for each location should be produced. Information to be included in the report should be as follows:

1. Operator's name.
2. Equipment type and geometry.
3. Test procedure.
4. General location.
5. Length of site.
6. Location of measurement point.
7. Date and time of test.
8. Ambient temperature.
9. Type and dimensions of markings.
10. Condition of marking.
11. Pre-treatment of marking (washed, brushed etc.).
12. Weather conditions at time of testing.
13. Road surface type.

Calculate an average of all the readings for a particular area. Readings for individual marks should be 'sub-totalled' before being combined.

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5. Optional Measurement of Retro Reflectivity in Condition of Wetness

In addition to the measurement method for the retro reflectivity, RL, of dry markings, EN 1436 also includes inspection methods for 'condition of wetness'. The inspection method for 'condition of wetness' is as follows:

The test condition is created using clean water poured from a bucket with an approximate capacity of 10 litres and from a height of approximately 0.5 m above the surface so that the measuring field and its surrounding area is momentarily flooded by a crest of water. The retro reflectivity shall be measured under the test condition one minute after the water has been poured.

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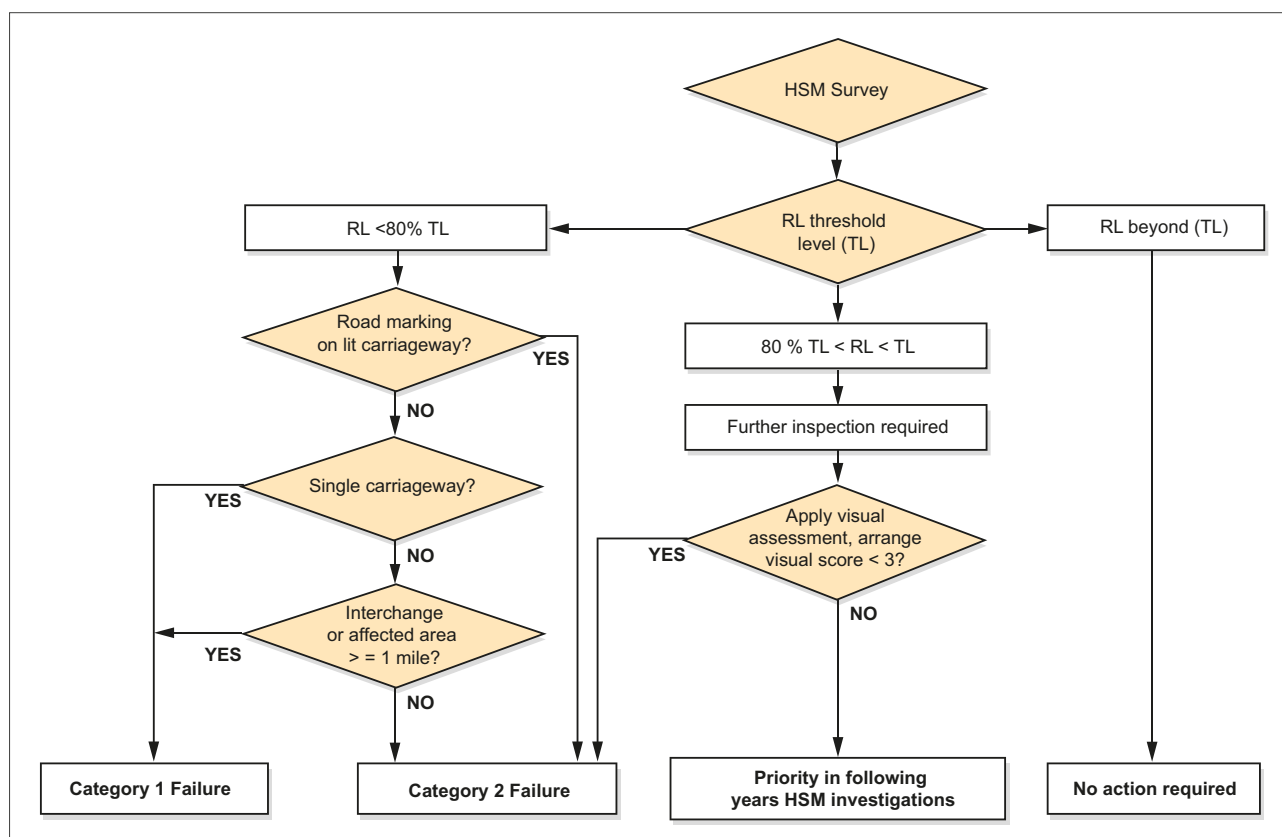
17

Appendices

Appendix F

Flow Design for Inspection Method

Figure F.1 Flow Diagram A: Assessment of Longitudinal Road Markings

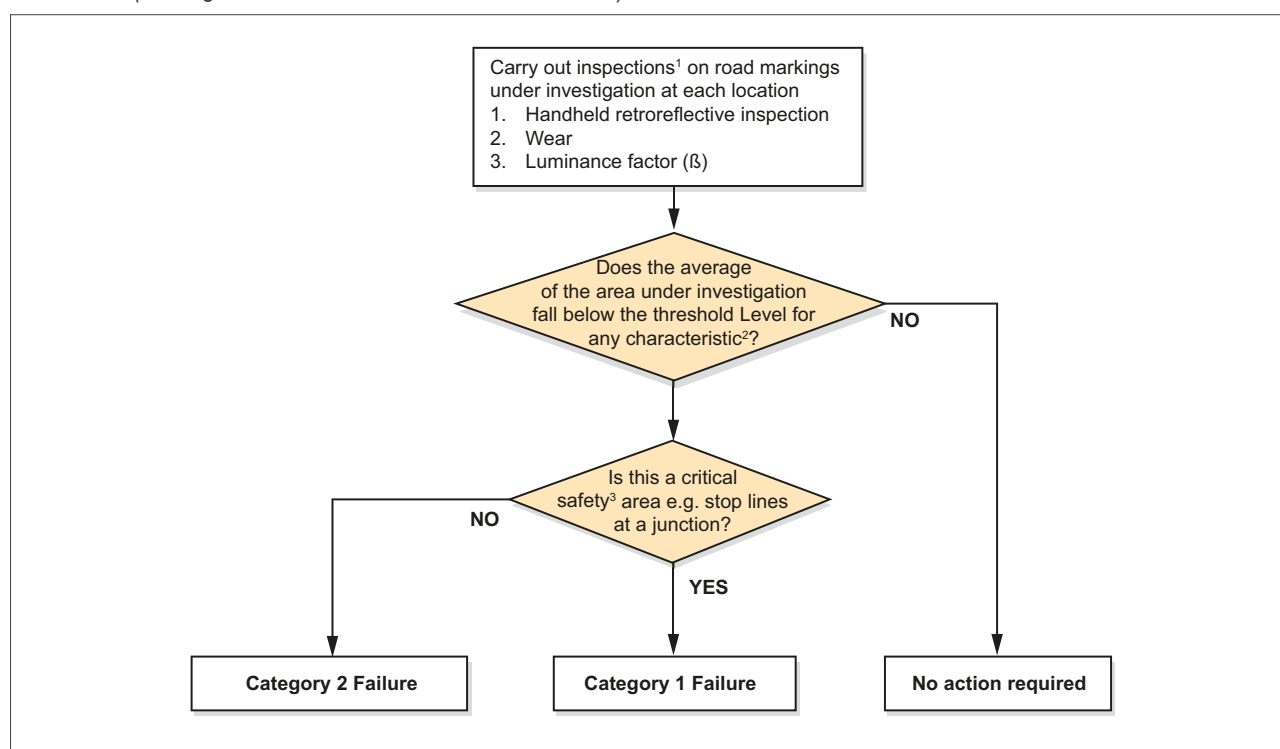


HSM – High Speed Monitor, RL – Dry Retro-reflectivity, TL- Threshold level

Note 1: HSM results should be aggregated and reported over 100m lengths.

Note 2: The visual assessment system is detailed in Appendix E and should be applied to each road marking and aggregated over 100 m length.

Figure F.2 Flow Diagram B: Assessment on areas not Accessible for Assessment by HSM
(For e.g. Side Road Junction, Exit Arrows etc.)



Note ¹: Measurements of retro reflectivity shall be carried out as set out in Appendix D. Measurements for wear shall be carried out on 50% of the road markings at each location or every 20 m for continuous road markings. Optionally, where “wet Night” markings have been laid, markings may be assessed by the inspection method.

Note ²: **Retro-reflectivity (RL):** 80 mcd/m²/lux for design speed up to 60 kmph – BS EN 1436 & Appendix D
110 mcd/m²/lux for design speed bet 60 -100 kmph – BS EN 1436 & Appendix D
150 mcd/m²/lux for design speed above 100 kmph – BS EN 1436 & Appendix D
< 35 mcd/m²/lux for optional 'wet night' assessment – BS EN 1436 & Appendix D

Wear: < 70 % of marking remaining – Visual assessment

Coefficient: < 100 mcd/m²/lux for asphalt road – BS EN 1436
< 130 mcd/m²/lux for yellow cement road – BS EN 1436

Note ³: Critical areas of the network refer to those areas of the network that may pose a risk to the road user if badly worn.

Source: Indian Roads Congress IRC 35: 2015 Code of Practice for Road Marking.

Appendix G

Measuring Method for Skid Resistance

1. Principle of the Test

The test equipment consists of a swinging pendulum fitted with a rubber slider at its free end. The energy loss caused by the friction of the slider over a specified length of the road surface is measured and the result is expressed as Skid Resistance Tester (SRT) units.

Note: The Skid Resistance Tester simulates the performance of a vehicle with patterned tyres braking with blocked wheels at 50 km/h on a wet road.

2. Description of the Skid Resistance Tester

The skid resistance tester consists of a base with three levelling screws, a vertical column with a 508 mm pendulum of 1.5 kg mass, with a spring-loaded rubber slider mounted on the end, giving a constant force of 22.2 N on the test surface. On the column, control knobs permit the vertical movement of the suspension axis. The operator is provided with means for holding and releasing the pendulum arm so that it falls freely from a horizontal position. A 300 mm long pointer indicates the position of the pendulum throughout its forward swing and indicates the measured value on a circular scale. Two friction rings are used to bring the result to the zero of this scale for a completely free swing of the pendulum arm.

3. Maintenance of the Rubber Slider

The rubber slider is of dimensions 76.2 mm 25.4 mm 6.3 mm and shall be made from rubber with the properties given in Table G.1.

Table G.1 Properties of the Rubber Slider

Temperature °C	Resilience % Lupke	Hardness IRDH ^b
0	43 to 49	55 ± 5
10	58 to 65	
20	66 to 73	
30	71 to 77	
40	74 to 79	

Note: a. Lupke rebound test in accordance with ISO 4622. b. International Rubber Hardness Degree in accordance with ISO 48

The slider can only be used for one year after the date mentioned on its side face. One slider edge can be used for at least 100 settings (500 swings). The wear of the edge should not exceed 3.2 mm horizontal and 1.6 mm vertical as shown on Figure. All new sliders should be roughened by swinging 5 times on a dry surface and 25 times on a wet surface (after adjusting the sliding length to 115 mm to 117 mm).

4. Adjustment of the Sliding Length

Prior to measurement, the sliding length shall be adjusted as follows.

Ensure that the base of the apparatus is level, with the central column in front of the centre of the area under test. Raise the head so that the pendulum arm swings free of the surface and check the zero setting in this position. If necessary, adjust the setting using the friction rings so that the pointer reading is zero. Check the sliding length (to be between 115 mm and 117 mm) by gently lowering the pendulum arm until it touches the surface on one side. Place the spacer so that the outer mark on the side corresponds with the contact line between the rubber and the surface. Raise the slider off

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the surface by means of the lifting handle, move it without friction to the other side and then gently lower onto the surface again. The contact line should be between the two marks on that side of the spacer. Adjust by raising or lowering the head. When the required height is obtained, lock the head, and place the pendulum in its release position.

5. Measuring the SRT Value

The SRT value shall be measured as follows.

The surface under test shall be cleaned with a smooth wet brush. Place the pendulum arm in its release position and the pointer in line with it. Wet the surface by applying 100 ml $\pm$ 20 ml of water on the contact area, release the arm immediately after applying the water and, after the maximum has been reached, catch the pendulum on its return swing with the left hand to avoid damage by striking the road surface. Read the position of the pointer. Replace the arm and the pointer in the release position. Repeat the same measurement five times. If the values obtained do not differ by more than 3 units, record the mean of the five readings as the SRT value. Otherwise, repeat the test until three successive readings are constant. Record the temperature of the water lying on the road just after the measurement.

6. Correction for the Temperature

The effect of temperature on rubber resilience exerts a perceptible influence in all skidding resistance measurements; it shows itself as a fall in skidding resistance as the temperature rises. In addition, the magnitude of the variation of skidding resistance with temperature varies considerably from road to road, because of the changes in road surface texture. The effect of the temperature only becomes important for tests made below 100°C, and then its main use is to give a more accurate assessment of the skidding resistance which the road is likely to offer to the tyres of vehicles, since they are likely to be running at temperatures rather higher than that of the slider rubber on the portable tester.

7. Temperature Corrections to Allow for Changes in Resilience of the Slider Rubber

To assist in the interpretation of the results, the temperature of the water lying on the road immediately after test should be recorded. It should be stressed that the change in state of polish of road surfaces throughout the year is a much bigger factor determining changes in skid resistance than is the change in temperature; the latter accounts for about one-quarter of the total seasonal change in skid resistance, which is primarily due to real and reversible changes in the road surface. In order to have an idea of the influence of all variable parameters such as temperature, slider wear etc., both before and after a series of measurements, the skid measurement should be conducted with the same slider on one or more standard samples, the value of which has previously been determined in the laboratory at 20 °C.

Note: The trueness of a measurement cannot be established as the characteristic is defined by the Skid Resistance Tester itself and not by physical laws.

Note 1: Critical areas of the network are those areas that pose a risk to the road user through skidding or potential accidents. These are:

1. GIVE WAY lines and /or STOP lines.
2. Large areas of paint, e.g., exit arrows to slips on the main line.
3. Transverse yellow bars.

The threshold levels of skid resistance are given in Table G.2.

Table G.2 Threshold Levels of Skid Resistance

Area	Threshold Level (BPN - British Pendulum Number)	Method
Normal	< 45 BPN	
Large surface areas, e.g., letters, numbers or arrows	< 55 BPN	BS EN 1436
Transverse yellow bar markings	< 55 BPN	

Source: Indian Roads Congress IRC 35: 2015

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Appendices